Electrical Circuit

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Contents

1 Electrical Circuit

I Introduction

i Convention

- Time t (s)
- Period T (s)
- Complex frequency (Laplace domain) s (Hz = 1 / s)
- Angular or radial frequency ω (Hz = 1 / s)
- Voltage v (time domain) **V** (phasor domain (in (peak) (magnitude))) $\mathbf{V}_{rms}$ (phasor domain in RMS) V (Laplace domain) (volt = V) across (跨壓)
- Current i (time domain) **I** (phasor domain) I (Laplace domain) (ampere = A) through
- Powe p (watt = W = A V)
- Real, average, or active power P (W)
- Apparent power S (volt-ampere = VA)
- Complex power **S** (VA)
- Reactive power Q (volt-ampere-reactive = VAR)
- Imaginary unit (since i is used by current) $j = \sqrt{-1}$
- Resistance R (ohm = Ω= V / A)
- Capacitance C (farad = F = A s / V = S s)
- Inductance L (henry = H = V s / A = Ωs)
- Reactance X (Ω)
- Impedance Z (Ω)
- Admittance Y (siemens = S = 1 / Ω)
- Conductance G (S)
- Susceptance B (S)
- Time constant τ (s)
- Conjugate *
- Dirac delta function $\delta(t)$
- Unit step function $u(t)$
- Impulse response function $h(t)$
- Transfer function $H(s)$

- Phase (angle) φ:

$$\varphi = 2\pi(\frac{t - t_0}{T} - \left\lfloor \frac{t - t_0}{T} \right\rfloor).$$

ii Electrical network

An electrical network or simply network is an interconnection of electrical components or a model of such an interconnection, consisting of electrical elements.

iii Electrical element

Electrical elements or simply elements are conceptual abstractions representing idealized electrical components, such as resistors, capacitors, and inductors, used in the analysis of electrical networks and denoted with circuit symbols, also called electronic symbol, in circuit diagrams.

iv Average as Cesàro mean

The term average used below is defined as Cesàro mean over $[0, \infty)$.

v Active element

An active element is an element capable of supplying a power with average over $[0, \infty)$ greater than zero to some external network.

vi Passive element

A passive element is an element that is not an active element.

vii Lumped-element model, lumped-parameter model, or lumped-component model

A simplified and idealized representation of an electrical network that assumes all components are concentrated at a single point and their behavior can be described by idealized mathematical models, that is, the lumped-matter discipline:

- The change of the magnetic flux in time through the surface surrounded by a loop is zero.
- The change of the charge in time inside conductor is zero.
- Signal timescales of interest are much larger than propagation delay of electromagnetic waves across the lumped element.

such that the system is a dynamic system whose state space is a finite dimension, called lumped-parameter system.

In contrast, distributed parameter systems have infinite-dimensional state space.

We will concentrate on lumped-parameter networks in this text.

viii Time-invariant network

Elements' properties and connections don't change over time. We will concentrate on time-invariant networks in this text.

ix Circuit element

A circuit element is a two terminal electrical network that can be completely characterized by its voltage-current relationship.

We label + and - on the two terminal repsectively and voltage v between them to define the voltage v of the element as the electric potential difference of + terminal relative to - terminal.

We label arrow and current i on it to define the reference direction of the current i of the element.

If the current arrow enters the + terminal, the element is said to follow the passive (polarity/sign) convention. Under this convention, the power absorbed (dissipated) by the element is

$$p(t) = v(t)i(t),$$

with negative power means supplying. This convention is used for passive elements.

If the current arrow enters the - terminal, the element is said to follow the active (polarity/sign) convention. Under this convention, the power supplied (generated) by the element is

$$p(t) = v(t)i(t),$$

with negative power means absorbing. This convention is used for active elements.

x Direct current (DC)

A direct current is a current that is one-directional and with constant voltage and current over time.

xi Effective value or root mean square (RMS)

The effective value or root mean square (RMS) of a voltage or current is defined as its RMS over $[0, \infty)$.

xii Steady-state sinusoidal alternating current (AC) and phasor domain

A steady-state sinusoidal alternating current (AC), often simply called alternating current (AC), is a current whose voltage $v(t)$ and current $i(t)$ are of the form:

$$v(t) = v_m \cos(\omega t + \varphi),$$

$$i(t) = i_m \cos(\omega t + \theta),$$

where ω is the angular frequency of them, and the period T and critical frequency f of them are

$$T = \frac{2\pi}{\omega},$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi}.$$

The RMSs are

$$v_{rms} = \sqrt{\frac{1}{T}\int_0^T (v(t))^2 \, \mathrm{d}t} = \frac{v_m}{\sqrt{2}},$$

$$i_{rms} = \sqrt{\frac{1}{T}\int_0^T (i(t))^2 \, \mathrm{d}t} = \frac{i_m}{\sqrt{2}}.$$

They can be represented in phasor (in (peak) (magnitude)) as

$$\mathbf{V} = v_m \angle \varphi = v_m e^{j\varphi},$$

$$\mathbf{I} = i_m \angle \theta = i_m e^{j\theta},$$

or phasor in RMS as

$$\mathbf{V}_{rms} = v_{rms} \angle \varphi = v_{rms} e^{j\varphi},$$

$$\mathbf{I}_{rms} = i_{rms} \angle \theta = i_{rms} e^{j\theta},$$

where phasors in RMS are used in power analysis, $v_m \angle \varphi$ notation is called phasor, and $v_m e^{j\varphi}$ notation is called complex number or complex magnitude. Lagging and leading mean current relative to voltage unless otherwise specified.

xiii General current and Laplace domain

Given a general current with voltage $v(t)$ and current $i(t)$ over time since $t = 0$, we can perform Laplace transforms,

$$V(s) = \mathcal{L}\{v(t)\}(s) = \int_0^\infty e^{-st} v(t) \, \mathrm{d}t \,,$$

$$I(s) = \mathcal{L}\{i(t)\}(s) = \int_0^\infty e^{-st} i(t) \, \mathrm{d}t \,,$$

to obtain its voltage $V(s)$ and current $I(s)$ as a function of Laplace domain s.

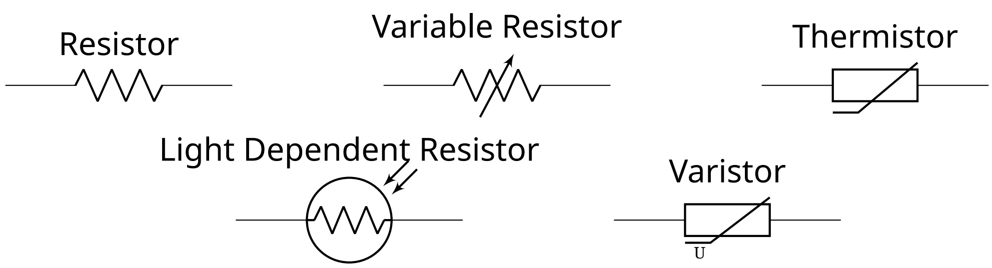
xiv Resistance and resistor

The resistance R of an element is the voltage across it over the current through it under DC:

$$v(t) = Ri(t).$$

$$V(s) = RI(s).$$

An ideal resistor is a circuit element with a resistance and no capacitance or inductance.



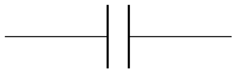
Power absorbed:

$$p(t) = \frac{(v(t))^2}{R} = (i(t))^2 R.$$

xv Capacitance and capacitor

An ideal capacitor is a circuit element with a capacitance and no resistance or inductance.

Capacitor



The capacitance C of an element is the current through it divided by the derivative of the voltage across it with respect to time, in which voltage must not be allowed to jump instantaneously from one value to another.

Time domain:

$$i(t) = C\frac{\mathrm{d}v(t)}{\mathrm{d}t}.$$

Phasor domain (steady-state sinusoidal AC with angular frequency ω):

$$\mathbf{I} = Cj\omega\mathbf{V}.$$

Laplace domain:

$$I(s) = CsV(s) - Cv(0^-).$$

$$V(s) = \frac{I(s)}{Cs} + \frac{v(0^-)}{s}.$$

A capacitor with capacitance C and initial voltage $v(0^-)$ is equivalent to a capacitance C without initial energy in parallel connection with an independent current source enforcing current $-Cv(0^-)\delta(t)$ (time domain), i.e., $-Cv(0^-)$ (Laplace domain), where the real current through the capacitor is the sum of that of the two, or a capacitance C without initial energy in series connection with an independent voltage source enforcing voltage $v(0^-)u(t)$ (time domain), i.e., $\frac{v(0^-)}{s}$ (Laplace domain), where the real voltage across the capacitor is the sum of that of the two.

Integral voltage-current relationship:

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,\mathrm{d}\tau + v(t_0{}^-) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,\mathrm{d}\tau\,.$$

Power absorbed:

$$p(t) = i(t)v(t) = Cv(t)\frac{\mathrm{d}v(t)}{\mathrm{d}t}.$$

Energy stored:

$$\int_{-\infty}^{t} p(t)\,\mathrm{d}t = \frac{1}{2}C(v(t))^2.$$

Charge in it:

$$q = CV$$

xvi Inductance and inductor

An ideal inductor is a circuit element with an inductance and no resistance or capacitance.

Inductor

The inductance L of an element is the voltage across it divided by the derivative of the current through it with respect to time, in which the current must not be allowed to jump instantaneously from one value to another.

Time domain:

$$v(t) = L\frac{\mathrm{d}i(t)}{\mathrm{d}t}.$$

Phasor domain (steady-state sinusoidal AC with angular frequency ω):

$$\mathbf{V} = Lj\omega\mathbf{I}.$$

Laplace domain:

$$V(s) = LsI(s) - Li(0^-).$$

$$I(s) = \frac{V(s)}{Ls} + \frac{i(0^-)}{s}.$$

An inductor with inductance L and initial current $i(0^-)$ is equivalent to a inductance L without initial energy in series connection with an independent voltage source enforcing voltage $-Li(0^-)\delta(t)$ (time domain), i.e., $-Li(0^-)$ (Laplace domain), where the real voltage across the inductor is the sum of that of the two, or a inductance L without initial energy in parallel connection with an independent current source enforcing current $i(0^-)u(t)$ (time domain), i.e., $\frac{i(0^-)}{s}$ (Laplace domain), where the real current through the inductor is the sum of that of the two.

Integral voltage-current relationship:

$$i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,\mathrm{d}\tau + i({t_0}^-) = \frac{1}{L}\int_{-\infty}^{t} v(\tau)\,\mathrm{d}\tau\,.$$

Power absorbed:

$$p(t) = i(t)v(t) = Li(t)\frac{\mathrm{d}i(t)}{\mathrm{d}t}.$$

Energy stored:

$$\int_{-\infty}^{t} p(t)\,\mathrm{d}t = \frac{1}{2}L(i(t))^2.$$

Flux linkage in it:

$$\lambda = LI$$

xvii Resistive element

A resistive element is an element with nonzero resistance.

xviii Reactive element

A reactive element is an element with either nonzero capacitance or nonzero inductance.

xix Purely resistive element

A pure resistive element is a resistive element with zero capacitance and zero inductance.

xx Purely reactive element

A pure reactive element is a reactive element with zero resistance.

xxi Reactance (phasor domain)

The (capacitive) reactance X_C of a one-port network with capacitance C under steady-state sinusoidal AC of angular frequency ω is defined as:

$$X_C = -\frac{1}{C\omega}.$$

The (inductive) reactance X_L of a one-port network with inductance L under steady-state sinusoidal AC of angular frequency ω is defined as:

$$X_L = L\omega.$$

The reactance X of a one-port network under steady-state sinusoidal AC of angular frequency ω is defined as:

$$X = X_L + X_C.$$

Another choice is to define X_C as $\frac{1}{C\omega}$ and $X = X_L - X_C$.

xxii Impedance (phasor domain)

The impedance Z of a one-port network under steady-state sinusoidal AC of angular frequency ω is defined as:

$$\mathbf{V} = Z\mathbf{I},$$

and is such that

$$Z = R + jX.$$

The triangle on the complex plane of Z, R, and jX showing

$$Z = R + jX$$

is called the impedance triangle.

xxiii Impedance (Laplace domain)

The impedance $Z(s)$ of a one-port network is defined as:

$$V(s) = Z(s)I(s),$$

and is such that for a one-port network with capacitance C:

$$Z(s) = \frac{1}{Cs};$$

and for a one-port network with inductance L:

$$Z(s) = Ls.$$

The impedance in phasor domain can be obtained by assigning $s = j\omega$.

xxiv Admittance (phasor domain)

The admittance Y of a one-port network is defined as the reciprocal of its impedance Z.

xxv Admittance (Laplace domain)

The admittance $Y(s)$ of a one-port network is defined as the reciprocal of its impedance $Z(s)$.

xxvi Conductance (phasor domain)

The conductance G of a circuit element is defined as the real part of its admittance.

xxvii Susceptance (phasor domain)

The susceptance B of a circuit element is defined as the imaginary part of its admittance.

xxviii Node

A node is a point in a network where two or more elements are connected.

xxix Pole or terminal

A pole or a terminal is a node that is available for connection to an external network.

xxx Path

A path is any route through which current can flow from one node to another.

xxxi Branch

A branch is a single element or a group of elements in series that connects two nodes.

xxxii Loop

A loop is any closed path in a network where you can start at a node, traverse branches, and return to the starting node without retracing any branch.

xxxiii (Essential) mesh

An (essential) mesh is a loop ℓ such that there is no other loop whose surface is a subset of the surface surrounded by ℓ.

xxxiv Electrical circuit or closed circuit

An electrical circuit is a network containing loops.

xxxv Open circuit (開路)

An open circuit is a path with infinite impedance. Open circuit voltage is denoted as v_{oc}.

xxxvi Short circuit (短路)

A short circuit is a path with zero impedance. Short circuit current is denoted as i_{sc}.

xxxvii Port (埠)

A port is a pair of terminals through which current can enter or leave a network and satisfies the port condition: the current flowing into one pole from outside is equal to the current flowing out of the other pole to outside.

An n-port network is a network with $2n$ poles forming n ports.

xxxviii Quality factor (品質因數)

The quality factor Q (or Q_0) of a network is 2π multiplied by the maximum energy stored in a period, E_m, divided by the total energy lost in the period, E_l, during natural response, that is,

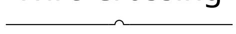
$$Q = \frac{2\pi E_m}{E_l}.$$

xxxix Wire or lead

Circuit elements in a network are connected with wires (aka leads), which are assumed to have zero impedance.

Wire

Wire Crossing



Adding a wire directly between two nodes is called adding a short circuit.

xl Kirchhoff's current law (KCL), Kirchhoff's first law, or Kirchhoff's junction rule

The algebraic sum of the currents entering any node is zero.

xli Kirchhoff's voltage law (KVL), Kirchhoff's second law, or Kirchhoff's loop rule

The algebraic sum of the voltages around any closed path is zero.

xlii Planar network

A network that is planar as a graph with nodes being vertices and branches as edges.

xliii Series connection

Two or more elements are connected in series if they are connected along a single path, and each component has the same current through it, equal to the current through the network. The voltage across the network is equal to the sum of the voltages across each element. The current through the network and the current through each element is the same.

xliv Parallel connection

Two or more elements are connected in parallel if they are connected along multiple paths from one node to another, and each component has the same voltage across it, equal to the voltage through the network. The current through the network is equal to the sum of the current through each element. The voltage across the network and the voltage across each element is the same.

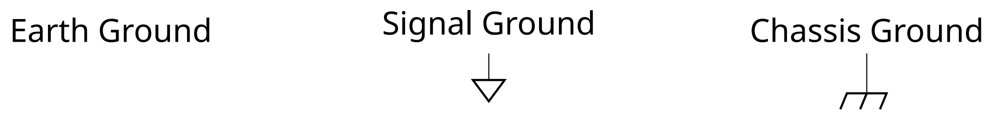
xlv Bilateral network

A bilateral electrical network is an electrical network whose behaviors remain the same regardless of the direction of current flow or applied voltage.

xlvi Ground

A ground is a node designated to have 0 electric potential. All other electric potential in the network are measured relative to the grounds.

- **Earth ground**: Earth ground is a point in the network that is physically connected to the Earth.
- **Signal ground**: Signal ground is a ground in the network provided by an external signal, which can be float relative to Earth and such circuit is called float. An example of signal ground is a GND pin on a microcontroller board.
- **Chassis ground**: Chassis ground is a point in the circuit that is physically connected to a devices's metal frame. The metal frame acts as a shield against electromagnetic interference and can be connected to Earth for safety in AC-powered devices.



xlvii Operating point, Q-point, quiescent point, or bias point

The steady-state DC voltage and current values of a circuit when no input signal is applied.

xlviii Biasing resistor

A biasing resistor is a resistor used to achieve a desired operating point.

xlix SPICE

SPICE (Simulation Program with Integrated Circuit Emphasis) is a general-purpose, open-source analog electronic circuit simulator. It is a program used in integrated circuit and board-level design to check the integrity of circuit designs and to predict circuit behavior.

II Souce

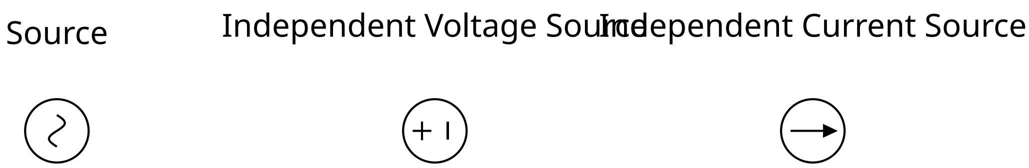
i Ideal sources

Ideal voltage sources are active elements with zero output resistance. To turn off an ideal voltage source means replace it with short circuit.

Ideal current sources are active elements with infinite output resistance. To turn off an ideal current source means replace it with open circuit.

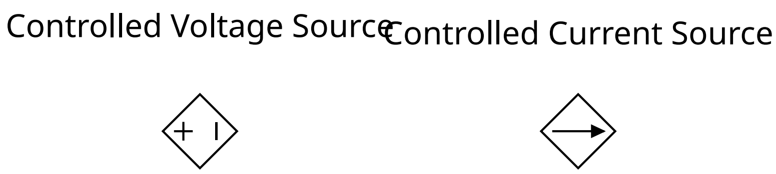
ii Independent source

- An independent voltage source enforces a prescribed DC or sinusoidal AC voltage v_s across its terminals.
- An independent current source enforces a prescribed DC or sinusoidal AC current i_s through its terminals.



iii Linear dependent or controlled source

- A voltage-controlled voltage source enforces a DC or sinusoidal AC voltage $v_s = \mu v_x$ across its terminals, where v_x is control voltage and real for DC or complex for AC constant μ is dimensionless voltage gain.
- A current-controlled voltage source enforces a DC or sinusoidal AC voltage $v_s = r i_x$ across its terminals, where i_x is control current and real for DC or complex for AC constant r is transresistance.
- A voltage-controlled current source enforces a DC or sinusoidal AC current $i_s = g v_x$ through its terminals, where v_x is control voltage and real for DC or complex for AC constant g is transconductance.
- A current-controlled current source enforces a DC or sinusoidal AC current $i_s = \beta i_x$ through its terminals, where i_x is control current and real for DC or complex for AC constant β is dimensionless current gain.



iv Practical souce

A practical independent voltage source can be approximated with an ideal independent voltage source in series connection with a resistor.

A practical independent current source can be approximated with an ideal independent current source in parallel connection with a resistor.

III Linear circuit

i Linear element

A linear element is a passive element whose inputs $\mathbf{x}$ and outputs $\mathbf{F}(\mathbf{x})$ satisfy linearity:

$$\mathbf{F}(a\mathbf{x} + b\mathbf{y}) = a\mathbf{F}(\mathbf{x}) + b\mathbf{F}(\mathbf{y})$$

ii Linear network

A linear network is a network whose inputs $\mathbf{x}$ and outputs $\mathbf{F}(\mathbf{x})$ satisfy linearity:

$$\mathbf{F}(a\mathbf{x} + b\mathbf{y}) = a\mathbf{F}(\mathbf{x}) + b\mathbf{F}(\mathbf{y})$$

A linear network consists of only linear elements, independent sources, and linear dependent sources.

iii Linear circuit

A linear network is a circuit whose inputs $\mathbf{x}$ and outputs $\mathbf{F}(\mathbf{x})$ satisfy linearity:

$$\mathbf{F}(a\mathbf{x} + b\mathbf{y}) = a\mathbf{F}(\mathbf{x}) + b\mathbf{F}(\mathbf{y})$$

A linear circuit consists of only linear elements, independent sources, and linear dependent sources.

iv One-port linear element

A one-port linear element is a passive element with resistance R, capacitance C, and inductance L that are independent of voltage and current, that is, equivalent to a one-port network consists of ideal resistors, capacitors, and inductors, that is, the voltage-current relationship is a Laplace-domain equation $V(s) = Z(s)I(s)$ with voltage V, current I, and impedance Z.

v Time-invariant circuit

A circuit whose outputs as a function of the inputs is time-invariant.

vi Response of linear time-invariant circuit

- (Total or complete) response: the reponse.
- Natural, homogeneous, unforced, or zero-input response (ZIR): The response with all independent sources in the linear circuit turned off and initial states preserved. The ZIR is the solution of an IVP of a linear homogenous ODE, called governing equation, with initial states given by the initial states of inductors and capacitors.

- Forced, driven, particular, or zero-state responses (ZSR): A response with only one of the independent sources in the linear circuit, called forcing function, is turned on, and with zero initial state. Each ZSR is the response of a causal continuous-time linear time-invariant (LTI) systems with its forcing function as forcing function and the complementary equation of its governing equation being the governing equation of ZIR.
- Superposition principle: The sum of ZIR and all ZSRs is the total response.
- Transient response: A response component that converges to an equilibrium as t gets large.

vii Time constant, rising time, and falling time

For a response function in the form $y(t) = y_0 e^{-\alpha t}$ with constants $y_0 \neq 0$ and $\alpha > 0$, the time constant τ is defined as $\frac{1}{\alpha}$, that is, $y(\tau) = y_0 e^{-1}$. For the smallest positive integer (or rational number to a number of decimals) k such that $\frac{y(k\tau)}{y_0} < 0.01$ (or other designated number), we typically call that $y(t)$ decays to 0 about k times time constants. The falling time or pulse decay time is defined as the time $y(t)$ changes from $0.9y_0$ to $0.1y_0$, which is approximately 22τ.

For a response function in the form $y(t) = y_0(1 - e^{-\alpha t})$ with constants $y_0 \neq 0$ and $\alpha > 0$, the time constant τ is defined as $\frac{1}{\alpha}$, that is, $y(\tau) = y_0(1 - e^{-1})$. For the smallest positive integer (or rational number to a number of decimals) k such that $\frac{y(k\tau)}{y_0} > 0.99$ (or other designated number), we typically call that $y(t)$ decays to 0 about k times time constants. The rising time or pulse grow time is defined as the time $y(t)$ changes from $0.1y_0$ to $0.9y_0$, which is approximately 22τ.

viii Series connection

For linear networks, the impedance, reciprocal of capacitance, and inductance of the network are equal to the sums of those of each element. When steady, the charges in each capacitor and the equivalent capacitor are the the same.

ix Parallel connection

For linear networks, the admittance, capacitance, and reciprocal of inductance of the network are equal to the sums of those of each element. When steady, the flux linkage in each inductor and the equivalent inductor are the the same.

x Linear circuit as graph

A linear circuit is a graph with nodes with all voltage sources enclosed as supernodes being vertices, branches without nodes inside except ends being edges, and admittances (or conductance for DC) being weights.

xi Duality

Two linear circuit are dual if their governing equations in terms of variables become identical after systematically exchanging voltage and current, capacitance and inductance, impedance and admittance, open circuit and short circuit, KCL and KVL, Thévenin's theorem and Norton's theorem, series and parallel, node and loop, etc.

Two linear circuit are exact dual if their governing equations in terms of numerical values become identical after systematically exchanging voltage and current, capacitance and inductance, impedance and admittance, open circuit and short circuit, KCL and KVL, Thévenin's theorem and Norton's theorem, series and parallel, node and loop, etc.

IV Thévenin's theorem and Norton's theorem

i Thévenin's theorem

Any linear network with two terminals is equivalent to a voltage source with a voltage called Thévenin voltage in a series connection with a linear element (or resistor for DC) with an impedance (or resistance for DC) called Thévenin, series, or equivalent impedance (or resistance for DC), such equivalent circuit is called Thévenin equivalent.

ii Norton's theorem

Any linear network with two terminals is equivalent to a current source with a current called Norton current in a parallel connection with a linear element (or resistor for DC) with an impedance (or resistance for DC) called Norton, parallel, or equivalent impedance (or resistance for DC), such equivalent circuit is called Norton equivalent.

iii Finding Thévenin voltage

Remove all loads, and the open-circuit voltage v_{oc} between the two terminals is the Thévenin voltage.

iv Finding Norton current

Add a short circuit between the two terminals, and the short circuit current i_{sc} on it is the Norton current.

v Finding Thévenin impedance or Norton impedance

Thévenin and Norton impedance Z_{th} (or resistance R_{th} for DC) are the same and can be found with:

- **Source deactivation method**: Replace all independent voltage sources with short circuits and independent current sources with open circuits,
  - If there are no dependent sources, the equivalent impedance between the two terminals of the dead network is the Thévenin impedance.
  - **Driving-point impedance method or test source method**: Otherwise, put an arbitrary nonzero-valued independent voltage V_t or current I_t source between the

two terminals, and the voltage divided by the current of the added source $\frac{V_t}{I_t}$, called driving-point impedance, is the Thévenin impedance.

- **Division method**: The Thévenin impedance or Norton impedance is the Thévenin voltage divided by the Norton current.

vi Thévenin parameters

Thévenin voltage, Norton current, and Thévenin impedance are called Thévenin parameters.

vii Maximum power theorem under DC

A one-port linear network delivers maximum power to a load resistance when the load resistance is equal to the Thévenin resistance R_{th} of the network, and the power delivered is

$$P = \frac{V_{oc}{}^2}{4R_{th}}.$$

V (Matrix) nodal analysis or node-voltage analysis of linear circuits

i (Nodal) admittance matrix

The (nodal) admittance (conductance for DC) matrix **Y** of a linear circuit that is a simple graph is the Laplacian matrix of it with all sources omitted, admittances as weights, and circuit as weighted undirected graph.

ii Method

1. Choose a reference node and assign a voltage for it (typically 0).
2. For each pair of nodes, if there exists an edge with zero impedance and no source, merge the two nodes, and discard all edges between them.
3. For each pair of nodes, use series and parallel connections to combine all edges without sources between them.
4. For each collection of voltage sources in series connection between two nodes, combine them to one.
5. For each collection of current sources in parallel connection between two nodes, combine them to one.
6. For each pair of node with a voltage source in series connection with an nonzero impedance (resistance for DC) connecting them, convert it to its Norton equivalent.
7. For each voltage source whose either endpoint is the reference node, the nodal voltage of the other endpoint is immediately known.
8. For each node with unknown voltage, assign a nodal voltage variable.

9. For each voltage source whose neither endpoint is the reference node, enclose both nodes and the source as a supernode.
10. Omitting all voltage sources, form a system of linear equations given by KCL:

$$\mathbf{Y}\mathcal{V} = \mathcal{I},$$

    where $\mathcal{V}$ is the column vector of the nodal voltage variables and known nodal voltages, $\mathcal{I}$ is the column vector of the sum of currents injected into the nodes from current sources with VCCSs and CCCSs directly replaced with the control functions, and $\mathbf{Y}$ is the (nodal) admittance (conductance for DC) matrix of the circuit. Nodes with known nodal voltage and current injected to it are omitted from the equation, where edges to them still count in $\mathbf{Y}$ but columns and rows of them are omitted.
11. For each voltage source, append the equation to the system with the enforced voltage under $\mathcal{I}$ and the row vector of weights of nodal voltage variables under $\mathbf{Y}$.
12. Solve the system of linear equations.

VI (Matrix) mesh analysis of planar linear circuit

i (Mesh) impedance matrix

The (mesh) impedance (resistance for DC) matrix $\mathbf{Z}$ of a linear circuit that is a simple planar graph is the matrix with entry Z_{ii} being the sum of the impedances of the branches in the ith mesh and Z_{ij} for $i \neq j$ being the sum of the impedances of the branches shared between the ith and jth meshes.

ii Method

1. For each pair of nodes, if there exists an edge with zero impedance and no source, merge the two nodes, and discard all edges between them.
2. For each pair of nodes, use series and parallel connections to combine all edges without sources between them.
3. For each collection of voltage sources in series connection between two nodes, combine them to one.
4. For each collection of current sources in parallel connection between two nodes, combine them to one.
5. For each pair of node with a current source in parallel connection with an nonzero impedance (resistance for DC) connecting them, convert it to its Thévenin equivalent.
6. Indentify all meshes.
7. Assign an arbitrary but consistent direction (usually clockwise) of current for all meshes, and assign a mesh current variable for each mesh.
8. For each current source, if it lies on two meshes, exclude the edge with the current source, and merge the two meshes into one, called a supermesh.

9. Omitting all current sources, form a system of linear equations given by KVL:

$$\mathbf{Z}\mathcal{I} = \mathcal{V},$$

where $\mathcal{I}$ is the column vector of the mesh current variables, $\mathcal{V}$ is the column vector of the sum of voltage delivered from the voltage sources in the mesh with mesh current variable direction as positive direction and with VCVSs and CCVSs directly replaced with the control functions, and **Z** is the (mesh) impedance (resistance for DC) matrix of the circuit.

10. For each current source, append the equation to the system with the enforced current under $\mathcal{V}$ and the row vector of weights of mesh current variables under **Z**.

11. Solve the system of linear equations.

VII Transformations

i Source transformation

An independent voltage source V in series connection with an impedance Z is equivalent to an independent current source $I = \dfrac{V}{Z}$ in parallel connection with the same impedance Z.

ii Y-Δ transformation and Δ-Y transformation

- Y (wye or star) network: a common central node N, three terminals N_1, N_2, N_3, and three circuit elements with impedances Z_1, Z_2, Z_3 connecting N_1 and N, N_2 and N, and N_3 and N respectively.
- Δ (delta or triangle) network: three terminals N_1, N_2, N_3, and three circuit elements with impedances Z_{12}, Z_{23}, Z_{31} connecting N_1 and N_2, N_2 and N_3, and N_3 and N_1 respectively.
- Y-Δ transformation: The two networks are equivalent if

$$Z_{12} = Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3},$$

$$Z_{23} = Z_2 + Z_3 + \frac{Z_2 Z_3}{Z_1},$$

$$Z_{31} = Z_3 + Z_1 + \frac{Z_3 Z_1}{Z_2}.$$

- Δ-Y transformation: The two networks are equivalent if

$$Z_1 = \frac{Z_{12} Z_{31}}{Z_{12} + Z_{23} + Z_{31}},$$

$$Z_2 = \frac{Z_{23} Z_{12}}{Z_{12} + Z_{23} + Z_{31}},$$

$$Z_3 = \frac{Z_{31} Z_{23}}{Z_{12} + Z_{23} + Z_{31}}.$$

iii Star-mesh transformation

For all integer $n \geq 3$:

- n-star network: a common central node N, n terminals $N_1, N_2, \dots, N_n$, and n circuit elements with impedances Z_i connecting N_i and N for all $i \in \mathbb{N} \wedge i \leq n$.
- n-mesh (complete graph) network: n terminals $N_1, N_2, \dots, N_n$, and $\frac{n(n-1)}{2}$ circuit elements with impedances Z_{ij} connecting N_i and N_j for all $i, j \in \mathbb{N} \wedge i < j \leq n$.
- Star-mesh transformation: The two networks are equivalent if

$$Z_{ij} = Z_i + Z_j + \frac{Z_i Z_j}{-Z_i - Z_j + \sum_{k=1}^{n} Z_k}.$$

VIII Transient response

i Capacitor transient response

Capacitor with initial voltage $v(0^-)$ under DC has time constant $\tau = R_{th}C$ where R_{th} is the Thévenin resistance seen by the capacitor, voltage $v(\infty)$ and current 0 at steady state, and transient response:

$$v(t) = v(\infty) + (v(0^-) - v(\infty))e^{-\frac{t}{\tau}}$$
$$i(t) = \frac{v(\infty) - v(0^-)}{R_{th}} e^{-\frac{t}{\tau}}$$

ii Inductor transient response

Inductor with initial current $i(0^-)$ under DC has time constant $\tau = \frac{L}{R_{th}}$ where R_{th} is the Thévenin resistance seen by the inductor, current $i(\infty)$ and voltage 0 at steady state, and transient response:

$$i(t) = i(\infty) + (i(0^-) - i(\infty))e^{-\frac{t}{\tau}}$$
$$v(t) = (i(\infty) - i(0^-))R_{th} e^{-\frac{t}{\tau}}$$

IX Steady-state sinusoidal AC response

i Capacitor response

For a capacitor, if

$$v(t) = v_m \cos(\omega t + \theta),$$
$$\mathbf{V} = v_m e^{j\theta},$$

then

$$i(t) = C\omega v_m \cos\left(\omega t + \theta + \frac{\pi}{2}\right),$$
$$\mathbf{I} = C\omega v_m e^{j(\theta + \frac{\pi}{2})}.$$

The current leads voltage by $\frac{\pi}{2}$, and the higher angular frequency ω is, the lower $\frac{v_m}{i_m} = \frac{1}{C\omega}$ is.

ii Inductor response

For an inductor, if

$$i(t) = i_m \cos(\omega t + \theta),$$

$$\mathbf{I} = i_m e^{j\theta},$$

then

$$v(t) = L\omega i_m \cos\left(\omega t + \theta + \frac{\pi}{2}\right),$$

$$\mathbf{V} = L\omega i_m e^{j(\theta + \frac{\pi}{2})}.$$

The current lags voltage by $\frac{\pi}{2}$, and the higher angular frequency ω is, the higher $\frac{v_m}{i_m} = L\omega$ is.

iii Second-order circuit

A natural response $f(t)$ of a second-order circuit has the form

$$f''(t) + 2\omega_0\zeta f'(t) + {\omega_0}^2 f(t) = 0,$$

and its characteristic equation is

$$r^2 + 2\omega_0\zeta r + {\omega_0}^2 = 0,$$

where $\alpha = \omega_0\zeta$ is called neper frequency (奈培頻率) or attenuation (衰減量) and ζ is called damping factor/ratio (阻尼係數/比).

The characteristic roots r are

$$r_{1,2} = -\omega_0(\zeta \pm \sqrt{\zeta^2 - 1}).$$

Natural response (i.e. $x(t) = 0$) is

- $\zeta > 1$ Overdamped (過阻尼) response:

$$f(t) = Ae^{-\omega_0(\zeta - \sqrt{\zeta^2 - 1})t} + Be^{-\omega_0(\zeta + \sqrt{\zeta^2 - 1})t}$$

- $\zeta = 1$ Critically damped (臨界阻尼) response:

$$f(t) = (A + Bt)e^{-\omega_0 t}$$

- $0 < \zeta < 1$ Underdamped (欠/輕/次阻尼) response:

$$f(t) = e^{-\omega_0\zeta t}(A\cos\left(\omega_0\sqrt{1 - \zeta^2}t\right) - B\sin\left(\omega_0\sqrt{1 - \zeta^2}t\right))$$

The damped oscillation of the underdamped response to a constant source has damped frequency ω_d:

$$\omega_d = \omega_0\sqrt{1 - \zeta^2}$$

and the response is

$$f(t) = Ae^{-\alpha t}\cos\left(\omega_d t + \varphi\right).$$

The energy stored must be in the form

$$E(t) = Ke^{-2\alpha t}\cos^2\left(\omega_d t + \theta\right)$$

$$E_m = Ke^{-2\alpha t_n}$$

$$E_l = Ke^{-2\alpha t_n} - Ke^{-2\alpha(t_n+\frac{2\pi}{\omega_d})} = E_m(1-e^{-\frac{4\pi\alpha}{\omega_d}}) = E_m(1-e^{-\frac{4\pi\zeta}{\sqrt{1-\zeta^2}}})$$

$$Q = \frac{2\pi E_m}{E_l} = \frac{2\pi}{1-e^{-\frac{4\pi\zeta}{\sqrt{1-\zeta^2}}}}$$

iv High-Q circuit

For circuit with large Q, i.e., small ζ, called high-Q circuit, which we usually take $Q > 5$, i.e., $\zeta < 0.1$, we have $\zeta \approx 0$, and $1 - e^{-x} \approx x$ for $x \approx 0$, thus

$$Q \approx \frac{2\pi}{\frac{4\pi\zeta}{\sqrt{1-\zeta^2}}} = \frac{\sqrt{1-\zeta^2}}{2\zeta} \approx \frac{1}{2\zeta}$$

X Steady-state sinusoidal AC power analysis

i Phasor in RMS

Suppose

$$v(t) = v_m \cos(\omega t + \theta),$$

$$i(t) = i_m \cos(\omega t + \theta - \varphi).$$

The phasors used here are

$$\mathbf{V}_{rms} = v_{rms}e^{j\theta} = \frac{v_m}{\sqrt{2}}e^{j\theta} = \frac{\mathbf{V}}{\sqrt{2}},$$

$$\mathbf{I}_{rms} = i_{rms}e^{j(\theta-\varphi)} = \frac{i_m}{\sqrt{2}}e^{j(\theta-\varphi)} = \frac{\mathbf{I}}{\sqrt{2}}.$$

ii Instantaneous power

$$\begin{aligned} p(t) = v(t)i(t) &= v_m i_m \cos(\omega t + \theta)\cos(\omega t + \theta - \varphi) \\ &= v_{rms}i_{rms}(\cos\varphi + \cos(2\omega t + 2\theta - \varphi)) \end{aligned}$$

iii Complex power

The complex power **S** with unit volt-ampere (VA) of a steady-state sinusoidal AC is defined as

$$\mathbf{S} = \mathbf{V}_{rms}\mathbf{I}_{rms}{}^* = v_{rms}i_{rms}e^{j\varphi}.$$

iv Real, average, or active power

The real, average, or active power P with unit watt (W) of a steady-state sinusoidal AC is defined as the average power of it, that is,

$$P = \langle p(t)\rangle = v_{rms}i_{rms}\cos\varphi = \Re(\mathbf{S}).$$

Average power of any pure reactive one-port network is 0.

v Apparent power

The apparent power S with unit volt-ampere (VA) of a steady-state sinusoidal AC is defined as

$$S = |\mathbf{S}| = v_{rms} i_{rms}.$$

vi Power factor (PF, pf, or P.F.)

The power factor (PF) of a steady-state sinusoidal AC is defined as

$$\text{PF} = \frac{P}{S} = \cos\varphi.$$

PF = 1 is called ideal. We also often write lagging or leading after power factor.

vii Reactive power

The reactive power Q with unit volt-ampere-reactive (VAR or VAr) of a steady-state sinusoidal AC is defined as

$$Q = \Im(\mathbf{S}) = v_{rms} i_{rms} \sin\varphi,$$

which is positive for lagging and negative for leading. It's the power that oscillates between sources and reactive elements.

viii Power triangle

The triangle on the complex plane of **S**, P, and jQ showing

$$\mathbf{S} = P + jQ$$

is called the power triangle.

ix Source power factor

A source's power factor is in $[-1, 1]$, and is ± 1 iff the sum of Q of all other elements is 0, and 0 iff the sum of P of all other elements is 0.

x Maximum power theorem under steady-state sinusoidal AC

A one-port linear network delivers maximum power to a load impedance when the load impedance is equal to the conjugate of the Thévenin impedance Z_{th} of the network, and the average power delivered is

$$P = \frac{|\mathbf{V}_{oc}|^2}{8\Re(Z_{th})} = \frac{|\mathbf{V}_{oc,rms}|^2}{4\Re(Z_{th})}.$$

xi Wattmeter

A wattmeter measuring a load Z contains a current-sensing coil with current $i_i(t)$ in series connection with the load, and a voltage-sensing coil with current $i_v(t)$ in series connection with an impedance Z_m, in parallel connection with the load. Electrodynamic interaction produce a torque proportional to $i_v(t)i_i(t)$ on the rotating pointor. The steady-state deflection angle γ of the rotating pointor is proportional to the average of $i_v(t)i_i(t)$:

$$\gamma = \frac{K_M}{T} \int_{t_1}^{t_1+T} i_v(t)i_i(t)\,\mathrm{d}t\,,$$

where K_M is a proportionality constant and T is the period of the current. The impedance Z_m is a large resistance $R_m \gg |Z|$. If the load has

$$v(t) = \sqrt{2}v_{rms}\cos(\omega t + \theta)$$

$$i(t) = \sqrt{2}i_{rms}\cos(\omega t + \theta - \varphi),$$

then

$$i_v(t) = \frac{\sqrt{2}v_{rms}}{R_M}\cos(\omega t + \theta)$$

$$i_i(t) \approx \sqrt{2}i_{rms}\cos(\omega t + \theta - \varphi).$$

Then,

$$\gamma \approx \frac{K_M}{R_M}v_{rms}i_{rms}\cos\varphi = \frac{K_M}{R_M}P,$$

and reads approximately P.

An ideal wattmeter reads exactly P and $i_v \to 0$.

By convention, the current input terminal is labeled $\pm$.

xii Balanced three-phase circuit

A power system supplied by just one sinusoidal AC source is called single-phase (1-φ). The oscillatory behavior of instantaneous power puts pulsating strain on the generating and load equipments. Polyphase circuit with multiple sources were developed in response to this problem. The most common one us the balanced three-phase (3-φ) circuit, which has three sources with $\frac{2\pi}{3}$ phase difference and has constant instantaneous power.

The equation

$$v_a(t) = \sqrt{2}v_{rms}\cos(\omega t + \theta)$$
$$v_b(t) = \sqrt{2}v_{rms}\cos\left(\omega t + \theta - \frac{2\pi}{3}\right)$$
$$v_c(t) = \sqrt{2}v_{rms}\cos\left(\omega t + \theta + \frac{2\pi}{3}\right)$$

defines a symmetrical three-phase set with phase sequence $a{-}b{-}c$, where the phase sequence indicates the time sequence of the waveform peaks. Their sum is zero.

A (balanced) wye (Y) generator consists of a node called neutral and three branches with phase voltages $\mathbf{V}_{rms,a}, \mathbf{V}_{rms,b}, \mathbf{V}_{rms,c}$ w.r.t. the neutral that have phase difference $\frac{2\pi}{3}$ among each other, and line voltages $\mathbf{V}_{rms,ab}, \mathbf{V}_{rms,bc}, \mathbf{V}_{rms,ca}$ are defined across each pairs of their terminals.

$$\mathbf{V}_{rms,ab} = \mathbf{V}_{rms,a} - \mathbf{V}_{rms,b}$$

$$\mathbf{V}_{rms,bc} = \mathbf{V}_{rms,b} - \mathbf{V}_{rms,c}$$

$$\mathbf{V}_{rms,ca} = \mathbf{V}_{rms,c} - \mathbf{V}_{rms,a}$$

An equivalent generator is (balanced) delta (Δ) generator obtained by applying Y-Δ transformation to the Y generator, in which the neutral and phase voltages do not physically exist.

Similarly, a (balanced) wye load has a neutral and three branches with identical phase impedances, and a (balanced) delta (Δ) load is it applied Y-Δ transformation.

A three-phase circuit with wye generator and loads with neutrals of all elements form a circuit, terminals of each branch of all elements form identical circuit, is called balanced (wye) (three-phase) circuit, and the circuit obtain by applying Y-Δ transformation to all wye generator and loads is called balanced (delta) (three-phase) circuit.

XI Frequency response

i DC gain

Low-pass gain is also called DC gain.

ii Resonance

A network is called in resonance or resonant if its voltage and current are in phase. The angular frequency and frequency at which are called resonance/resonant frequency and denoted ω_r and f_r.

In series resonance, the phenomenon that amplitude of voltage of element is larger than that of source is called resonant voltage rise. In parallel resonance, the phenomenon that amplitude of voltage of element is larger than that of source is called resonant current rise.

iii Natural frequency

The natural frequency is the magnitude of poles on s-plane if they are the same.

iv dB or decibels

For a filter circuit power response P_{out}, its power gain in dB at an angular frequency ω is

$$G = 10 \log_{10}(\frac{P_{out}}{P_{in}}).$$

For a filter circuit voltage or current response with a transfer function H, its magnitude gain in dB at an angular frequency ω is

$$20 \log_{10}(|H(j\omega)|).$$

v Octave (oct) and decade (dec)

In logarithmic scale, changing by 8 times is called an octave or oct, and changing by 10 times is called a decade or dec.

vi Cutoff (截止), break, corner, or half-power frequency

An angular or non-angular frequency such that the output is -3dB, that is, $|H(j\omega)| = \frac{1}{\sqrt{2}}$ or $P_{out} = \frac{1}{2}P_{in}$, is called break, corner, or half-power angular frequency ω_c, $\omega_{3\text{dB}}$, $\omega_{-3\text{dB}}$ and non-angular frequency f_c, $f_{3\text{dB}}$, or $f_{-3\text{dB}}$ respectively.

vii Bandwidth

Bandwidth is an interval of ω (or f) ≥ -3dB, or the length $\mathcal{B}$ of it.

viii Bode plot (波德圖)

- Bode plot of magnitude: magnitude gain in dB (dB)-$\log_{10}\omega$ (decade) or $\log_{10} f$ (decade)
- Bode plot of phase: Arg$(H(j\omega))$ (°)-$\log_{10}\omega$ (decade) or $\log_{10} f$ (decade)

TODO how to get Bode plot TODO first-order TODO First-orde filter, RLC add High Q, all circuits 增 Y / Z, add general Thevening and Norton second order circuit, Lissajous curve.

ix Second-order canonical low-pass circuit

A second-order canonical low-pass filter has the form

$$s^2F(s) + 2\zeta\omega_0 sF(s) + {\omega_0}^2F(s) = {\omega_0}^2X(s)$$

$$F(s) = \frac{{\omega_0}^2X(s)}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}$$

$$H(s) = \frac{F(s)}{X(s)} = \frac{{\omega_0}^2}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}.$$

The poles on the complex plane are

$$r_{1,2} = -\alpha \pm j\omega_d.$$

The frequency response is

$$H(j\omega) = \frac{{\omega_0}^2}{{\omega_0}^2 - \omega^2 + j2\zeta\omega_0\omega}.$$

Magnitude response is

$$|H(j\omega)|^2 = \frac{{\omega_0}^4}{({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2}.$$

Filter:

$$\lim_{\omega\to 0}|H(j\omega)| = 1$$

$$\lim_{\omega\to\infty}|H(j\omega)| = 0.$$

When $\zeta < \frac{1}{\sqrt{2}}$, the resonance frequency ω_r is

$$\omega_r = \arg_{\omega}\max_{\omega}|H(j\omega)| = \omega_0\sqrt{1-2\zeta^2},$$

and

$$\max_{\omega} |H(j\omega)| = \left|H(j\omega_r)\right| = \frac{1}{2\zeta\sqrt{1-\zeta^2}}.$$

Proof.

$$\frac{\mathrm{d}\,|H(j\omega)|^2}{\mathrm{d}\omega} = -\frac{{\omega_0}^4}{(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2)^2}\frac{\mathrm{d}}{\mathrm{d}\omega}\left(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2\right)$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\omega}\left(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2\right) =& \frac{\mathrm{d}}{\mathrm{d}\omega}\left({\omega_0}^4-2{\omega_0}^2\omega^2+\omega^4+4\zeta^2{\omega_0}^2\omega^2\right)\\
=& -4{\omega_0}^2\omega+4\omega^3+8\zeta^2{\omega_0}^2\omega\\
=& 4\omega(\omega^2-(1-2\zeta^2){\omega_0}^2)
\end{aligned}$$

$$\omega = 0 \vee \omega_0\sqrt{1-2\zeta^2}$$

$$|H(j0)|^2 = 1$$

$$\begin{aligned}
\left|H(j\omega_0\sqrt{1-2\zeta^2})\right|^2 =& \frac{1}{(2\zeta^2)^2+(2\zeta\sqrt{1-2\zeta^2})^2}\\
=& \frac{1}{4\zeta^2-4\zeta^4}
\end{aligned}$$

□

The cutoff frequency $\omega_{1,2}$ such that

$$\left|H(j\omega_{1,2})\right| = \frac{1}{\sqrt{2}}$$

are

$$\omega_{1,2} = \omega_0,$$

where ω_2 is called upper half-power frequency and ω_1 is called lower half-power frequency.

Proof. Let $\omega_{1,2} = a\omega_0$.

$$\left|H(j\omega_{1,2})\right|^2 = \frac{1}{(1-a^2)^2+(2\zeta a)^2} = \frac{1}{1-2a^2+a^4+4\zeta^2a^2}$$

$$1-2a^2+a^4+4\zeta^2a^2 = 2$$

$$a^4-2a^2(1-2\zeta^2)-1 = 0$$

$$a^2 = 1-2\zeta^2 \pm \sqrt{(1-2\zeta^2)^2+1} = 1-2\zeta^2 \pm \sqrt{2-4\zeta^2+4\zeta^4} = ??$$

?? TODO □

For high-Q circuit,

$$H(j\omega) = \frac{{\omega_0}^2}{{\omega_0}^2-\omega^2+j\frac{\omega_0}{Q}\omega},$$

since $\sqrt{1\mp 2x} \approx 1\mp x$,

$$\omega_{1,2} \approx (1\mp\zeta)\omega_0 = \omega_0 \mp \alpha \approx \omega_0(1\mp\frac{1}{2Q}),$$

$$\mathcal{B} = \omega_2-\omega_1 \approx 2\zeta\omega_0 = 2\alpha \approx \frac{\omega_0}{Q}$$

x Second-order canonical band-pass circuit

A second-order canonical band-pass filter has the form

$$s^2F(s) + 2\zeta\omega_0 sF(s) + {\omega_0}^2F(s) = 2\zeta\omega_0 sX(s)$$

$$F(s) = \frac{2\zeta\omega_0 sX(s)}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}$$

$$H(s) = \frac{F(s)}{X(s)} = \frac{2\zeta\omega_0 s}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}.$$

The poles on the complex plane are

$$r_{1,2} = -\alpha \pm j\omega_d.$$

The frequency response is

$$H(j\omega) = \frac{j2\zeta\omega_0\omega}{{\omega_0}^2 - \omega^2 + j2\zeta\omega_0\omega}.$$

Magnitude response is

$$|H(j\omega)|^2 = \frac{(2\zeta\omega_0\omega)^2}{({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2}.$$

Band-pass filter whose resonance frequency is natural frequency ω_0 and has magnitude gain 1:

$$\arg_\omega \max_\omega |H(j\omega)| = \omega_0$$

$$\max_\omega |H(j\omega)| = \left|H(j\omega_0)\right| = 1.$$

Proof.

$$\frac{1}{(2\zeta\omega_0)^2}\frac{\mathrm{d}\,|H(j\omega)|^2}{\mathrm{d}\omega} = \frac{2\omega(({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2) - \omega^2\frac{\mathrm{d}}{\mathrm{d}\omega}(({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2)}{(({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2)^2}$$

$$\frac{\mathrm{d}}{\mathrm{d}\omega}(({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2) = 4\omega^3 - 4{\omega_0}^2\omega + 8\zeta^2{\omega_0}^2\omega$$

$${\omega_0}^4 - 2{\omega_0}^2\omega^2 + \omega^4 + 4\zeta^2{\omega_0}^2\omega^2 - 2\omega^4 + 2{\omega_0}^2\omega^2 - 4\zeta^2{\omega_0}^2\omega^2 = 0$$

$${\omega_0}^4 - \omega^4 = 0$$

$$\omega = \omega_0$$

□

The cutoff frequency $\omega_{1,2}$ such that

$$\left|H(j\omega_{1,2})\right| = \frac{1}{\sqrt{2}}$$

are

$$\omega_{1,2} = \sqrt{1 + 2\zeta^2 \mp 2\zeta\sqrt{1+\zeta^2}}\omega_0,$$

where ω_2 is called upper half-power frequency and ω_1 is called lower half-power frequency.

Proof. Let $\omega_{1,2} = a\omega_0$.

$$\left|H(j\omega_{1,2})\right|^2 = \frac{(2\zeta a)^2}{(1-a^2)^2 + (2\zeta a)^2} = \frac{1}{2}$$
$$1 - 2a^2 + a^4 + 4\zeta^2 a^2 = 8\zeta^2 a^2$$
$$a^4 - 2(1+2\zeta^2)a^2 + 1 = 0$$
$$a^2 = 1 + 2\zeta^2 \pm \sqrt{(1+2\zeta^2)^2 - 1} = 1 + 2\zeta^2 \pm 2\zeta\sqrt{1+\zeta^2}$$

□

For high-Q circuit,

$$H(j\omega) = \frac{j\frac{\omega_0}{Q}\omega}{{\omega_0}^2 - \omega^2 + j\frac{\omega_0}{Q}\omega},$$

since $\sqrt{1 \mp 2x} \approx 1 \mp x$,

$$\omega_{1,2} \approx (1 \mp \zeta)\omega_0 = \omega_0 \mp \alpha \approx \omega_0(1 \mp \frac{1}{2Q}),$$
$$\mathcal{B} = \omega_2 - \omega_1 \approx 2\zeta\omega_0 = 2\alpha \approx \frac{\omega_0}{Q}$$

xi Second-order canonical high-pass circuit

A second-order canonical high-pass filter has the form

$$s^2F(s) + 2\zeta\omega_0 sF(s) + {\omega_0}^2F(s) = s^2X(s)$$
$$F(s) = \frac{s^2X(s)}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}$$
$$H(s) = \frac{F(s)}{X(s)} = \frac{s^2}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}.$$

The poles on the complex plane are

$$r_{1,2} = -\alpha \pm j\omega_d.$$

The frequency response is

$$H(j\omega) = \frac{-\omega^2}{{\omega_0}^2 - \omega^2 + j2\zeta\omega_0\omega}.$$

Magnitude response is

$$|H(j\omega)|^2 = \frac{\omega^4}{({\omega_0}^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2}.$$

High-pass filter with high-pass gain TODO:

$$\lim_{\omega\to 0}|H(j\omega)| = 0$$
$$\lim_{\omega\to\infty}|H(j\omega)| = TODO.$$

The resonance frequency is the natural frequency ω_0:

$$\arg_\omega \max_\omega |H(j\omega)| = \omega_0$$
$$\max_\omega |H(j\omega)| = \left|H(j\omega_0)\right| = \frac{1}{2\zeta}.$$

Proof.

$$\frac{1}{\mathrm{d}\omega}\frac{\mathrm{d}\,|H(j\omega)|^2}{} = \frac{4\omega^3(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2)-\omega^4\frac{\mathrm{d}}{\mathrm{d}\omega}(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2)}{(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2)^2}$$

$$\frac{\mathrm{d}}{\mathrm{d}\omega}(({\omega_0}^2-\omega^2)^2+(2\zeta\omega_0\omega)^2) = 4\omega^3-4{\omega_0}^2\omega+8\zeta^2{\omega_0}^2\omega$$

$$4{\omega_0}^4-8{\omega_0}^2\omega^2+4\omega^4+8\zeta^2{\omega_0}^2\omega^2-4\omega^4+4{\omega_0}^2\omega^2-8\zeta^2{\omega_0}^2\omega^2=0$$

$$\omega=\omega_0$$

□

The cutoff frequency $\omega_{1,2}$ such that

$$\left|H(j\omega_{1,2})\right| = \frac{1}{2\sqrt{2}\zeta}$$

are

$$\omega_{1,2} = \sqrt{1+2\zeta^2\mp 2\zeta\sqrt{1+\zeta^2}}\omega_0,$$

where ω_2 is called upper half-power frequency and ω_1 is called lower half-power frequency.

Proof. Let $\omega_{1,2} = a\omega_0$.

$$\left|H(j\omega_{1,2})\right|^2 = \frac{a^4}{(1-a^2)^2+(2\zeta a)^2} = \frac{1}{8\zeta^2}$$

$$1-2a^2+a^4+4\zeta^2a^2 = 8\zeta^2a^4$$

$$a^4-2a^2(1+2\zeta^2)+1=0$$

$$a^2 = 1+2\zeta^2\pm\sqrt{(1+2\zeta^2)^2-1} = 1+2\zeta^2\pm 2\zeta\sqrt{1+\zeta^2}$$

□

For high-Q circuit,

$$H(j\omega) = \frac{-\omega^2}{{\omega_0}^2-\omega^2+j\frac{\omega_0}{Q}\omega},$$

since $\sqrt{1\mp 2x}\approx 1\mp x$,

$$\omega_{1,2}\approx(1\mp\zeta)\omega_0 = \omega_0\mp\alpha\approx\omega_0(1\mp\frac{1}{2Q}),$$

$$\mathcal{B} = \omega_2-\omega_1\approx 2\zeta\omega_0 = 2\alpha\approx\frac{\omega_0}{Q}$$

XII R, L, C, RL, RC, LC, and RLC Circuits

i Resistor with voltage source

A resistor with resistance R and an independent voltage source v_{in} in series connection.

Natural response (time domain):

$$i=0$$

Natural response (Laplace domain):

$$I=0$$

Forced/total response (time domain):

$$i = \frac{v_{in}}{R}$$

Forced/total response (Laplace domain):

$$I = \frac{V_{in}}{R}$$

Impulse response:

$$h_i = \frac{\delta(t)}{R}$$

Transfer function:

$$H_I = \frac{1}{R}$$

ii Resistor with current source

A resistor with resistance R and an independent current source i_{in} in series connection.

Natural response (time domain):

$$v = 0$$

Natural response (Laplace domain):

$$V = 0$$

Forced/total response (time domain):

$$v = i_{in}R$$

Forced/total response (Laplace domain):

$$V = I_{in}R$$

Impulse response:

$$h_v = \delta(t)R$$

Transfer function:

$$H_V = R$$

iii Inductor with voltage source

An inductor with inductance L and an independent voltage source v_{in} in series connection.

Natural response (time domain):

$$i = i(0^-)$$

Natural response (Laplace domain):

$$I = i(0^-)$$

Forced response (time domain):

$$i = \int_0^t \frac{v_{in}(\tau)}{L}\,\mathrm{d}\tau$$

Forced response (Laplace domain):

$$I = \frac{V_{in}}{Ls}$$

Total response (time domain):

$$i = i(0^-) + \frac{1}{L}\int_0^t v_{in}(\tau)\,\mathrm{d}\tau$$

Total response (Laplace domain):

$$I = \frac{Li(0^-) + V_{in}}{Ls}$$

Constant source forced response (time domain):

$$i = \frac{v_{in}t}{L}$$

Constant source total response (time domain):

$$i = i(0^-) + \frac{v_{in}t}{L}$$

Steady-state sinusoidal AC (time domain): Source angular frequency ω, maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$v_{in} = v_m \cos(\omega t + \theta)$$

$$i = \frac{v_m}{L\omega}\sin(\omega t + \theta)$$

$$E_m = \frac{{v_m}^2}{2L\omega^2}$$

$$E_l = 0$$

$$Q = \infty$$

Steady-state sinusoidal AC (phasor domain):

$$\mathbf{V}_{in} = v_m e^{j\theta}$$

$$\mathbf{I} = \frac{v_m}{L\omega}e^{j(\theta - \frac{\pi}{2})}$$

Impulse response:

$$h_i = \frac{u(t)}{L}$$

Transfer function:

$$H_I = \frac{1}{Ls}$$

iv Inductor with current source

An inductor with inductance L and an independent current source i_{in} in series connection.

Natural response (time domain):

$$v = 0$$

Natural response (Laplace domain):

$$V = 0$$

Forced/total response (time domain):

$$v = L\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

Forced/total response (Laplace domain):

$$V = I_{in}Ls$$

Constant source forced/total response/steady state (time domain):

$$v = 0$$

Steady-state sinusoidal AC (time domain): Source angular frequency ω, maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$i_{in} = i_m \cos(\omega t + \theta)$$

$$v = -i_m L\omega \sin(\omega t + \theta)$$

$$E_m = \frac{L{i_m}^2}{2}$$

$$E_l = 0$$

$$Q = \infty$$

Steady-state sinusoidal AC (phasor domain):

$$\mathbf{I}_{in} = i_m e^{j\theta}$$

$$\mathbf{V} = i_m L\omega e^{j(\theta+\frac{\pi}{2})}$$

Impulse response:

$$h_v = \delta'(t)L$$

Transfer function:

$$H_I = Ls$$

v Capacitor with voltage source

A capacitor with capacitance C and an independent voltage source v_{in} in series connection.

Natural response (time domain):

$$i = 0$$

Natural response (Laplace domain):

$$I = 0$$

Forced/total response (time domain):

$$i = C\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

Forced/total response (Laplace domain):

$$I = V_{in}Cs$$

Constant source forced/total response/steady state (time domain):

$$i = 0$$

Steady-state sinusoidal AC (time domain): Source angular frequency ω, maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$v_{in} = v_m \cos(\omega t + \theta)$$

$$i = v_m C\omega \sin(\omega t + \theta)$$

$$E_m = \frac{C{v_m}^2}{2}$$

$$E_l = 0$$

$$Q = \infty$$

Steady-state sinusoidal AC (phasor domain):

$$\mathbf{V}_{in} = v_m e^{j\theta}$$

$$\mathbf{I} = v_m C\omega e^{j(\theta+\frac{\pi}{2})}$$

Impulse response:

$$h_i = \delta'(t)C$$

Transfer function:

$$H_I = Cs$$

vi Capacitor with current source

A capacitor with capacitance C and an independent current source i_{in} in series connection.

Natural response (time domain):

$$v = v(0^-)$$

Natural response (Laplace domain):

$$V = v(0^-)$$

Forced response (time domain):

$$v = \frac{1}{C}\int_0^t i_{in}(\tau)\,\mathrm{d}\tau$$

Forced response (Laplace domain):

$$V = \frac{I_{in}}{Cs}$$

Total response (time domain):

$$v = v(0^-) + \frac{1}{C}\int_0^t i_{in}(\tau)\,\mathrm{d}\tau$$

Total response (Laplace domain):

$$V = \frac{Cv(0^-) + I_{in}}{Cs}$$

Constant source forced response (time domain):

$$v = \frac{i_{in}t}{C}$$

Constant source total response (time domain):

$$v = v(0^-) + \frac{i_{in}t}{C}$$

Steady-state sinusoidal AC (time domain): Source angular frequency ω, maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$i_{in} = i_m \cos(\omega t + \theta)$$

$$v = \frac{i_m}{C\omega} \sin(\omega t + \theta)$$

$$E_m = \frac{{i_m}^2}{2C\omega^2}$$

$$E_l = 0$$

$$Q = \infty$$

Steady-state sinusoidal AC (phasor domain):

$$\mathbf{I}_{in} = i_m e^{j\theta}$$

$$\mathbf{V} = \frac{i_m}{C\omega} e^{j(\theta - \frac{\pi}{2})}$$

Impulse response:

$$h_v = \frac{u(t)}{C}$$

Transfer function:

$$H_V = \frac{1}{Cs}$$

vii RL circuit

A (series) RL circuit or filter consists of a resistor with resistance R, an inductor with inductance L, and an independent voltage source v_{in} in series connection.

Natural response (time domain):

$$Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} = 0$$

$$i = i(0^-)e^{-\frac{R}{L}t}$$

$$v_R = -v_L = Ri(0^-)e^{-\frac{R}{L}t}$$

$$\tau_i = \tau_{v_R} = \tau_{v_L} = \frac{L}{R}$$

$$p_R = -p_L = R(i(0^-))^2 e^{-2\frac{R}{L}t}$$

$$\int_0^\infty p_R \,\mathrm{d}t = -\int_0^\infty p_L \,\mathrm{d}t = \frac{1}{2}L(i(0^-))^2$$

Proof.

$$Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} = 0$$

$$Lr + R = 0$$

$$r = -\frac{R}{L}$$

$$i = Ae^{-\frac{R}{L}t}$$

$$i = i(0^-)e^{-\frac{R}{L}t}$$

$$
\begin{aligned}
v_R = -v_L &= Ri(0^-)e^{-\frac{R}{L}t} \\
p_R = -p_L &= R(i(0^-))^2 e^{-2\frac{R}{L}t} \\
\int_0^\infty p_R \, \mathrm{d}t &= R(i(0^-))^2 \left(-\frac{L}{2R} e^{-2\frac{R}{L}t} \right|_0^\infty \\
&= \frac{1}{2} L(i(0^-))^2
\end{aligned}
$$

□

Natural response (Laplace domain):

$$
\begin{aligned}
I &= \frac{Li(0^-)}{R+Ls} \\
V_R = -V_L &= \frac{RLi(0^-)}{R+Ls}
\end{aligned}
$$

Proof. Check consistency:

$$
\begin{aligned}
\mathcal{L}\{i\} &= \frac{i(0^-)}{s+\frac{R}{L}} = \frac{Li(0^-)}{R+Ls} \\
\mathcal{L}\{v_R\} &= Ri(0^-)\frac{1}{s+\frac{R}{L}} = \frac{RLi(0^-)}{R+Ls}
\end{aligned}
$$

□

Forced response (time domain):

$$
\begin{aligned}
Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} &= v_{in} \\
i &= \frac{1}{L}\int_0^t e^{\frac{R}{L}\tau} v_{in}(\tau)\, \mathrm{d}\tau \, e^{-\frac{R}{L}t} \\
v_R = v_{in} - v_L &= \frac{R}{L}\int_0^t e^{\frac{R}{L}\tau} v_{in}(\tau)\, \mathrm{d}\tau \, e^{-\frac{R}{L}t}
\end{aligned}
$$

Proof.

$$
\begin{aligned}
Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} &= v_{in} \\
\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{R}{L}i &= \frac{v_{in}}{L} \\
\mu(t) = e^{\int \frac{R}{L}\mathrm{d}t} &= e^{\frac{R}{L}t} \\
\frac{\mathrm{d}}{\mathrm{d}t}(e^{\frac{R}{L}t} i) &= e^{\frac{R}{L}t}\frac{v_{in}}{L} \\
i &= \frac{1}{L}\int_0^t e^{\frac{R}{L}\tau} v_{in}(\tau)\, \mathrm{d}\tau \, e^{-\frac{R}{L}t}
\end{aligned}
$$

□

Forced response (Laplace domain):

$$
\begin{aligned}
I &= \frac{V_{in}}{R+Ls} \\
V_R &= \frac{RV_{in}}{R+Ls} \\
V_L &= \frac{LsV_{in}}{R+Ls}
\end{aligned}
$$

Proof. Check consistency:

$$\begin{aligned}
\mathcal{L}\{i\} &= \frac{1}{L}\mathcal{L}\{\int_0^t e^{\frac{R}{L}\tau} v_{in}(\tau)u(t-\tau)\,\mathrm{d}\tau\}(s+\frac{R}{L}) \\
&= \frac{1}{L}\mathcal{L}\{e^{\frac{R}{L}t} v_{in}(t)\}(s+\frac{R}{L})\mathcal{L}\{u(t)\}(s+\frac{R}{L}) \\
&= \frac{1}{L}V_{in}(s-\frac{R}{L}+\frac{R}{L})\frac{1}{s+\frac{R}{L}} \\
&= \frac{V_{in}}{R+Ls}
\end{aligned}$$

□

Total response (time domain):

$$i = (i(0^-) + \frac{1}{L}\int_0^t e^{\frac{R}{L}\tau} v_{in}(\tau)\,\mathrm{d}\tau)e^{-\frac{R}{L}t}$$

$$v_R = v_{in} - v_L = R(i(0^-) + \frac{1}{L}\int_0^t e^{\frac{R}{L}\tau} v_{in}(\tau)\,\mathrm{d}\tau)e^{-\frac{R}{L}t}$$

Total response (Laplace domain):

$$I = \frac{Li(0^-) + V_{in}}{R+Ls}$$

$$V_R = \frac{RLi(0^-) + RV_{in}}{R+Ls}$$

$$V_L = \frac{LsV_{in} - RLi(0^-)}{R+Ls}$$

Steady state (time domain):

$$i = \frac{v_{in}}{R}$$

$$v_R = v_{in}$$

$$v_L = 0$$

Constant source forced response (time domain):

$$i = \frac{v_{in}}{R}(1 - e^{-\frac{R}{L}t}) = i(\infty) - i(\infty)e^{-\frac{R}{L}t}$$

$$v_L = v_{in} - v_R = v_{in}e^{-\frac{R}{L}t} = i(\infty)Re^{-\frac{R}{L}t}$$

Constant source total response (time domain):

$$i = \frac{v_{in}}{R} - (\frac{v_{in}}{R} - i(0^-))e^{-\frac{R}{L}t} = i(\infty) + (i(0^-) - i(\infty))e^{-\frac{R}{L}t}$$

$$v_L = v_{in} - v_R = (v_{in} - i(0^-)R)e^{-\frac{R}{L}t} = -(i(0^-) - i(\infty))Re^{-\frac{R}{L}t}$$

Steady-state sinusoidal AC (time domain): Source angular frequency ω, maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$v_{in} = v_m \cos(\omega t + \theta)$$

$$i = \frac{v_m}{\sqrt{R^2 + L^2\omega^2}} \cos\Big(\omega t + \theta - \arctan\Big(\frac{L\omega}{R}\Big)\Big)$$

$$v_R = \frac{v_m R}{\sqrt{R^2 + L^2\omega^2}} \cos\left(\omega t + \theta - \arctan\left(\frac{L\omega}{R}\right)\right)$$

$$v_L = -\frac{v_m L\omega}{\sqrt{R^2 + L^2\omega^2}} \sin\left(\omega t + \theta - \arctan\left(\frac{L\omega}{R}\right)\right)$$

$$E_m = \frac{L {v_m}^2}{2(R^2 + L^2\omega^2)}$$

$$E_l = \frac{R {v_m}^2 \pi}{\omega(R^2 + L^2\omega^2)}$$

$$Q = \frac{L\omega}{R}$$

Steady-state sinusoidal AC (phasor domain):

$$\mathbf{V}_{in} = v_m e^{j\theta}$$

$$\mathbf{I} = \frac{v_m}{\sqrt{R^2 + L^2\omega^2}} e^{j(\theta - \arctan\left(\frac{L\omega}{R}\right))}$$

$$\mathbf{V}_R = \frac{v_m R}{\sqrt{R^2 + L^2\omega^2}} e^{j(\theta - \arctan\left(\frac{L\omega}{R}\right))}$$

$$\mathbf{V}_L = \frac{v_m L\omega}{\sqrt{R^2 + L^2\omega^2}} e^{j(\theta - \arctan\left(\frac{L\omega}{R}\right) + \frac{\pi}{2})}$$

Impulse response:

$$h_i = \frac{1}{L} u(t) e^{-\frac{R}{L}t}$$

$$h_{v_R} = \frac{R}{L} u(t) e^{-\frac{R}{L}t}$$

$$h_{v_L} = \delta(t) - \frac{R}{L} u(t) e^{-\frac{R}{L}t}$$

Proof.

$$h_i = \frac{1}{L} \int_0^t e^{\frac{R}{L}\tau} \delta(\tau)\, \mathrm{d}\tau\, e^{-\frac{R}{L}t} = \frac{1}{L} u(t) e^{-\frac{R}{L}t}$$

□

Transfer function:

$$H_I = \frac{1}{R + Ls}$$

$$H_{V_R} = \frac{R}{R + Ls}$$

$$H_{V_L} = \frac{Ls}{R + Ls}$$

Proof. Check consistency:

$$\mathcal{L}\{h_i\} = \frac{1}{L(s + \frac{R}{L})} = \frac{1}{R + Ls}$$

$$\mathcal{L}\{h_{V_L}\} = 1 - \frac{R}{R + Ls} = \frac{Ls}{R + Ls}$$

□

V_R being low-pass filter:

$$\lim_{\omega \to 0} \left| H_{V_R}(j\omega) \right| = 1$$

$$\lim_{\omega \to \infty} \left| H_{V_R}(j\omega) \right| = 0$$

Proof.

$$\lim_{\omega \to 0} \left| H_{V_R}(j\omega) \right| = \lim_{\omega \to 0} \left| \frac{R}{R + Lj\omega} \right| = 1$$

$$\lim_{\omega \to \infty} \left| H_{V_R}(j\omega) \right| = \lim_{\omega \to \infty} \left| \frac{R}{R + Lj\omega} \right| = 0$$

□

V_L being high-pass filter:

$$\lim_{\omega \to 0} \left| H_{V_L}(j\omega) \right| = 0$$

$$\lim_{\omega \to \infty} \left| H_{V_L}(j\omega) \right| = 1$$

Proof.

$$\lim_{\omega \to 0} \left| H_{V_L}(j\omega) \right| = \lim_{\omega \to 0} \left| \frac{Lj\omega}{R + Lj\omega} \right| = 0$$

$$\lim_{\omega \to \infty} \left| H_{V_L}(j\omega) \right| = \lim_{\omega \to \infty} \left| \frac{Lj\omega}{R + Lj\omega} \right| = 1$$

□

viii RC circuit

A (series) RC circuit or filter consists of a resistor with resistance R, a capacitor with capacitance C, and an independent voltage source v_{in} in series connection.

Natural response (time domain):

$$\frac{v_C}{R} + C\frac{\mathrm{d}v_C}{\mathrm{d}t} = 0$$

$$v_C = -v_R = v_C(0^-)e^{-\frac{1}{RC}t}$$

$$i = -\frac{1}{R}v_C(0^-)e^{-\frac{1}{RC}t}$$

$$\tau_{v_C} = \tau_{v_R} = \tau_i = \frac{1}{RC}$$

$$p_R = -p_C = \frac{1}{R}(v_C(0^-))^2 e^{-\frac{2}{RC}t}$$

$$\int_0^\infty p_R \,\mathrm{d}t = -\int_0^\infty p_C \,\mathrm{d}t = \frac{1}{2}C(v_C(0^-))^2$$

Proof.

$$v_R + v_C = 0$$

$$\frac{v_R}{R} - C\frac{\mathrm{d}v_C}{\mathrm{d}t} = 0$$

$$\frac{v_C}{R} + C\frac{\mathrm{d}v_C}{\mathrm{d}t} = 0$$

$$Cr + \frac{1}{R} = 0$$

$$r = -\frac{1}{RC}$$

$$v_C = Ae^{-\frac{1}{RC}t}$$

$$v_C = v_C(0^-)e^{-\frac{1}{RC}t}$$

$$i = -\frac{1}{R}v_C(0^-)e^{-\frac{1}{RC}t}$$

$$p_R = -p_C = \frac{1}{R}(v_C(0^-))^2 e^{-\frac{2}{RC}t}$$

$$\begin{aligned}\int_0^\infty p_R \,\mathrm{d}t &= \frac{1}{R}(v_C(0^-))^2 - \frac{RC}{2}e^{-\frac{2}{RC}t}\Big|_0^\infty \\ &= \frac{1}{2}C(v_C(0^-))^2\end{aligned}$$

□

Natural response (Laplace domain):

$$V_C = -V_R = \frac{RCv_C(0^-)}{1 + RCs}$$

$$I = -\frac{Cv_C(0^-)}{1 + RCs}$$

Proof.

$$I = -\frac{v_C(0^-)}{s(\frac{1}{Cs} + R)} = -\frac{Cv_C(0^-)}{1 + RCs}$$

$$V_C = \frac{I + Cv_C(0^-)}{Cs} = \frac{RCv_C(0^-)}{1 + RCs}$$

Check consistency:

$$\mathcal{L}\{v_C\} = \frac{v_C(0^-)}{s + \frac{1}{RC}} = \frac{RCv_C(0^-)}{1 + RCs}$$

$$\mathcal{L}\{i\} = -\frac{1}{R}v_C(0^-)\frac{1}{s + \frac{1}{RC}} = -\frac{Cv_C(0^-)}{1 + RCs}$$

□

Forced response (time domain):

$$\frac{v_C - v_{in}}{R} + C\frac{\mathrm{d}v_C}{\mathrm{d}t} = 0$$

$$v_C = v_{in} - v_R = \frac{1}{RC}\int_0^t e^{\frac{1}{RC}\tau}v_{in}(\tau)\,\mathrm{d}\tau\, e^{-\frac{1}{RC}t}$$

$$i = \frac{1}{R}(v_{in} - \frac{1}{RC}\int_0^t e^{\frac{1}{RC}\tau}v_{in}(\tau)\,\mathrm{d}\tau\, e^{-\frac{1}{RC}t})$$

Proof.

$$v_R + v_C = v_{in}$$
$$\frac{v_R}{R} - C\frac{\mathrm{d}v_C}{\mathrm{d}t} = 0$$
$$\frac{v_C - v_{in}}{R} + C\frac{\mathrm{d}v_C}{\mathrm{d}t} = 0$$
$$\frac{\mathrm{d}v_C}{\mathrm{d}t} + \frac{v_C}{RC} = \frac{v_{in}}{RC}$$
$$\mu(t) = e^{\int \frac{1}{RC}\mathrm{d}t} = e^{\frac{1}{RC}t}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}(e^{\frac{1}{RC}t}v_C) = e^{\frac{1}{RC}t}\frac{v_{in}}{RC}$$
$$v_C = \frac{1}{RC}\int_0^t e^{\frac{1}{RC}\tau}v_{in}(\tau)\,\mathrm{d}\tau\, e^{-\frac{1}{RC}t}$$

□

Forced response (Laplace domain):

$$V_C = \frac{V_{in}}{1 + RCs}$$
$$V_R = \frac{RCsV_{in}}{1 + RCs}$$
$$I = -\frac{CsV_{in}}{1 + RCs}$$

Proof.

$$V_C = \frac{1}{\frac{1}{R} + Cs}CV_{in} = \frac{RCV_{in}}{1 + RCs}$$

Check consistency:

$$\begin{aligned}
\mathcal{L}\{v_C\} &= \frac{1}{RC}\mathcal{L}\{\int_0^t e^{\frac{1}{RC}\tau}v_{in}(\tau)u(t-\tau)\,\mathrm{d}\tau\}(s + \frac{1}{RC}) \\
&= \frac{1}{RC}\mathcal{L}\{e^{\frac{1}{RC}t}v_{in}(t)\}(s + \frac{1}{RC})\mathcal{L}\{u(t)\}(s + \frac{1}{RC}) \\
&= \frac{1}{RC}V_{in}(s - \frac{1}{RC} + \frac{1}{RC})\frac{1}{s + \frac{1}{RC}} \\
&= \frac{V_{in}}{1 + RCs}
\end{aligned}$$

□

Total response (time domain):

$$v_C = v_{in} - v_R = (v_C(0^-) + \frac{1}{RC}\int_0^t e^{\frac{1}{RC}\tau}v_{in}(\tau)\,\mathrm{d}\tau)e^{-\frac{1}{RC}t}$$

$$i = \frac{1}{R}(v_{in} - (v_C(0^-) + \frac{1}{RC}\int_0^t e^{\frac{1}{RC}\tau}v_{in}(\tau)\,\mathrm{d}\tau)e^{-\frac{1}{RC}t})$$

Total response (Laplace domain):

$$V_C = \frac{RCv_C(0^-) + V_{in}}{1 + RCs}$$

$$V_R = \frac{-RCv_C(0^-) + RCsV_{in}}{1 + RCs}$$
$$I = -\frac{Cv_C(0^-) + CsV_{in}}{1 + RCs}$$

Steady state (time domain):

$$v_C = v_{in}$$
$$v_R = 0$$
$$i = 0$$

Constant source forced response (time domain):

$$v_C = v_{in} - v_R = v_{in}(1 - e^{-\frac{1}{RC}t}) = v_C(\infty) - v_C(\infty)e^{-\frac{1}{RC}t}$$
$$i = \frac{v_{in}}{R}e^{-\frac{1}{RC}t} = \frac{v_C(\infty)}{R}e^{-\frac{1}{RC}t}$$

Constant source total response (time domain):

$$v_C = v_{in} - v_R = v_{in} - (v_{in} - v_C(0^-))e^{-\frac{1}{RC}t} = v_C(\infty) + (v_C(0^-) - v_C(\infty))e^{-\frac{1}{RC}t}$$
$$i = \frac{v_{in} - v_C(0^-)}{R}e^{-\frac{1}{RC}t} = -\frac{v_C(0^-) - v_C(\infty)}{R}e^{-\frac{1}{RC}t}$$

Steady-state sinusoidal AC (time domain): Source angular frequency ω, maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$v_{in} = v_m \cos(\omega t + \theta)$$
$$i = \frac{v_m C\omega}{\sqrt{R^2C^2\omega^2 + 1}} \cos\left(\omega t + \theta + \arctan\left(\frac{1}{RC\omega}\right)\right)$$
$$v_C = \frac{v_m}{\sqrt{R^2C^2\omega^2 + 1}} \sin\left(\omega t + \theta + \arctan\left(\frac{1}{RC\omega}\right)\right)$$
$$v_R = \frac{v_m RC\omega}{\sqrt{R^2C^2\omega^2 + 1}} \cos\left(\theta + \arctan\left(\frac{1}{RC\omega}\right)\right)$$
$$E_m = \frac{C{v_m}^2}{2(R^2C^2\omega^2 + 1)}$$
$$E_l = \frac{RC^2\omega^2{v_m}^2\pi}{\omega(R^2C^2\omega^2 + 1)}$$
$$Q = \frac{1}{RC\omega}$$

Steady-state sinusoidal AC (phasor domain):

$$\mathbf{V}_{in} = v_m e^{j\theta}$$
$$\mathbf{I} = \frac{v_m C\omega}{\sqrt{R^2C^2\omega^2 + 1}} e^{j(\theta + \arctan\left(\frac{1}{RC\omega}\right))}$$
$$\mathbf{V}_C = \frac{v_m}{\sqrt{R^2C^2\omega^2 + 1}} e^{j(\theta + \arctan\left(\frac{1}{RC\omega}\right) - \frac{\pi}{2})}$$

$$\mathbf{V}_R = \frac{v_m RC\omega}{\sqrt{R^2C^2\omega^2+1}} e^{j(\theta+\arctan\left(\frac{1}{RC\omega}\right))}$$

Impulse response:

$$h_{v_C} = \frac{1}{RC} u(t) e^{-\frac{1}{RC}t}$$

$$h_{v_R} = \delta(t) - \frac{1}{RC} u(t) e^{-\frac{1}{RC}t}$$

$$h_i = \frac{1}{R}(\delta(t) - \frac{1}{RC} u(t) e^{-\frac{1}{RC}t})$$

Proof.

$$h_{v_C} = \frac{1}{RC}\int_0^t e^{\frac{1}{RC}\tau}\delta(\tau)\,\mathrm{d}\tau\, e^{-\frac{1}{RC}t} = \frac{1}{RC}u(t)e^{-\frac{1}{RC}t}$$

□

Transfer function:

$$H_{V_C} = \frac{1}{1+RCs}$$

$$H_{V_R} = \frac{RCs}{1+RCs}$$

$$H_I = -\frac{Cs}{1+RCs}$$

Proof. Check consistency:

$$\mathcal{L}\{h_{v_C}\} = \frac{1}{RC(s+\frac{1}{RC})} = \frac{1}{1+RCs}$$

$$\mathcal{L}\{h_i\} = \frac{1}{R}(1 - \frac{1}{1+RCs}) = -\frac{Cs}{1+RCs}$$

□

V_C being low-pass filter:

$$\lim_{\omega\to 0}\left|H_{V_C}(j\omega)\right| = 1$$

$$\lim_{\omega\to \infty}\left|H_{V_C}(j\omega)\right| = 0$$

Proof.

$$\lim_{\omega\to 0}\left|H_{V_C}(j\omega)\right| = \lim_{\omega\to 0}\left|\frac{1}{1+RCj\omega}\right| = 1$$

$$\lim_{\omega\to \infty}\left|H_{V_C}(j\omega)\right| = \lim_{\omega\to \infty}\left|\frac{1}{1+RCj\omega}\right| = 0$$

□

V_R being high-pass filter:

$$\lim_{\omega\to 0}\left|H_{V_R}(j\omega)\right| = 0$$

$$\lim_{\omega\to \infty}\left|H_{V_R}(j\omega)\right| = 1$$

Proof.

$$\lim_{\omega\to 0}\left|H_{V_R}(j\omega)\right| = \lim_{\omega\to 0}\left|\frac{RCj\omega}{1+RCj\omega}\right| = 0$$

$$\lim_{\omega\to \infty}\left|H_{V_R}(j\omega)\right| = \lim_{\omega\to \infty}\left|\frac{RCj\omega}{1+RCj\omega}\right| = 1$$

□

ix Series LC circuit

A (lossless) series LC (resonant, oscillator, tank, or tuned) circuit consists of an inductor with inductance L, a capacitor with capacitance C, and an independent voltage source v_{in} in series connection. Angular resonance/resonant frequency (角共振頻率) ω_0.

Natural response (time domain): Total energy stored in inductor and capacitor E.

$$LC\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + i = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + {\omega_0}^2 i = 0$$

$$\begin{aligned} i &= i(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - v_C(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\ &= i(0^-)\cos(\omega_0 t) - v_C(0^-)C\omega_0\sin(\omega_0 t) \end{aligned}$$

$$\begin{aligned} v_C &= -v_L \\ &= i(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + v_C(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) \\ &= i(0^-)L\omega_0\sin(\omega_0 t) + v_C(0^-)\cos(\omega_0 t) \end{aligned}$$

$$\begin{aligned} p_C &= -p_L \\ &= \frac{1}{2}((i(0^-))^2\sqrt{\frac{L}{C}} - (v_C(0^-))^2\sqrt{\frac{C}{L}})\sin\left(\frac{2}{\sqrt{LC}}t\right) + i(0^-)v_C(0^-)\cos\left(\frac{2}{\sqrt{LC}}t\right) \\ &= \frac{1}{2}((i(0^-))^2 L\omega_0 - (v_C(0^-))^2 C\omega_0)\sin(2\omega_0 t) + i(0^-)v_C(0^-)\cos(2\omega_0 t) \end{aligned}$$

$$\lim_{n\to\infty}\frac{1}{n}\int_0^n p_C\,\mathrm{d}t = -\lim_{n\to\infty}\frac{1}{n}\int_0^n p_L\,\mathrm{d}t = 0$$

$$E = \frac{1}{2}(C(v_C(0^-))^2 + L(I(0^-))^2)$$

Proof.

$$v_C + v_L = 0$$

$$i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$$

$$v_C = -v_L = -L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$LC\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + i = 0$$

$$LCr^2 + 1 = 0$$

$$r = \pm\frac{1}{\sqrt{LC}}j$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + {\omega_0}^2 i = 0$$

$$i = A\cos\left(\frac{1}{\sqrt{LC}}t\right) + B\sin\left(\frac{1}{\sqrt{LC}}t\right)$$

$$A = i(0^-)$$

$$L\frac{\mathrm{d}i}{\mathrm{d}t}(0^-) + v_C(0^-) = 0$$

$$LB\frac{1}{\sqrt{LC}} + v_C(0^-) = 0$$

$$B = -v_C(0^-)\sqrt{\frac{C}{L}} = -v_C(0^-)\frac{1}{L\omega_0}$$

$$\begin{aligned} i &= i(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - v_C(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\ &= i(0^-)\cos\left(\omega_0 t\right) - v_C(0^-)C\omega_0\sin\left(\omega_0 t\right) \\ v_C &= -v_L = -L\frac{\mathrm{d}i}{\mathrm{d}t} \\ &= i(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + v_C(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) \\ &= i(0^-)L\omega_0\sin\left(\omega_0 t\right) + v_C(0^-)\cos\left(\omega_0 t\right) \end{aligned}$$

$$\begin{aligned} p_C &= -p_L \\ &= (i(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - v_C(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right))(i(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + v_C(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right)) \\ &= (i(0^-))^2\sqrt{\frac{L}{C}}\cos\left(\frac{1}{\sqrt{LC}}t\right)\sin\left(\frac{1}{\sqrt{LC}}t\right) + i(0^-)v_C(0^-)\cos^2(\frac{1}{\sqrt{LC}}t) - i(0^-)v_C(0^-)\sin^2(\frac{1}{\sqrt{LC}}t) - (v_C(0^- \\ &= \frac{1}{2}((i(0^-))^2\sqrt{\frac{L}{C}} - (v_C(0^-))^2\sqrt{\frac{C}{L}})\sin\left(\frac{2}{\sqrt{LC}}t\right) + i(0^-)v_C(0^-)\cos\left(\frac{2}{\sqrt{LC}}t\right) \\ &= \frac{1}{2}((i(0^-))^2L\omega_0 - (v_C(0^-))^2C\omega_0)\sin\left(2\omega_0 t\right) + i(0^-)v_C(0^-)\cos\left(2\omega_0 t\right) \end{aligned}$$

□

Natural response (Laplace domain):

$$\begin{aligned} I &= \frac{LCsi(0^-) - Cv_C(0^-)}{LCs^2 + 1} \\ &= \frac{si(0^-) - C{\omega_0}^2 v_C(0^-)}{s^2 + {\omega_0}^2} \end{aligned}$$

$$\begin{aligned}
V_C &= -V_L \\
&= \frac{LCsv_C(0^-) + Li(0^-)}{LCs^2 + 1} \\
&= \frac{sv_C(0^-) + L{\omega_0}^2 i(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}$$

Proof.

$$\begin{aligned}
I &= (Li(0^-) - \frac{v_C(0^-)}{s})\frac{1}{Ls + \frac{1}{Cs}} \\
&= \frac{LCsi(0^-) - Cv_C(0^-)}{LCs^2 + 1} \\
&= \frac{{\omega_0}^{-2} si(0^-) - Cv_C(0^-)}{{\omega_0}^{-2} s^2 + 1} \\
&= \frac{si(0^-) - C{\omega_0}^2 v_C(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
V_C &= -V_L = -LsI + Li(0^-) \\
&= \frac{-L^2Cs^2 i(0^-) + LCsv_C(0^-) + L^2Cs^2 i(0^-) + Li(0^-)}{LCs^2 + 1} \\
&= \frac{LCsv_C(0^-) + Li(0^-)}{LCs^2 + 1} \\
&= \frac{{\omega_0}^{-2} sv_C(0^-) + Li(0^-)}{{\omega_0}^{-2} s^2 + 1} \\
&= \frac{sv_C(0^-) + L{\omega_0}^2 i(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}$$

Check consistency:

$$\begin{aligned}
\mathcal{L}\{i\} &= i(0^-)\frac{s}{s^2 + \frac{1}{LC}} - v_C(0^-)\sqrt{\frac{C}{L}}\frac{\frac{1}{\sqrt{LC}}}{s^2 + \frac{1}{LC}} \\
&= \frac{LCsi(0^-) - Cv_C(0^-)}{LCs^2 + 1}
\end{aligned}$$

$$\begin{aligned}
\mathcal{L}\{v_C\} &= i(0^-)\sqrt{\frac{L}{C}}\frac{\frac{1}{\sqrt{LC}}}{s^2 + \frac{1}{LC}} + v_C(0^-)\frac{s}{s^2 + \frac{1}{LC}} \\
&= \frac{LCsv_C(0^-) + Li(0^-)}{LCs^2 + 1}
\end{aligned}$$

□

Forced response (time domain):

$$\begin{aligned}
LC\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + i &= C\frac{\mathrm{d}v_{in}}{\mathrm{d}t} \\
\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + {\omega_0}^2 i &= \frac{1}{L}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}
\end{aligned}$$

$$
\begin{aligned}
i &= \frac{1}{L}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau \\
&= \frac{1}{L}\int_0^t \cos\big(\omega_0(t-\tau)\big)v_{in}(\tau)\,\mathrm{d}\tau \\
v_C &= v_{in} - v_L \\
&= \frac{1}{\sqrt{LC}}\int_0^t \sin\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau \\
&= \omega_0\int_0^t \sin\big(\omega_0(t-\tau)\big)v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

Proof.

$$v_L + v_C = v_{in}$$

$$i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$$

$$v_C = v_{in} - v_L = v_{in} - L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$i = C\frac{\mathrm{d}v_{in}}{\mathrm{d}t} - LC\frac{\mathrm{d}^2 i}{\mathrm{d}t^2}$$

$$LC\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + i = C\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + \frac{1}{LC}i = \frac{1}{L}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + {\omega_0}^2 i = \frac{1}{L}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

$$i_1 = \cos\left(\frac{1}{\sqrt{LC}}t\right)$$

$$i_2 = \sin\left(\frac{1}{\sqrt{LC}}t\right)$$

$$\frac{1}{i_1 i_2' - i_2 i_1'} = \sqrt{LC}$$

$$
\begin{aligned}
i &= \sqrt{\frac{C}{L}}(-\cos\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \sin\left(\frac{1}{\sqrt{LC}}\tau\right)\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau + \sin\left(\frac{1}{\sqrt{LC}}t\right)\int \cos\left(\frac{1}{\sqrt{LC}}\tau\right)\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau) \\
&= \sqrt{\frac{C}{L}}(-\cos\left(\frac{1}{\sqrt{LC}}t\right)(\sin\left(\frac{1}{\sqrt{LC}}t\right)(v_{in} - v_{in}(0^-)) - \frac{1}{\sqrt{LC}}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}\tau\right)v_{in}(\tau)\,\mathrm{d}t) + \sin\left(\frac{1}{\sqrt{LC}}t\right)(\mathrm{co} \\
&= \frac{1}{L}(\cos\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \cos\left(\frac{1}{\sqrt{LC}}\tau\right)v_{in}(\tau)\,\mathrm{d}\tau + \sin\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \sin\left(\frac{1}{\sqrt{LC}}\tau\right)v_{in}(\tau)\,\mathrm{d}\tau) \\
&= \frac{1}{L}\int_0^t (\cos\left(\frac{1}{\sqrt{LC}}t\right)\cos\left(\frac{1}{\sqrt{LC}}\tau\right) + \sin\left(\frac{1}{\sqrt{LC}}t\right)\sin\left(\frac{1}{\sqrt{LC}}\tau\right))v_{in}(\tau)\,\mathrm{d}\tau \\
&= \frac{1}{L}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau \\
&= \frac{1}{L}\int_0^t \cos\big(\omega_0(t-\tau)\big)v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

$$
\begin{aligned}
v_C &= v_{in} - v_L = v_{in} - L\frac{\mathrm{d}i}{\mathrm{d}t} \\
&= v_{in} - \cos(0^-)v_{in} - \int_0^t \frac{\partial}{\partial t}(\cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau))\,\mathrm{d}\tau \\
&= \frac{1}{\sqrt{LC}}\int_0^t \sin\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau \\
&= \omega_0 \int_0^t \sin\big(\omega_0(t-\tau)\big)v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

□

Forced response (Laplace domain):

$$
\begin{aligned}
I &= \frac{CsV_{in}}{LCs^2+1} \\
&= \frac{{\omega_0}^2 CsV_{in}}{s^2+{\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_C &= \frac{V_{in}}{LCs^2+1} \\
&= \frac{{\omega_0}^2 V_{in}}{s^2+{\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_L &= \frac{LCs^2V_{in}}{LCs^2+1} \\
&= \frac{s^2V_{in}}{s^2+{\omega_0}^2}
\end{aligned}
$$

Proof.

$$
\begin{aligned}
I &= \frac{V_{in}}{Ls+\frac{1}{Cs}} \\
&= \frac{CsV_{in}}{LCs^2+1} \\
&= \frac{CsV_{in}}{{\omega_0}^{-2}s^2+1} \\
&= \frac{{\omega_0}^2 CsV_{in}}{s^2+{\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_L &= \frac{LCs^2V_{in}}{LCs^2+1} \\
&= \frac{s^2V_{in}}{s^2+{\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_C &= \frac{V_{in}}{LCs^2+1} \\
&= \frac{{\omega_0}^2 V_{in}}{s^2+{\omega_0}^2}
\end{aligned}
$$

Check consistency:

$$
\mathcal{L}\{i\} = \frac{1}{L}\frac{s}{s^2+\frac{1}{LC}}V_{in} = \frac{CsV_{in}}{LCs^2+1}
$$

$$\mathcal{L}\{v_C\} = \frac{1}{\sqrt{LC}} \frac{\frac{1}{\sqrt{LC}}}{s^2 + \frac{1}{LC}} V_{in} = \frac{V_{in}}{LCs^2 + 1}$$

□

Total response (time domain):

$$\begin{aligned} i &= i(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - v_C(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + \frac{1}{L}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau \\ &= i(0^-)\cos(\omega_0 t) - v_C(0^-)C\omega_0\sin(\omega_0 t) + \frac{1}{L}\int_0^t \cos(\omega_0(t-\tau))v_{in}(\tau)\,\mathrm{d}\tau \end{aligned}$$

$$\begin{aligned} v_C &= v_{in} - v_L \\ &= i(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + v_C(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) + \frac{1}{\sqrt{LC}}\int_0^t \sin\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau \\ &= i(0^-)L\omega_0\sin(\omega_0 t) + v_C(0^-)\cos(\omega_0 t) + \omega_0\int_0^t \sin(\omega_0(t-\tau))v_{in}(\tau)\,\mathrm{d}\tau \end{aligned}$$

Total response (Laplace domain):

$$\begin{aligned} I &= \frac{LCsi(0^-) - Cv_C(0^-) + CsV_{in}}{LCs^2 + 1} \\ &= \frac{si(0^-) - C\omega_0{}^2 v_C(0^-) + \omega_0{}^2 CsV_{in}}{s^2 + \omega_0{}^2} \end{aligned}$$

$$\begin{aligned} V_C &= \frac{LCsv_C(0^-) + Li(0^-) + V_{in}}{LCs^2 + 1} \\ &= \frac{sv_C(0^-) + L\omega_0{}^2 i(0^-) + \omega_0{}^2 V_{in}}{s^2 + \omega_0{}^2} \end{aligned}$$

$$\begin{aligned} V_L &= \frac{LCs^2V_{in} - LCsv_C(0^-) - Li(0^-)}{LCs^2 + 1} \\ &= \frac{s^2V_{in} - sv_C(0^-) - L\omega_0{}^2 i(0^-)}{s^2 + \omega_0{}^2} \end{aligned}$$

Steady state (time domain):

$$v_C = v_{in}$$

$$v_L = 0$$

$$i = 0$$

Constant source forced response (time domain):

$$\begin{aligned} i &= v_{in}\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\ &= v_{in}C\omega_0\sin(\omega_0 t) \end{aligned}$$

$$\begin{aligned} v_C &= v_{in} - v_L \\ &= v_{in}\cos\left(\frac{1}{\sqrt{LC}}t\right) \\ &= v_{in}\cos(\omega_0 t) \end{aligned}$$

Constant source total response (time domain):

$$\begin{aligned} i &= i(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) + (v_{in} - v_C(0^-))\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\ &= i(0^-)\cos(\omega_0 t) + (v_{in} - v_C(0^-))C\omega_0\sin(\omega_0 t) \end{aligned}$$

$$\begin{aligned} v_C &= v_{in} - v_L \\ &= i(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + (v_{in} + v_C(0^-))\cos\left(\frac{1}{\sqrt{LC}}t\right) \\ &= i(0^-)L\omega_0\sin(\omega_0 t) + (v_{in} + v_C(0^-))\cos(\omega_0 t) \end{aligned}$$

Resonance state (time domain): Maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$v_{in} = v_m\cos(\omega_0 t + \theta)$$

$$i_m = i(0^-) = \infty$$

$$v_C(0^-) = 0$$

$$i = i_m\cos(\omega_0 t + \theta)$$

$$v_C = -v_L = \frac{i_m}{C\omega_0}\sin(\omega_0 t + \theta) = i_m L\omega_0\sin(\omega_0 t + \theta)$$

$$E_m = \frac{L{i_m}^2}{2}$$

$$E_l = \frac{v_m i_m \pi}{\omega_0}$$

$$Q = \frac{L i_m \omega_0}{v_m} = \frac{i_m}{v_m}\sqrt{\frac{L}{C}}$$

Resonance state (phasor domain):

$$\mathbf{V}_{in} = v_m e^{j\theta}$$

$$\mathbf{I} = i_m e^{j\theta}, \quad i_m = \infty$$

$$\mathbf{V}_C = -\mathbf{V}_L = -\frac{j i_m}{C\omega_0}e^{j\theta} = -j i_m L\omega_0 e^{j\theta}$$

Impulse response:

$$\begin{aligned} h_i &= \frac{1}{L}\cos\left(\frac{1}{\sqrt{LC}}t\right)u(t) \\ &= \frac{1}{L}\cos(\omega_0 t)u(t) \end{aligned}$$

$$\begin{aligned} h_{v_C} &= \delta(t) - h_{v_L} \\ &= \frac{1}{\sqrt{LC}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\ &= \omega_0\sin(\omega_0 t) \end{aligned}$$

Transfer function:

$$\begin{aligned} H_I &= \frac{Cs}{LCs^2 + 1} \\ &= \frac{{\omega_0}^2 Cs}{s^2 + {\omega_0}^2} \end{aligned}$$

$$
\begin{aligned}
H_{V_C} &= \frac{1}{LCs^2+1} \\
&= \frac{{\omega_0}^2}{s^2+{\omega_0}^2} \\
H_{V_L} &= \frac{LCs^2}{LCs^2+1} \\
&= \frac{s^2}{s^2+{\omega_0}^2}
\end{aligned}
$$

V_C being low-pass filter:

$$\lim_{\omega\to 0}\left|H_{V_C}(j\omega)\right| = 1$$

$$\lim_{\omega\to\infty}\left|H_{V_C}(j\omega)\right| = 0$$

Proof.

$$\lim_{\omega\to 0}\left|H_{V_C}(j\omega)\right| = \lim_{\omega\to 0}\left|\frac{1}{1-LC\omega^2}\right| = 1$$

$$\lim_{\omega\to\infty}\left|H_{V_C}(j\omega)\right| = \lim_{\omega\to\infty}\left|\frac{1}{1-LC\omega^2}\right| = 0$$

□

V_L being high-pass filter:

$$\lim_{\omega\to 0}\left|H_{V_L}(j\omega)\right| = 0$$

$$\lim_{\omega\to\infty}\left|H_{V_L}(j\omega)\right| = 1$$

Proof.

$$\lim_{\omega\to 0}\left|H_{V_L}(j\omega)\right| = \lim_{\omega\to 0}\left|\frac{-LC\omega^2}{1-LC\omega^2}\right| = 0$$

$$\lim_{\omega\to\infty}\left|H_{V_L}(j\omega)\right| = \lim_{\omega\to\infty}\left|\frac{-LC\omega^2}{1-LC\omega^2}\right| = 1$$

□

x Parallel LC circuit

A (lossless) parallel LC (resonant, oscillator, tank, or tuned) circuit consists of an inductor with inductance L, a capacitor with capacitance C, and an independent current source i_{in} in parallel connection. Angular resonance/resonant frequency ω_0.

Natural response (time domain): Total energy stored in inductor and capacitor E.

$$LC\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + v = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + {\omega_0}^2 v = 0$$

$$
\begin{aligned}
v &= v(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - i_L(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\
&= v(0^-)\cos(\omega_0 t) - i_L(0^-)L\omega_0\sin(\omega_0 t)
\end{aligned}
$$

$$
\begin{aligned}
i_L &= -i_C \\
&= v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) \\
&= v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right)
\end{aligned}
$$

$$
\begin{aligned}
p_L &= -p_C \\
&= \frac{1}{2}((v(0^-))^2\sqrt{\frac{C}{L}} - (i_L(0^-))^2)\sin\left(\frac{2}{\sqrt{LC}}t\right) + v(0^-)i_L(0^-)\cos\left(\frac{2}{\sqrt{LC}}t\right) \\
&= \frac{1}{2}((v(0^-))^2C\omega_0 - (i_L(0^-))^2)\sin(2\omega_0 t) + v(0^-)i_L(0^-)\cos(2\omega_0 t)
\end{aligned}
$$

$$
\lim_{n\to\infty}\frac{1}{n}\int_0^n p_C\,\mathrm{d}t = -\lim_{n\to\infty}\frac{1}{n}\int_0^n p_L\,\mathrm{d}t = 0
$$

$$
E = \frac{1}{2}(C(v_C(0^-))^2 + L(I(0^-))^2)
$$

Proof.

$$
i_L + i_C = 0
$$

$$
v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}
$$

$$
i_L = -i_C = -C\frac{\mathrm{d}v}{\mathrm{d}t}
$$

$$
LC\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + v = 0
$$

$$
LCr^2 + 1 = 0
$$

$$
r = \pm\frac{1}{\sqrt{LC}}j
$$

$$
\omega_0 = \frac{1}{\sqrt{LC}}
$$

$$
\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + {\omega_0}^2 v = 0
$$

$$
v = A\cos\left(\frac{1}{\sqrt{LC}}t\right) + B\sin\left(\frac{1}{\sqrt{LC}}t\right)
$$

$$
A = v(0^-)
$$

$$
i_L = v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) - B\sqrt{\frac{C}{L}}\cos\left(\frac{1}{\sqrt{LC}}t\right)
$$

$$
B = -i_L(0^-)\sqrt{\frac{L}{C}}
$$

$$
\begin{aligned}
v &= v(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - i_L(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\
&= v(0^-)\cos(\omega_0 t) - i_L(0^-)L\omega_0\sin(\omega_0 t) \\
i_L &= -i_C = -C\frac{\mathrm{d}v}{\mathrm{d}t} \\
&= v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) \\
&= v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right)
\end{aligned}
$$

$$
\begin{aligned}
p_L &= -p_C \\
&= (v(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - i_L(0^-)\sqrt{\frac{L}{C}}\sin\left(\frac{1}{\sqrt{LC}}t\right))(v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right)) \\
&= (v(0^-))^2\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right)\cos\left(\frac{1}{\sqrt{LC}}t\right) + v(0^-)i_L(0^-)\cos^2(\frac{1}{\sqrt{LC}}t) - v(0^-)i_L(0^-)\sin^2(\frac{1}{\sqrt{LC}}t) - (i_L(0^- \\
&= \frac{1}{2}((v(0^-))^2\sqrt{\frac{C}{L}} - (i_L(0^-))^2)\sin\left(\frac{2}{\sqrt{LC}}t\right) + v(0^-)i_L(0^-)\cos\left(\frac{2}{\sqrt{LC}}t\right) \\
&= \frac{1}{2}((v(0^-))^2C\omega_0 - (i_L(0^-))^2)\sin(2\omega_0 t) + v(0^-)i_L(0^-)\cos(2\omega_0 t)
\end{aligned}
$$

□

Natural response (Laplace domain):

$$
\begin{aligned}
V &= \frac{LCsv(0^-) - Li_L(0^-)}{LCs^2 + 1} \\
&= \frac{sv(0^-) - L{\omega_0}^2 i_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
I_L &= -I_C \\
&= \frac{Cv(0^-) + LCsi_L(0^-)}{LCs^2 + 1} \\
&= \frac{C{\omega_0}^2 v(0^-) + si_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

Proof.

$$
\begin{aligned}
I_L + I_C &= 0 \\
V &= \frac{Cv(0^-) - \frac{i_L(0^-)}{s}}{\frac{1}{Ls} + Cs} \\
&= \frac{LCsv(0^-) - Li_L(0^-)}{LCs^2 + 1} \\
&= \frac{{\omega_0}^{-2}sv(0^-) - Li_L(0^-)}{{\omega_0}^{-2}s^2 + 1} \\
&= \frac{sv(0^-) - L{\omega_0}^2 i_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
I_L &= -I_C \\
&= \frac{V}{Ls} + \frac{i_L(0^-)}{s} \\
&= \frac{Csv(0^-) - i_L(0^-) + LCs^2 i_L(0^-) + i_L(0^-)}{LCs^3 + s} \\
&= \frac{Cv(0^-) + LCsi_L(0^-)}{LCs^2 + 1} \\
&= \frac{Cv(0^-) + {\omega_0}^{-2} s i_L(0^-)}{{\omega_0}^{-2} s^2 + 1} \\
&= \frac{C{\omega_0}^2 v(0^-) + s i_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

Check consistency:

$$
\begin{aligned}
\mathcal{L}\{V\} &= v(0^-)\frac{s}{s^2 + \frac{1}{LC}} - i_L(0^-)\sqrt{\frac{L}{C}}\frac{\frac{1}{\sqrt{LC}}}{s^2 + \frac{1}{LC}} \\
&= \frac{LCsv(0^-) - Li_L(0^-)}{LCs^2 + 1}
\end{aligned}
$$

$$
\mathcal{L}\{i_L\} = v(0^-)\sqrt{\frac{C}{L}}\frac{\frac{1}{\sqrt{LC}}}{s^2 + \frac{1}{LC}} + i_L(0^-)\frac{s}{s^2 + \frac{1}{LC}}
$$

□

Forced response (time domain):

$$
LC\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + v = \frac{\mathrm{d}i_{in}}{\mathrm{d}t}
$$

$$
\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + {\omega_0}^2 v = {\omega_0}^2 \frac{\mathrm{d}i_{in}}{\mathrm{d}t}
$$

$$
\begin{aligned}
v &= \frac{1}{C}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right) i_{in}(\tau)\,\mathrm{d}\tau \\
&= \frac{1}{C}\int_0^t \cos\big(\omega_0(t-\tau)\big) i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

$$
\begin{aligned}
i_L &= i_{in} - i_C \\
&= \frac{1}{\sqrt{LC}}\int_0^t \sin\left(\frac{1}{\sqrt{LC}}(t-\tau)\right) i_{in}(\tau)\,\mathrm{d}\tau \\
&= \omega_0 \int_0^t \sin\big(\omega_0(t-\tau)\big) i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

Proof.

$$
i_L + i_C = i_{in}
$$

$$
v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}
$$

$$
i_L = i_{in} - i_C = i_{in} - C\frac{\mathrm{d}v}{\mathrm{d}t}
$$

$$v = L\frac{\mathrm{d}i_{in}}{\mathrm{d}t} - LC\frac{\mathrm{d}^2 v}{\mathrm{d}t^2}$$

$$LC\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + v = \frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \frac{v}{LC} = \frac{1}{C}\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + {\omega_0}^2 v = {\omega_0}^2\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

$$v_1 = \cos\left(\frac{1}{\sqrt{LC}}t\right)$$

$$v_2 = \sin\left(\frac{1}{\sqrt{LC}}t\right)$$

$$\frac{1}{v_1 v_2' - v_2 v_1'} = \sqrt{LC}$$

$$\begin{aligned}
v &= \sqrt{\frac{L}{C}}(-\cos\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \sin\left(\frac{1}{\sqrt{LC}}\tau\right)\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau + \sin\left(\frac{1}{\sqrt{LC}}t\right)\int \cos\left(\frac{1}{\sqrt{LC}}\tau\right)\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau)\\
&= \sqrt{\frac{L}{C}}(-\cos\left(\frac{1}{\sqrt{LC}}t\right)(\sin\left(\frac{1}{\sqrt{LC}}t\right)(i_{in} - i_{in}(0^-)) - \frac{1}{\sqrt{LC}}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}\tau\right)i_{in}(\tau)\,\mathrm{d}t) + \sin\left(\frac{1}{\sqrt{LC}}t\right)(\cos\\
&= \frac{1}{C}\cos\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \cos\left(\frac{1}{\sqrt{LC}}\tau\right)i_{in}(\tau)\,\mathrm{d}\tau + \frac{1}{C}\sin\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \sin\left(\frac{1}{\sqrt{LC}}\tau\right)i_{in}(\tau)\,\mathrm{d}\tau\\
&= \frac{1}{C}(\cos\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \cos\left(\frac{1}{\sqrt{LC}}\tau\right)i_{in}(\tau)\,\mathrm{d}\tau + \sin\left(\frac{1}{\sqrt{LC}}t\right)\int_0^t \sin\left(\frac{1}{\sqrt{LC}}\tau\right)i_{in}(\tau)\,\mathrm{d}\tau)\\
&= \frac{1}{C}\int_0^t (\cos\left(\frac{1}{\sqrt{LC}}t\right)\cos\left(\frac{1}{\sqrt{LC}}\tau\right) + \sin\left(\frac{1}{\sqrt{LC}}t\right)\sin\left(\frac{1}{\sqrt{LC}}\tau\right))i_{in}(\tau)\,\mathrm{d}\tau\\
&= \frac{1}{C}\int_0^t \cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)i_{in}(\tau)\,\mathrm{d}\tau\\
&= \frac{1}{C}\int_0^t \cos\big(\omega_0(t-\tau)\big)i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$\begin{aligned}
i_L = i_{in} - i_C &= i_{in} - C\frac{\mathrm{d}v}{\mathrm{d}t}\\
&= i_{in} - \cos(0^-)i_{in} - \int_0^t \frac{\partial}{\partial t}(\cos\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)i_{in}(\tau))\,\mathrm{d}\tau\\
&= \frac{1}{\sqrt{LC}}\int_0^t \sin\left(\frac{1}{\sqrt{LC}}(t-\tau)\right)i_{in}(\tau)\,\mathrm{d}\tau\\
&= \omega_0\int_0^t \sin\big(\omega_0(t-\tau)\big)i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

□

Forced response (Laplace domain):

$$\begin{aligned}
V &= \frac{LsI_{in}}{LCs^2+1}\\
&= \frac{{\omega_0}^2 LsI_{in}}{s^2+{\omega_0}^2}
\end{aligned}$$

$$I_L = \frac{I_{in}}{LCs^2 + 1}$$
$$= \frac{{\omega_0}^2 I_{in}}{s^2 + {\omega_0}^2}$$

$$I_C = \frac{LCs^2 I_{in}}{LCs^2 + 1}$$
$$= \frac{s^2 I_{in}}{s^2 + {\omega_0}^2}$$

Proof.

$$V = \frac{I_{in}}{\frac{1}{Ls} + Cs}$$
$$= \frac{LsI_{in}}{LCs^2 + 1}$$
$$= \frac{LsI_{in}}{{\omega_0}^{-2} s^2 + 1}$$
$$= \frac{{\omega_0}^2 LsI_{in}}{s^2 + {\omega_0}^2}$$

$$I_L = \frac{I_{in}}{LCs^2 + 1}$$
$$= \frac{{\omega_0}^2 I_{in}}{s^2 + {\omega_0}^2}$$

$$I_C = \frac{LCs^2 I_{in}}{LCs^2 + 1}$$
$$= \frac{s^2 I_{in}}{s^2 + {\omega_0}^2}$$

Check consistency:

$$\mathcal{L}\{v\} = \frac{1}{C} \frac{s}{s^2 + \frac{1}{LC}} I_{in} = \frac{LsI_{in}}{LCs^2 + 1}$$

$$\mathcal{L}\{i_L\} = \frac{1}{\sqrt{LC}} \frac{\frac{1}{\sqrt{LC}}}{s^2 + \frac{1}{LC}} I_{in} = \frac{I_{in}}{LCs^2 + 1}$$

□

Total response (time domain):

$$v = \frac{1}{C} \int_0^t \cos\left(\frac{1}{\sqrt{LC}}(t - \tau)\right) i_{in}(\tau)\,\mathrm{d}\tau + v(0^-) \cos\left(\frac{1}{\sqrt{LC}} t\right) - i_L(0^-) \sqrt{\frac{L}{C}} \sin\left(\frac{1}{\sqrt{LC}} t\right)$$
$$= \frac{1}{C} \int_0^t \cos\big(\omega_0 (t - \tau)\big) i_{in}(\tau)\,\mathrm{d}\tau + v(0^-) \cos\big(\omega_0 t\big) - i_L(0^-) L\omega_0 \sin\big(\omega_0 t\big)$$

$$
\begin{aligned}
i_L &= i_{in} - i_C \\
&= \frac{1}{\sqrt{LC}} \int_0^t \sin\left(\frac{1}{\sqrt{LC}}(t-\tau)\right) i_{in}(\tau)\,\mathrm{d}\tau + v(0^-)\sqrt{\frac{C}{L}} \sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) \\
&= \omega_0 \int_0^t \sin\big(\omega_0(t-\tau)\big) i_{in}(\tau)\,\mathrm{d}\tau + v(0^-)\sqrt{\frac{C}{L}} \sin\left(\frac{1}{\sqrt{LC}}t\right) + i_L(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right)
\end{aligned}
$$

Total response (Laplace domain):

$$
\begin{aligned}
V &= \frac{LsI_{in} + LCsv(0^-) - Li_L(0^-)}{LCs^2+1} \\
&= \frac{{\omega_0}^2 LsI_{in} + sv(0^-) - L{\omega_0}^2 i_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
I_L &= \frac{I_{in} + Cv(0^-) + LCsi_L(0^-)}{LCs^2+1} \\
&= \frac{{\omega_0}^2 I_{in} + C{\omega_0}^2 v(0^-) + si_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
I_C &= \frac{LCs^2 I_{in} - Cv(0^-) - LCsi_L(0^-)}{LCs^2+1} \\
&= \frac{s^2 I_{in} - C{\omega_0}^2 v(0^-) - si_L(0^-)}{s^2 + {\omega_0}^2}
\end{aligned}
$$

Steady state (time domain):

$$i_L = i_{in}$$

$$i_C = 0$$

$$v = 0$$

Constant source forced response (time domain):

$$
\begin{aligned}
v &= i_{in}\sqrt{\frac{L}{C}} \sin\left(\frac{1}{\sqrt{LC}}t\right) \\
&= i_{in} L\omega_0 \sin\big(\omega_0 t\big)
\end{aligned}
$$

$$
\begin{aligned}
i_L &= i_{in} - i_C \\
&= -i_{in}\cos\left(\frac{1}{\sqrt{LC}}t\right) \\
&= -i_{in}\cos\big(\omega_0 t\big)
\end{aligned}
$$

Constant source total response (time domain):

$$
\begin{aligned}
v &= v(0^-)\cos\left(\frac{1}{\sqrt{LC}}t\right) - (i_{in} - i_L(0^-))\sqrt{\frac{L}{C}} \sin\left(\frac{1}{\sqrt{LC}}t\right) \\
&= v(0^-)\cos\big(\omega_0 t\big) + (i_{in} - i_L(0^-))L\omega_0 \sin\big(\omega_0 t\big)
\end{aligned}
$$

$$\begin{aligned} i_L &= i_{in} - i_C \\ &= v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) - (i_{in} - i_L(0^-))\cos\left(\frac{1}{\sqrt{LC}}t\right) \\ &= v(0^-)\sqrt{\frac{C}{L}}\sin\left(\frac{1}{\sqrt{LC}}t\right) + (i_{in} - i_L(0^-))\cos\left(\omega_0 t\right) \end{aligned}$$

Resonance state (time domain): Maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$i_{in} = i_m \cos\left(\omega_0 t + \theta\right)$$

$$v(0^-) = v_m = \infty$$

$$v = v_m \cos\left(\omega_0 t + \theta\right)$$

$$i_L(0^-) = 0$$

$$i_C = -i_L = -v_m C\omega_0 \sin\left(\omega_0 t + \theta\right) = \frac{v_m}{L\omega_0}\sin\left(\omega_0 t + \theta\right)$$

$$E_m = \frac{C{v_m}^2}{2}$$

$$E_l = \frac{v_m i_m \pi}{\omega_0}$$

$$Q = \frac{C v_m \omega_0}{i_m} = \frac{v_m}{i_m}\sqrt{\frac{C}{L}}$$

Resonance state (phasor domain):

$$\mathbf{I}_{in} = i_m e^{j\theta}$$

$$\mathbf{V} = v_m e^{j\theta}, \quad v_m = \infty$$

$$\mathbf{I}_C = -\mathbf{I}_L = -jv_m C\omega_0 e^{j\theta} = -\frac{jv_m}{L\omega_0}e^{j\theta}$$

Impulse response:

$$\begin{aligned} h_i &= \frac{1}{C}\cos\left(\frac{1}{\sqrt{LC}}t\right)u(t) \\ &= \frac{1}{C}\cos\left(\omega_0 t\right)u(t) \end{aligned}$$

$$\begin{aligned} h_{i_L} &= \delta(t) - h_{i_C} \\ &= \frac{1}{\sqrt{LC}}\sin\left(\frac{1}{\sqrt{LC}}t\right) \\ &= \omega_0 \sin\left(\omega_0 t\right) \end{aligned}$$

Transfer function:

$$\begin{aligned} H_V &= \frac{Ls}{LCs^2 + 1} \\ &= \frac{{\omega_0}^2 Ls}{s^2 + {\omega_0}^2} \end{aligned}$$

$$\begin{aligned} H_{I_L} &= \frac{1}{LCs^2 + 1} \\ &= \frac{{\omega_0}^2}{s^2 + {\omega_0}^2} \end{aligned}$$

$$
\begin{aligned}
H_{I_C} &= \frac{LCs^2}{LCs^2+1} \\
&= \frac{s^2}{s^2+{\omega_0}^2}
\end{aligned}
$$

I_L being low-pass filter:

$$\lim_{\omega\to 0}\left|H_{I_L}(j\omega)\right| = 1$$

$$\lim_{\omega\to\infty}\left|H_{I_L}(j\omega)\right| = 0$$

Proof.

$$\lim_{\omega\to 0}\left|H_{I_L}(j\omega)\right| = \lim_{\omega\to 0}\left|\frac{1}{1-LC\omega^2}\right| = 1$$

$$\lim_{\omega\to\infty}\left|H_{I_L}(j\omega)\right| = \lim_{\omega\to\infty}\left|\frac{1}{1-LC\omega^2}\right| = 0$$

□

I_C being high-pass filter:

$$\lim_{\omega\to 0}\left|H_{I_C}(j\omega)\right| = 0$$

$$\lim_{\omega\to\infty}\left|H_{I_C}(j\omega)\right| = 1$$

Proof.

$$\lim_{\omega\to 0}\left|H_{I_C}(j\omega)\right| = \lim_{\omega\to 0}\left|\frac{-LC\omega^2}{1-LC\omega^2}\right| = 0$$

$$\lim_{\omega\to\infty}\left|H_{I_C}(j\omega)\right| = \lim_{\omega\to\infty}\left|\frac{-LC\omega^2}{1-LC\omega^2}\right| = 1$$

□

xi Series RLC circuit

A series RLC (oscillator) circuit consists of a resistor with resistance R, an inductor with inductance L, a capacitor with capacitance C, and an independent voltage source v_{in} in series connection. Neper frequency or attenuation α, angular resonance/resonant frequency ω_0, damping factor/ratio ζ, characteristic roots $r_{1,2}$, critical resistance R_c (the resistance for critically damped response), damping angular frequency ω_d, $t_n = \dfrac{2\pi n}{\omega}$, stored energy in the inductor and capacitor E_L and E_C, $E = E_L + E_C$, dissipated energy of the resistor in E_R.

Natural response (time domain):

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{LC} = 0$$

$$\alpha = \frac{R}{2L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

$$R_c = 2\sqrt{\frac{L}{C}}$$

$$\omega_d = \omega_0\sqrt{1-\zeta^2}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}i}{\mathrm{d}t} + {\omega_0}^2 i = 0$$

$$r_{1,2} = -\omega_0(\zeta \pm \sqrt{\zeta^2-1})$$

Proof.

$$v_R + v_C + v_L = 0$$

$$v_R = Ri$$

$$i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$$

$$v_L = L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$R\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{C} + L\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} = 0$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{LC} = 0$$

$$\alpha = \frac{R}{2L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\zeta = \frac{\alpha}{\omega_0} = \frac{R}{2}\sqrt{\frac{C}{L}}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}i}{\mathrm{d}t} + {\omega_0}^2 i = 0$$

$$r^2 + 2\omega_0\zeta r + {\omega_0}^2 = 0$$

$$\begin{aligned} r_{1,2} &= \frac{-2\omega_0\zeta \pm \sqrt{4{\omega_0}^2\zeta^2 - 4{\omega_0}^2}}{2} \\ &= -\omega_0(\zeta \pm \sqrt{\zeta^2-1}) \end{aligned}$$

□

- $\zeta > 1$ Overdamped response:

$$i = -\frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$v_R = -\frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$v_L = \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta}{4\zeta\sqrt{\zeta^2-1}}$$

$$v_C = \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

Proof.

$$i = Ae^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + Be^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$i(0^-) = A + B$$

$$v_R = ARe^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + BRe^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$\begin{aligned} v_L &= L\frac{\mathrm{d}i}{\mathrm{d}t} \\ &= -AR\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - BR\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t} \end{aligned}$$

$$\begin{aligned} v_C &= -v_R - v_L \\ &= -ARe^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - BRe^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t} + AR\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + BR\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta+\sqrt{} \\ &= -AR\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - BR\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t} \end{aligned}$$

$$\begin{aligned} v_C(0^-) &= -AR\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta} - BR\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta} \\ &= -AR\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta} - i(0^-)R\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta} + AR\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta} \\ &= \frac{-i(0^-)R(\zeta-\sqrt{\zeta^2-1}) - 2AR\sqrt{\zeta^2-1}}{2\zeta} \end{aligned}$$

$$A = -\frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}$$

$$B = i(0^-) - A = \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}$$

$$i = -\frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$\begin{aligned} v_C &= \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}R\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}R\frac{\zeta \\ &= \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t} \end{aligned}$$

Check $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$\begin{aligned} C\frac{\mathrm{d}v_C}{\mathrm{d}t} &= -\frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}\frac{2\zeta}{R}(\zeta-\sqrt{\zeta^2-1})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2}}{4\zeta\sqrt{\zeta^2-1}} \\ &= -\frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-}} \end{aligned}$$

$$\begin{aligned} v_L &= \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2\sqrt{\zeta^2-1}}\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2\sqrt{\zeta^2-1}}\frac{\zeta+}{} \\ &= \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta}{4\zeta\sqrt{\zeta^2-1}} \end{aligned}$$

□

- $\zeta = 1$ Critically damped response:

$$i = (i(0^-) - (i(0^-) + 2\frac{v_C(0^-)}{R})\omega_0 t)e^{-\omega_0 t}$$

$$v_R = (i(0^-)R - (i(0^-)R + 2v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$v_L = -(i(0^-)R + v_C(0^-) + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$v_C = (v_C(0^-) + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

Proof.

$$i = (A + Bt)e^{-\omega_0 t}$$

$$A = i(0^-)$$

$$i = (i(0^-) + Bt)e^{-\omega_0 t}$$

$$v_R = (i(0^-) + Bt)Re^{-\omega_0 t}$$

$$\begin{aligned} v_L &= L\frac{\mathrm{d}i}{\mathrm{d}t} \\ &= L(-(i(0^-) + Bt)\omega_0 + B)e^{-\omega_0 t} \\ &= (-i(0^-)\frac{R}{2} + \frac{BR}{2\omega_0} - \frac{BR}{2}t)e^{-\omega_0 t} \end{aligned}$$

$$\begin{aligned} v_C &= -v_R - v_L \\ &= (-i(0^-)R - BRt + i(0^-)\frac{R}{2} - \frac{BR}{2\omega_0} + \frac{BR}{2}t)e^{-\omega_0 t} \\ &= (-i(0^-)\frac{R}{2} - \frac{BR}{2\omega_0} - \frac{BR}{2}t)e^{-\omega_0 t} \end{aligned}$$

$$v_C(0^-) = -i(0^-)\frac{R}{2} - \frac{BR}{2\omega_0}$$

$$\begin{aligned} B &= -(i(0^-)\frac{R}{2} + v_C(0^-))\frac{2\omega_0}{R} \\ &= -\frac{(i(0^-)R + 2v_C(0^-))\omega_0}{R} \\ &= -(i(0^-) + 2\frac{v_C(0^-)}{R})\omega_0 \end{aligned}$$

$$i = (i(0^-) - (i(0^-) + 2\frac{v_C(0^-)}{R})\omega_0 t)e^{-\omega_0 t}$$

$$\begin{aligned} v_C &= (-i(0^-)\frac{R}{2} + \frac{i(0^-)R\omega_0 + 2v_C(0^-)\omega_0}{2\omega_0} + \frac{i(0^-)R\omega_0 + 2v_C(0^-)\omega_0}{2}t)e^{-\omega_0 t} \\ &= (v_C(0^-) + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t} \end{aligned}$$

Check $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$\begin{aligned} C\frac{\mathrm{d}v_C}{\mathrm{d}t} &= (-v_C(0^-)\frac{2}{R} + (\frac{1}{2}i(0^-)R + v_C(0^-))\frac{2}{R} - (\frac{1}{2}i(0^-)R + v_C(0^-))\frac{2}{R}\omega_0 t)e^{-\omega_0 t} \\ &= (i(0^-) - (\frac{1}{2}i(0^-)R + v_C(0^-))\frac{2}{R}\omega_0 t)e^{-\omega_0 t} \\ &= (i(0^-) - (i(0^-) + 2\frac{v_C(0^-)}{R})\omega_0 t)e^{-\omega_0 t} \end{aligned}$$

$$v_R = (i(0^-)R - (i(0^-)R + 2v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$
$$v_L = -(i(0^-)R + v_C(0^-) + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

□

- $0 < \zeta < 1$ Underdamped response:

$$i = e^{-\omega_0 \zeta t}(i(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (2\frac{v_C(0^-)}{R} + i(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$v_R = e^{-\omega_0 \zeta t}(i(0^-)R\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (2v_C(0^-) + i(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$v_L = e^{-\omega_0 \zeta t}(-(v_C(0^-) + i(0^-)R)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \frac{v_C(0^-)\zeta + \frac{1}{2}i(0^-)R\frac{2\zeta^2-1}{\zeta}}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$v_C = e^{-\omega_0 \zeta t}(v_C(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (v_C(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}} + \frac{1}{2}\frac{i(0^-)R}{\zeta\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$\omega = \omega_0\sqrt{1-\zeta^2}$$
$$t_n = \frac{2\pi n}{\omega_0\sqrt{1-\zeta^2}}$$
$$i(t_n) = e^{-\omega_0\zeta t_n}i(0^-)$$
$$v_C(t_n) = e^{-\omega_0\zeta t_n}v_C(0^-)$$
$$E_L(t_n) = \frac{1}{2}L(i(0^-))^2 e^{-2\omega_0\zeta t_n}$$
$$E_C(t_n) = \frac{1}{2}C(v_C(0^-))^2 e^{-2\omega_0\zeta t_n}$$
$$E(t_n) = (\frac{1}{2}L(i(0^-))^2 + \frac{1}{2}C(v_C(0^-))^2)e^{-2\omega_0\zeta t_n}$$
$$E_R(t_n) = (\frac{1}{2}L(i(0^-))^2 + \frac{1}{2}C(v_C(0^-))^2)(1 - e^{-2\omega_0\zeta t_n})$$

Proof.

$$i = e^{-\omega_0\zeta t}(A\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$A = i(0^-)$$
$$i = e^{-\omega_0\zeta t}(i(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$v_R = Re^{-\omega_0\zeta t}(i(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$\begin{aligned}
v_L &= L\frac{\mathrm{d}i}{\mathrm{d}t} \\
&= -\frac{R}{2\zeta}\zeta e^{-\omega_0\zeta t}(i(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)) + \frac{R}{2\zeta}\sqrt{1-\zeta^2}e^{-\omega_0\zeta t}(-i(0^-)\sin\left(\omega_0\sqrt{1-\right. \\
&= \frac{R}{2\zeta}e^{-\omega_0\zeta t}((B\sqrt{1-\zeta^2} - i(0^-)\zeta)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (B\zeta + i(0^-)\sqrt{1-\zeta^2})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))
\end{aligned}$$

$$\begin{aligned}
v_C &= -v_R - v_L \\
&= \frac{R}{2\zeta}e^{-\omega_0\zeta t}((-B\sqrt{1-\zeta^2} + i(0^-)\zeta - 2i(0^-)\zeta)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (B\zeta + i(0^-)\sqrt{1-\zeta^2} - 2B\zeta)\sin\left(\omega_0 \sqrt{\right. \\
&= \frac{R}{2\zeta}e^{-\omega_0\zeta t}(-(B\sqrt{1-\zeta^2} + i(0^-)\zeta)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (-B\zeta + i(0^-)\sqrt{1-\zeta^2})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))
\end{aligned}$$

$$v_C(0^-) = -\frac{R}{2\zeta}(B\sqrt{1-\zeta^2} + i(0^-)\zeta)$$

$$B = -(2\frac{v_C(0^-)}{R} + i(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}$$

$$i = e^{-\omega_0\zeta t}(i(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) - (2\frac{v_C(0^-)}{R} + i(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big))$$

$$\begin{aligned}
v_C &= \frac{R}{2\zeta}e^{-\omega_0\zeta t}(((2\frac{v_C(0^-)}{R} + i(0^-)) - i(0^-))\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + ((2\frac{v_C(0^-)}{R} + i(0^-))\frac{\zeta^2}{\sqrt{1-\zeta^2}} + i(0^-)\sqrt{1-} \\
&= e^{-\omega_0\zeta t}(v_C(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + (2\frac{v_C(0^-)}{R}\frac{\zeta^2}{\sqrt{1-\zeta^2}} + i(0^-)\frac{\zeta^2}{\sqrt{1-\zeta^2}} + i(0^-)\frac{1-\zeta^2}{\sqrt{1-\zeta^2}})\frac{R}{2\zeta}\sin\Big(\omega_0\sqrt{1} \\
&= e^{-\omega_0\zeta t}(v_C(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + (v_C(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}} + \frac{1}{2}\frac{i(0^-)R}{\zeta\sqrt{1-\zeta^2}})\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big))
\end{aligned}$$

Check $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$\begin{aligned}
C\frac{\mathrm{d}v_C}{\mathrm{d}t} &= -\frac{2\zeta^2}{R}e^{-\omega_0\zeta t}(v_C(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + (v_C(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}} + \frac{1}{2}\frac{i(0^-)R}{\zeta\sqrt{1-\zeta^2}})\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big)) + \frac{2\zeta}{} \\
&= \frac{2\zeta}{R}e^{-\omega_0\zeta t}(-v_C(0^-)\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) - (v_C(0^-)\frac{\zeta^2}{\sqrt{1-\zeta^2}} + i(0^-)R\frac{1}{2\sqrt{1-\zeta^2}})\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big) \\
&= \frac{2\zeta}{R}e^{-\omega_0\zeta t}(i(0^-)R\frac{1}{2\zeta}\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) - (v_C(0^-)\frac{1}{\sqrt{1-\zeta^2}} + i(0^-)R\frac{1}{2\sqrt{1-\zeta^2}})\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big)) \\
&= e^{-\omega_0\zeta t}(i(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) - (2\frac{v_C(0^-)}{R} + i(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big))
\end{aligned}$$

$$v_R = e^{-\omega_0\zeta t}(i(0^-)R\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) - (2v_C(0^-) + i(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big))$$

$$v_L = e^{-\omega_0\zeta t}(-(v_C(0^-) + i(0^-)R)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + \frac{v_C(0^-)\zeta + \frac{1}{2}i(0^-)R\frac{2\zeta^2-1}{\zeta}}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}t\Big))$$

□

Natural response (Laplace domain):

$$\begin{aligned}
I &= \frac{i(0^-)LCs - v_C(0^-)C}{LCs^2 + RCs + 1} \\
&= \frac{i(0^-)s - \frac{2v_C(0^-)\omega_0\zeta}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
V_R &= \frac{i(0^-)RLCs - v_C(0^-)RC}{LCs^2 + RCs + 1} \\
&= \frac{i(0^-)Rs - 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
V_L &= -\frac{(i(0^-)R + v_C(0^-))LCs + i(0^-)L}{LCs^2 + RCs + 1} \\
&= -\frac{(i(0^-)R + v_C(0^-))s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$
\begin{aligned}
V_C &= \frac{v_C(0^-)LCs + i(0^-)L + v_C(0^-)RC}{LCs^2 + RCs + 1} \\
&= \frac{v_C(0^-)s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta} + 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

Proof.

$$
\begin{aligned}
I &= \frac{i(0^-)L - \frac{v_C(0^-)}{s}}{Ls + R + \frac{1}{Cs}} \\
&= \frac{i(0^-)LCs - v_C(0^-)C}{LCs^2 + RCs + 1} \\
&= \frac{i(0^-)s - \frac{v_C(0^-)}{L}}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \\
&= \frac{i(0^-)s - \frac{2v_C(0^-)\omega_0\zeta}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_R &= \frac{i(0^-)RLCs - v_C(0^-)RC}{LCs^2 + RCs + 1} \\
&= \frac{i(0^-)Rs - 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_L &= LsI - Li(0^-) \\
&= \frac{i(0^-)L^2Cs^2 - v_C(0^-)LCs - i(0^-)L^2Cs^2 - i(0^-)RLCs - i(0^-)L}{LCs^2 + RCs + 1} \\
&= -\frac{(i(0^-)R + v_C(0^-))LCs + i(0^-)L}{LCs^2 + RCs + 1} \\
&= -\frac{(i(0^-)R + v_C(0^-))s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_C &= \frac{I}{Cs} + \frac{v_C(0^-)}{s} \\
&= \frac{i(0^-)Ls - v_C(0^-) + v_C(0^-)(LCs^2 + RCs + 1)}{s(LCs^2 + RCs + 1)} \\
&= \frac{v_C(0^-)LCs + i(0^-)L + v_C(0^-)RC}{LCs^2 + RCs + 1} \\
&= \frac{\frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta}s - v_C(0^-)\omega_0^2 + v_C(0^-)(s^2 + 2\omega_0\zeta s + {\omega_0}^2)}{s(s^2 + 2\omega_0\zeta s + {\omega_0}^2)} \\
&= \frac{v_C(0^-)s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta} + 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

Check consistency:

- $\zeta > 1$ Overdamped response:

$$\mathcal{L}\{i\} = -\frac{i(0^-)R(\zeta - \sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}\frac{1}{s+\omega_0(\zeta-\sqrt{\zeta^2-1})} + \frac{i(0^-)R(\zeta + \sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}\frac{}{s+\omega_0(}$$

$$= \frac{(-i(0^-)R(\zeta-\sqrt{\zeta^2-1}) - 2v_C(0^-)\zeta)(s+\omega_0(\zeta+\sqrt{\zeta^2-1})) + (i(0^-)R(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta)(s+}{2R\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s+{\omega_0}^2)}$$

$$= \frac{-i(0^-)R(\zeta-\sqrt{\zeta^2-1})s - i(0^-)R(\zeta-\sqrt{\zeta^2-1})\omega_0(\zeta+\sqrt{\zeta^2-1}) - 2v_C(0^-)\zeta s - 2v_C(0^-)\omega_0\zeta(\zeta+\sqrt{}}{2R\sqrt{\zeta^2}}$$

$$= \frac{2i(0^-)R\sqrt{\zeta^2-1}s - 4v_C(0^-)\omega_0\zeta\sqrt{\zeta^2-1}}{2R\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s+{\omega_0}^2)}$$

$$= \frac{i(0^-)s - \frac{2v_C(0^-)\omega_0\zeta}{R}}{s^2+2\omega_0\zeta s+{\omega_0}^2}$$

$$\mathcal{L}\{v_L\} = \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}\frac{1}{s+\omega_0(\zeta-\sqrt{\zeta^2-1})} - \frac{i(0^-)R(\zeta+\sqrt{\zeta^2-1})^2 + 2}{4\zeta\sqrt{\zeta^2}}$$

$$= \frac{(i(0^-)R(\zeta-\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1}))(s+\omega_0(\zeta+\sqrt{\zeta^2-1}))(i(0^-)R(\zeta+\sqrt{\zeta^2-1})^2 +}{4\zeta\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s+{\omega_0}^2)}$$

$$= \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1})^2 s + i(0^-)R\omega_0(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})s + 2v_C(0^-)\zeta - i(0^-)R}{4\zeta\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s}$$

$$= \frac{-4i(0^-)R\zeta\sqrt{\zeta^2-1}s - 2i(0^-)R\omega_0\sqrt{\zeta^2-1} - 4v_C(0^-)\zeta\sqrt{\zeta^2-1}s}{4\zeta\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s+{\omega_0}^2)}$$

$$= -\frac{(i(0^-)R + v_C(0^-))s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta}}{s^2+2\omega_0\zeta s+{\omega_0}^2}$$

$$\mathcal{L}\{v_C\} = \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}\frac{1}{s+\omega_0(\zeta-\sqrt{\zeta^2-1})} - \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}\frac{}{s+\omega_0}$$

$$= \frac{(i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1}))(s+\omega_0(\zeta+\sqrt{\zeta^2-1})) - (i(0^-)R + 2v_C(0^-)\zeta(\zeta-\sqrt{\zeta^2-1}))(s+}{4\zeta\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s+{\omega_0}^2)}$$

$$= \frac{i(0^-)Rs + i(0^-)R\omega_0(\zeta+\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})s + 2v_C(0^-)\omega_0\zeta(\zeta+\sqrt{\zeta^2-1})^2 - i(0}{4\zeta\sqrt{\zeta^2-1}(s^2+2\omega}$$

$$= \frac{2i(0^-)R\omega_0\sqrt{\zeta^2-1} + 4v_C(0^-)\zeta\sqrt{\zeta^2-1}s + 8v_C(0^-)\omega_0\zeta^2\sqrt{\zeta^2-1}}{4\zeta\sqrt{\zeta^2-1}(s^2+2\omega_0\zeta s+{\omega_0}^2)}$$

$$= \frac{v_C(0^-)s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta} + 2v_C(0^-)\omega_0\zeta}{s^2+2\omega_0\zeta s+{\omega_0}^2}$$

- $\zeta = 1$ Critically damped response:

$$I = \frac{i(0^-)s - 2\frac{v_C(0^-)\omega_0}{R}}{(s+\omega_0)^2}$$

$$V_L = -\frac{(i(0^-)R + v_C(0^-))s + \frac{1}{2}i(0^-)R\omega_0}{(s+\omega_0)^2}$$

$$V_C = \frac{v_C(0^-)s + \frac{1}{2}i(0^-)R\omega_0 + 2v_C(0^-)\omega_0}{(s+\omega_0)^2}$$

$$\mathcal{L}\{i\} = \frac{i(0^-)}{s+\omega_0} - (i(0^-) + 2\frac{v_C(0^-)}{R})\omega_0\frac{1}{(s+\omega_0)^2}$$

$$= \frac{i(0^-)s - 2\frac{v_C(0^-)\omega_0}{R}}{(s+\omega_0)^2}$$

$$\mathcal{L}\{v_L\} = -\frac{i(0^-)R + v_C(0^-)}{s+\omega_0} - (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0\frac{1}{(s+\omega_0)^2}$$

$$= -\frac{(i(0^-)R + v_C(0^-))s + \frac{1}{2}i(0^-)R\omega_0}{(s+\omega_0)^2}$$

$$\mathcal{L}\{v_C\} = \frac{v_C(0^-)}{s+\omega_0} + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0\frac{1}{(s+\omega_0)^2}$$

$$= \frac{v_C(0^-)s + \frac{1}{2}i(0^-)R\omega_0 + 2v_C(0^-)\omega_0}{(s+\omega_0)^2}$$

- $0 < \zeta < 1$ Underdamped response:

$$\mathcal{L}\{i\} = i(0^-)\frac{(s+\omega_0\zeta)}{(s+\omega_0\zeta)^2 + \omega_0^2(1-\zeta^2)} - (2\frac{v_C(0^-)}{R} + i(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\frac{\omega_0\sqrt{1-\zeta^2}}{(s+\omega_0\zeta)^2 + \omega_0^2(1-\zeta^2)}$$

$$= \frac{i(0^-)(s+\omega_0\zeta) - (2\frac{v_C(0^-)}{R} + i(0^-))\omega_0\zeta}{s^2 + 2\omega_0\zeta s + \omega_0^2}$$

$$= \frac{i(0^-)s - \frac{2v_C(0^-)\omega_0\zeta}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$\mathcal{L}\{v_L\} = \frac{-(v_C(0^-) + i(0^-)R)(s+\omega_0\zeta) + \frac{v_C(0^-)\zeta + \frac{1}{2}i(0^-)R\frac{2\zeta^2-1}{\zeta}}{\sqrt{1-\zeta^2}}\omega_0\sqrt{1-\zeta^2}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$= \frac{-(v_C(0^-) + i(0^-)R)s - v_C(0^-)\omega_0\zeta - i(0^-)R\omega_0\zeta + v_C(0^-)\omega_0\zeta + \frac{1}{2}i(0^-)\omega_0R\frac{2\zeta^2-1}{\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$= -\frac{(i(0^-)R + v_C(0^-))s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$\mathcal{L}\{v_C\} = \frac{v_C(0^-)(s+\omega_0\zeta) + (v_C(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}} + \frac{1}{2}\frac{i(0^-)R}{\zeta\sqrt{1-\zeta^2}})\omega_0\sqrt{1-\zeta^2}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$= \frac{v_C(0^-)s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta} + 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

□

Forced response (time domain):

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{LC} = \frac{1}{L}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}i}{\mathrm{d}t} + {\omega_0}^2 = 2\frac{\omega_0\zeta}{R}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

Proof.

$$v_R + v_C + v_L = v_{in}$$

$$v_R = Ri$$

$$i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$$

$$v_L = L\frac{\mathrm{d}i}{\mathrm{d}t}$$

$$R\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{C} + L\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} = \frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + \frac{R}{L}\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{LC} = \frac{1}{L}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}i}{\mathrm{d}t} + {\omega_0}^2 = 2\frac{\omega_0\zeta}{R}\frac{\mathrm{d}v_{in}}{\mathrm{d}t}$$

□

- $\zeta > 1$ Overdamped response:

$$i = 2\frac{\omega_0\zeta}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_R = 2\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_L = v_{in}+\omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-2\zeta\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)+\frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_C = \frac{\omega_0}{\sqrt{\zeta^2-1}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau$$

Proof.

$$i_1 = e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}$$

$$i_2 = e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$\frac{1}{i_1 i_2' - i_2 i_1'} = -\frac{1}{2\omega_0\sqrt{\zeta^2-1}}e^{2\omega_0\zeta t}$$

$$
\begin{aligned}
i &= \frac{\zeta}{R\sqrt{\zeta^2-1}}(e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}\int_0^t e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})\tau}\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}e^{2\omega_0\zeta\tau}\,\mathrm{d}\tau - e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}\int_0^t e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})\tau}\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}e^{2\omega_0\zeta \ldots} \\
&= \frac{\zeta}{R\sqrt{\zeta^2-1}}(e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}\int_0^t e^{\omega_0(\zeta-\sqrt{\zeta^2-1})\tau}\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau - e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}\int_0^t e^{\omega_0(\zeta+\sqrt{\zeta^2-1})\tau}\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau) \\
&= \frac{\zeta}{R\sqrt{\zeta^2-1}}(e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}(e^{\omega_0(\zeta-\sqrt{\zeta^2-1})t}(v_{in}-v_{in}(0^-)) - \omega_0(\zeta-\sqrt{\zeta^2-1})\int_0^t e^{\omega_0(\zeta-\sqrt{\zeta^2-1})\tau}v_{in}(\tau)\,\mathrm{d}\tau) - e \ldots \\
&= \frac{\omega_0\zeta}{R\sqrt{\zeta^2-1}}(-(\zeta-\sqrt{\zeta^2-1})\int_0^t e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau + (\zeta+\sqrt{\zeta^2-1})\int_0^t e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})(t-\tau)}v_{in}(\tau)\,\mathrm{d} \ldots \\
&= \frac{\omega_0\zeta}{R\sqrt{\zeta^2-1}}\int_0^t(-(\zeta-\sqrt{\zeta^2-1})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})(t-\tau)} + (\zeta+\sqrt{\zeta^2-1})e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})(t-\tau)})v_{in}(\tau)\,\mathrm{d}\tau \\
&= \frac{\omega_0\zeta}{R}\int_0^t(-\frac{\zeta}{\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})(t-\tau)} + e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})(t-\tau)} + \frac{\zeta}{\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})(t-\tau)} + e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})(\ldots} \\
&= 2\frac{\omega_0\zeta}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

$$
v_R = 2\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau
$$

$$
\begin{aligned}
v_L &= L\frac{\mathrm{d}i}{\mathrm{d}t} \\
&= 2\frac{L\omega_0\zeta}{R}(v_{in} + \int_0^t \frac{\partial}{\partial t}(e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big)\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big)))v_{in}(\tau)\,\mathrm{d}\tau \\
&= v_{in} + \int_0^t(-\omega_0\zeta e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big)) + \omega_0\sqrt{\zeta^2-} \ldots \\
&= v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-\zeta\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) + \frac{\zeta^2}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) + \sqrt{\zeta^2-1}\,\mathrm{s}\ldots \\
&= v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-2\zeta\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) + \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

$$
\begin{aligned}
v_C &= v_{in} - v_R - v_L \\
&= -2\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau - \omega_0\int \ldots \\
&= \frac{\omega_0}{\sqrt{\zeta^2-1}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big)v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

Check $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$
\begin{aligned}
C\frac{\mathrm{d}v_C}{\mathrm{d}t} &= \frac{C\omega_0}{\sqrt{\zeta^2-1}}\int_0^t \frac{\partial}{\partial t}(e^{-\omega_0\zeta(t-\tau)}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau \\
&= 2\frac{\zeta}{R\sqrt{\zeta^2-1}}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\omega_0\sqrt{\zeta^2-1}\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \omega_0\zeta\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v \ldots \\
&= 2\frac{\omega_0\zeta}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

□

- $\zeta = 1$ Critically damped response:

$$i = 2\frac{\omega_0}{R}\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_R = 2\omega_0\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_L = v_{in} - \omega_0\int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_C = {\omega_0}^2\int_0^t (t-\tau)e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau$$

Proof.

$$i_1 = e^{-\omega_0 t}$$

$$i_2 = te^{-\omega_0 t}$$

$$\frac{1}{i_1i_2' - i_2i_1'} = e^{2\omega_0 t}$$

$$\begin{aligned}
i &= 2\frac{\omega_0\zeta}{R}(-e^{-\omega_0 t}\int_0^t \tau e^{\omega_0\tau}\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau + te^{-\omega_0 t}\int_0^t e^{\omega_0\tau}\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau) \\
&= 2\frac{\omega_0}{R}(-e^{-\omega_0 t}(te^{\omega_0 t}(v_{in} - v_{in}(0^-)) - \int_0^t (\omega_0\tau + 1)e^{\omega_0\tau}v_{in}(\tau)\,\mathrm{d}\tau) + te^{-\omega_0 t}(e^{\omega_0 t}(v_{in} - v_{in}(0^-)) - \omega_0\int_0^t e^{\omega_0\tau}v_i \\
&= 2\frac{\omega_0}{R}\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$v_R = 2\omega_0\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau$$

$$\begin{aligned}
v_L &= L\frac{\mathrm{d}i}{\mathrm{d}t} \\
&= 2\frac{L\omega_0}{R}\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau \\
&= v_{in} + \int_0^t \frac{\partial}{\partial t}((1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)})v_{in}(\tau)\,\mathrm{d}\tau \\
&= v_{in} + \int_0^t -\omega_0 e^{-\omega_0(t-\tau)} - (-\omega_0(t-\tau)+1)\omega_0 e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau \\
&= v_{in} - \omega_0\int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$\begin{aligned}
v_C &= v_{in} - v_R - v_L \\
&= -2\omega_0\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau + \omega_0\int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau \\
&= {\omega_0}^2\int_0^t (t-\tau)e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

Check $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$\begin{aligned}
C\frac{\mathrm{d}v_C}{\mathrm{d}t} &= 2\frac{\omega_0}{R}\int_0^t \frac{\partial}{\partial t}((t-\tau)e^{-\omega_0(t-\tau)})v_{in}(\tau)\,\mathrm{d}\tau \\
&= 2\frac{\omega_0}{R}\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

□

- $0 < \zeta < 1$ Underdamped response:

$$i = 2\frac{\omega_0}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_R = 2\omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_L = v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-2\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) + \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_C = \frac{\omega_0}{\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau$$

Proof.

$$i_1 = e^{-\omega_0\zeta t}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)$$

$$i_2 = e^{-\omega_0\zeta t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)$$

$$i_1 i_2' = e^{-2\omega_0\zeta t}(-\omega_0\zeta\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \omega_0\sqrt{1-\zeta^2}\cos^2(\omega_0\sqrt{1-\zeta^2}t))$$

$$i_2 i_1' = e^{-2\omega_0\zeta t}(-\omega_0\zeta\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \omega_0\sqrt{1-\zeta^2}\sin^2(\omega_0\sqrt{1-\zeta^2}t))$$

$$\begin{aligned}\frac{1}{i_1 i_2' - i_2 i_1'} &= \frac{1}{e^{-2\omega_0\zeta t}(-\omega_0\zeta\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \omega_0\sqrt{1-\zeta^2}\cos^2(\omega_0\sqrt{1-\zeta^2}t) + \omega_0\zeta\sin} \\ &= \frac{1}{\omega_0\sqrt{1-\zeta^2}}e^{2\omega_0\zeta t}\end{aligned}$$

$$\begin{aligned}i &= 2\frac{\zeta}{R\sqrt{1-\zeta^2}}(-e^{-\omega_0\zeta t}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\int_0^t e^{\omega_0\zeta\tau}\sin\left(\omega_0\sqrt{1-\zeta^2}\tau\right)\frac{\mathrm{d}v_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau + e^{-\omega_0\zeta t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right) \\ &= 2\frac{\zeta}{R\sqrt{1-\zeta^2}}(-e^{-\omega_0\zeta t}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)(e^{\omega_0\zeta t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)(v_{in} - v_{in}(0^-)) - \int_0^t e^{\omega_0\zeta\tau}(\omega_0\sqrt{1-\zeta^2}\,\mathrm{co} \\ &= 2\frac{\zeta}{R\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\omega_0\sqrt{1-\zeta^2}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}\tau\right) + \omega_0\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\mathrm{s} \\ &= 2\frac{\omega_0\zeta}{R\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\sqrt{1-\zeta^2}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}\tau\right) + \zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\sin(\omega \\ &= 2\frac{\omega_0\zeta}{R\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\sqrt{1-\zeta^2}\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \zeta\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau \\ &= 2\frac{\omega_0}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau\end{aligned}$$

$$v_R = 2\omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau$$

$$
\begin{aligned}
v_L &= L\frac{\mathrm{d}i}{\mathrm{d}t} \\
&= v_{in} + \frac{1}{\sqrt{1-\zeta^2}}\int_0^t \frac{\partial}{\partial t}(e^{-\omega_0\zeta(t-\tau)}(\sqrt{1-\zeta^2}\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \zeta\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)))v_{in}(\tau) \\
&= v_{in} + \int_0^t (-\omega_0\zeta e^{-\omega_0\zeta(t-\tau)}(\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)) + \omega_0\sqrt{1-\zeta^2} \\
&= v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) + \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \sqrt{1-\zeta^2}\sin \\
&= v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-2\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) + \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

$$
\begin{aligned}
v_C &= v_{in} - v_R - v_L \\
&= -2\omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau - \omega_0\int_0^t \\
&= -\omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{2\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - 2\zeta\cos\left(\omega_0\sqrt{1-\right. \\
&= \frac{\omega_0}{\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

Check $i = C\dfrac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$
\begin{aligned}
C\frac{\mathrm{d}v_C}{\mathrm{d}t} &= 2\frac{\zeta}{R\sqrt{1-\zeta^2}}\int_0^t \frac{\partial}{\partial t}(e^{-\omega_0\zeta(t-\tau)}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau \\
&= 2\frac{\omega_0}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))v_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}
$$

□

Forced response (Laplace domain):

$$
\begin{aligned}
I &= \frac{V_{in}Cs}{LCs^2 + RCs + 1} \\
&= \frac{2V_{in}\frac{\omega_0\zeta s}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_R &= \frac{V_{in}RCs}{LCs^2 + RCs + 1} \\
&= \frac{2V_{in}\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_L &= \frac{V_{in}LCs^2}{LCs^2 + RCs + 1} \\
&= \frac{V_{in}s^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
V_C &= \frac{V_{in}}{LCs^2 + RCs + 1} \\
&= \frac{V_{in}{\omega_0}^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

Proof.

$$
\begin{aligned}
I &= \frac{V_{in}}{Ls + R + \frac{1}{Cs}} \\
&= \frac{V_{in} Cs}{LCs^2 + RCs + 1} \\
&= \frac{V_{in}\frac{s}{L}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= \frac{2V_{in}\frac{\omega_0\zeta s}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
V_R &= \frac{V_{in} RCs}{LCs^2 + RCs + 1} \\
&= \frac{2V_{in}\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
V_L &= LsI \\
&= \frac{V_{in} LCs^2}{LCs^2 + RCs + 1} \\
&= \frac{V_{in} s^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
V_C &= \frac{I}{Cs} \\
&= \frac{V_{in}}{LCs^2 + RCs + 1} \\
&= \frac{V_{in}{\omega_0}^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

Check consistency:

- $\zeta > 1$ Overdamped response:

$$
\begin{aligned}
\mathcal{L}\{i\} &= 2\frac{\omega_0\zeta}{R}\left(\frac{s + \omega_0\zeta}{(s+\omega_0\zeta)^2 - {\omega_0}^2(\zeta^2-1)} - \frac{\zeta}{\sqrt{\zeta^2-1}}\frac{\omega_0\sqrt{\zeta^2-1}}{(s+\omega_0\zeta)^2 - {\omega_0}^2(\zeta^2-1)}\right)V_{in} \\
&= \frac{2V_{in}\frac{\omega_0\zeta s}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

- $\zeta = 1$ Critically damped response:

$$
\begin{aligned}
\mathcal{L}\{i\} &= 2\frac{\omega_0}{R}\left(\frac{1}{s+\omega_0} - \frac{\omega_0}{(s+\omega_0)^2}\right)V_{in} \\
&= \frac{2V_{in}\frac{\omega_0 s}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

- $0 < \zeta < 1$ Underdamped response:

$$
\begin{aligned}
\mathcal{L}\{i\} &= 2\frac{\omega_0}{R}\left(\zeta(s+\omega_0\zeta) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\omega_0\sqrt{1-\zeta^2}\right)\frac{V_{in}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= \frac{2V_{in}\frac{\omega_0\zeta s}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$\square$

Total response (time domain):

- $\zeta > 1$ Overdamped response:

$$i = 2\frac{\omega_0\zeta}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau - \frac{i(0^-)R(\zeta -}{}$$

$$v_R = 2\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau - \frac{i(0^-)R(\zeta}{}$$

$$v_L = v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-2\zeta\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) + \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau + \frac{i(0}{}$$

$$v_C = \frac{\omega_0}{\sqrt{\zeta^2-1}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big)v_{in}(\tau)\,\mathrm{d}\tau + \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta}}$$

- $\zeta = 1$ Critically damped response:

$$i = 2\frac{\omega_0}{R}\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau$$

$$v_R = 2\omega_0\int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau + (i(0^-) - (i(0^-) + 2\frac{v_C(0^-)}{R})\omega_0 t)e^{-\omega_0 t}$$

$$v_L = v_{in} - \omega_0\int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau - (i(0^-)R + v_C(0^-) + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$v_C = {\omega_0}^2\int_0^t (t-\tau)e^{-\omega_0(t-\tau)}v_{in}(\tau)\,\mathrm{d}\tau + (v_C(0^-) + (\frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

- $0 < \zeta < 1$ Underdamped response:

$$i = 2\frac{\omega_0}{R}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}(i(0^-)\,c$$

$$v_R = 2\omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}(i(0^-)$$

$$v_L = v_{in} + \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(-2\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) + \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big))v_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta}$$

$$v_C = \frac{\omega_0}{\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big)v_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}(v_C(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + (v_C(0^-)\frac{}{\sqrt{}}$$

Total response (Laplace domain):

$$\begin{aligned} I &= \frac{V_{in}Cs + i(0^-)LCs - v_C(0^-)C}{LCs^2 + RCs + 1} \\ &= \frac{2V_{in}\frac{\omega_0\zeta s}{R} + i(0^-)s - \frac{2v_C(0^-)\omega_0\zeta}{R}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \end{aligned}$$

$$V_R = \frac{V_{in}RCs + i(0^-)RLCs - v_C(0^-)RC}{LCs^2 + RCs + 1}$$
$$= \frac{2V_{in}\omega_0\zeta s + i(0^-)Rs - 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + \omega_0^{\ 2}}$$

$$V_L = \frac{V_{in}LCs^2 - (i(0^-)R + v_C(0^-))LCs - i(0^-)L}{LCs^2 + RCs + 1}$$
$$= \frac{V_{in}s^2 - (i(0^-)R + v_C(0^-))s - \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta}}{s^2 + 2\omega_0\zeta s + \omega_0^{\ 2}}$$

$$V_C = \frac{V_{in} + v_C(0^-)LCs + i(0^-)L + v_C(0^-)RC}{LCs^2 + RCs + 1}$$
$$= \frac{V_{in}\omega_0^{\ 2} + v_C(0^-)s + \frac{1}{2}\frac{i(0^-)R\omega_0}{\zeta} + 2v_C(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + \omega_0^{\ 2}}$$

Steady state (time domain):

$$v_C = v_{in}$$

$$v_R = v_L = 0$$

$$i = 0$$

Constant source forced response (time domain):

- $\zeta > 1$ Overdamped response:

$$i = 2\frac{v_{in}}{R}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2 - 1}}\,\text{sinh}\left(\omega_0\sqrt{\zeta^2 - 1}t\right)$$

$$v_R = 2v_{in}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2 - 1}}\,\text{sinh}\left(\omega_0\sqrt{\zeta^2 - 1}t\right)$$

$$v_L = v_{in}e^{-\omega_0\zeta t}(\text{cosh}\left(\omega_0\sqrt{\zeta^2 - 1}t\right) - \frac{\zeta}{\sqrt{\zeta^2 - 1}}\,\text{sinh}\left(\omega_0\sqrt{\zeta^2 - 1}t\right))$$

$$v_C = v_{in}(1 - e^{-\omega_0\zeta t}(\text{cosh}\left(\omega_0\sqrt{\zeta^2 - 1}t\right) + \frac{\zeta}{\sqrt{\zeta^2 - 1}}\,\text{sinh}\left(\omega_0\sqrt{\zeta^2 - 1}t\right)))$$

- $\zeta = 1$ Critically damped response:

$$i = 2v_{in}\frac{\omega_0}{R}te^{-\omega_0 t}$$

$$v_R = 2v_{in}\omega_0 te^{-\omega_0 t}$$

$$v_L = v_{in}(1 - \omega_0 t)e^{-\omega_0 t}$$

$$v_C = v_{in}(1 - (1 + \omega_0 t)e^{-\omega_0 t})$$

- $0 < \zeta < 1$ Underdamped response:

$$i = 2\frac{v_{in}}{R}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{1 - \zeta^2}}\,\text{sin}\left(\omega_0\sqrt{1 - \zeta^2}t\right)$$

$$v_R = 2v_{in}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{1 - \zeta^2}}\,\text{sin}\left(\omega_0\sqrt{1 - \zeta^2}t\right)$$

$$v_L = v_{in}e^{-\omega_0\zeta t}(\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$v_C = v_{in}(1 - e^{-\omega_0\zeta t}(\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)))$$

Constant source total response (time domain):

- $\zeta > 1$ Overdamped response:

$$i = 2\frac{v_{in}}{R}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right) - \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2R\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R(\zeta + \cdots}{2\cdots}$$

$$v_R = 2v_{in}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right) - \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1}) + 2v_C(0^-)\zeta}{2\sqrt{\zeta^2-1}}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{i(0^-)R(\zeta + \cdots}{\cdots}$$

$$v_L = v_{in}e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right)) + \frac{i(0^-)R(\zeta-\sqrt{\zeta^2-1})^2 + 2v_C(0^-)\zeta(\zeta - \cdots}{4\zeta\sqrt{\zeta^2-1}}\cdots$$

$$v_C = v_{in}(1 - e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right) + \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right))) + \frac{i(0^-)R + 2v_C(0^-)\zeta(\zeta+\sqrt{\zeta^2-1})}{4\zeta\sqrt{\zeta^2-1}}e\cdots$$

- $\zeta = 1$ Critically damped response:

$$i = (i(0^-) + (-i(0^-) + 2\frac{v_{in} - v_C(0^-)}{R})\omega_0 t)e^{-\omega_0 t}$$

$$v_R = (i(0^-)R + (-Ri(0^-) + 2(v_{in} - v_C(0^-)))\omega_0 t)e^{-\omega_0 t}$$

$$v_L = (v_{in} - i(0^-)R - v_C(0^-) - (v_{in} + \frac{1}{2}i(0^-)R + v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$v_C = v_{in} - (v_{in} - v_C(0^-) + (v_{in} - \frac{1}{2}i(0^-)R - v_C(0^-))\omega_0 t)e^{-\omega_0 t}$$

- $0 < \zeta < 1$ Underdamped response:

$$i = e^{-\omega_0\zeta t}(i(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (2\frac{v_{in} - v_C(0^-)}{R} - i(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$v_R = e^{-\omega_0\zeta t}(i(0^-)R\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (2(v_{in} - v_C(0^-)) - i(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$v_L = e^{-\omega_0\zeta t}((v_{in} - v_C(0^-) - i(0^-)R)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \frac{v_{in} - v_C(0^-)\zeta - \frac{1}{2}i(0^-)R\frac{2\zeta^2-1}{\zeta}}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$v_C = v_{in} - e^{-\omega_0\zeta t}((v_{in} - v_C(0^-))\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + ((v_{in} - v_C(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}} - \frac{1}{2}\frac{i(0^-)R}{\zeta\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

Resonance state (time domain): Maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$v_{in} = v_m\cos(\omega_0 t + \theta)$$

$$i(0^-) = \frac{v_m}{R}$$

$$v_C(0^-) = 0$$

$$i = \frac{v_m}{R}\cos\left(\omega_0 t + \theta\right)$$

$$v_C = -v_L = \frac{v_m}{RC\omega_0}\sin\left(\omega_0 t + \theta\right) = \frac{v_m L\omega_0}{R}\sin\left(\omega_0 t + \theta\right)$$

$$E_m = \frac{L{v_m}^2}{2R^2}$$

$$E_l = \frac{{v_m}^2\pi}{R\omega_0}$$

$$Q = \frac{L\omega_0}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Resonance state (phasor domain):

$$\mathbf{V}_{in} = v_m e^{j\theta}$$

$$\mathbf{I} = \frac{v_m}{R}e^{j\theta}$$

$$\mathbf{V}_C = -\mathbf{V}_L = -\frac{jv_m}{RC\omega_0}e^{j\theta} = -\frac{jv_m L\omega_0}{R}e^{j\theta}$$

Impulse response:

- $\zeta > 1$ Overdamped response:

$$h_i = 2\frac{\omega_0\zeta}{R}e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right))$$

$$h_{v_R} = 2\omega_0\zeta e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right))$$

$$h_{v_L} = v_{in} + \omega_0 e^{-\omega_0\zeta t}(-2\zeta\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right) + \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right))$$

$$h_{v_C} = \frac{\omega_0}{\sqrt{\zeta^2-1}}e^{-\omega_0\zeta t}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right)$$

- $\zeta = 1$ Critically damped response:

$$h_i = 2\frac{\omega_0}{R}(1-\omega_0 t)e^{-\omega_0 t}$$

$$h_{v_R} = 2\omega_0(1-\omega_0 t)e^{-\omega_0 t}$$

$$h_{v_L} = v_{in} - \omega_0(2-\omega_0 t)e^{-\omega_0 t}$$

$$h_{v_C} = {\omega_0}^2 t e^{-\omega_0 t}$$

- $0 < \zeta < 1$ Underdamped response:

$$h_i = 2\frac{\omega_0}{R}e^{-\omega_0\zeta t}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$h_{v_R} = 2\omega_0 e^{-\omega_0\zeta t}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$h_{v_L} = v_{in} + \omega_0 e^{-\omega_0\zeta t}(-2\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$h_{v_C} = \frac{\omega_0}{\sqrt{1-\zeta^2}}e^{-\omega_0\zeta t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)$$

Transfer function:

$$
\begin{aligned}
H_I &= \frac{Cs}{LCs^2 + RCs + 1} \\
&= \frac{2\frac{\omega_0 \zeta s}{R}}{s^2 + 2\omega_0 \zeta s + {\omega_0}^2} \\
H_{V_R} &= \frac{RCs}{LCs^2 + RCs + 1} \\
&= \frac{2\omega_0 \zeta s}{s^2 + 2\omega_0 \zeta s + {\omega_0}^2} \\
H_{V_L} &= \frac{LCs^2}{LCs^2 + RCs + 1} \\
&= \frac{s^2}{s^2 + 2\omega_0 \zeta s + {\omega_0}^2} \\
H_{V_C} &= \frac{1}{LCs^2 + RCs + 1} \\
&= \frac{{\omega_0}^2}{s^2 + 2\omega_0 \zeta s + {\omega_0}^2}
\end{aligned}
$$

V_L being high-pass filter:

$$\lim_{\omega \to 0} \left| H_{V_L}(j\omega) \right| = 0$$

$$\lim_{\omega \to \infty} \left| H_{V_L}(j\omega) \right| = 1$$

V_C being low-pass filter:

$$\lim_{\omega \to 0} \left| H_{V_C}(j\omega) \right| = 1$$

$$\lim_{\omega \to \infty} \left| H_{V_C}(j\omega) \right| = 0$$

xii Parallel RLC circuit

A parallel RLC (oscillator) circuit consists of a resistor with resistance R, an inductor with inductance L, a capacitor with capacitance C, and an independent current source i_{in} in parallel connection. Neper frequency or attenuation α, angular resonance/resonant frequency ω_0, damping factor/ratio ζ, characteristic roots $r_{1,2}$, critical resistance R_c (the resistance for critically damped response), damping angular frequency ω_d, $t_n = \frac{2\pi n}{\omega}$, stored energy in the inductor and capacitor E_L and E_C, $E = E_L + E_C$, dissipated energy of the resistor in E_R.

Natural response (time domain):

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{LC} = 0$$

$$\alpha = \frac{1}{2RC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$$

$$R_c = \frac{1}{2}\sqrt{\frac{L}{C}}$$

$$\omega_d = \omega_0\sqrt{1-\zeta^2}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}v}{\mathrm{d}t} + {\omega_0}^2 v = 0$$

$$r_{1,2} = -\omega_0(\zeta \pm \sqrt{\zeta^2-1})$$

Proof.

$$i_R + i_L + i_C = 0$$

$$v = Ri_R$$

$$i_C = C\frac{\mathrm{d}v}{\mathrm{d}t}$$

$$v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$$

$$\frac{1}{R}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{L} + C\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} = 0$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{LC} = 0$$

$$\alpha = \frac{1}{2RC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\zeta = \frac{\alpha}{\omega_0} = \frac{1}{2R}\sqrt{\frac{L}{C}}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}v}{\mathrm{d}t} + {\omega_0}^2 v = 0$$

$$r^2 + 2\omega_0\zeta r + {\omega_0}^2 = 0$$

$$\begin{aligned} r_{1,2} &= \frac{-2\omega_0\zeta \pm \sqrt{4{\omega_0}^2\zeta^2 - 4{\omega_0}^2}}{2} \\ &= -\omega_0(\zeta \pm \sqrt{\zeta^2-1}) \end{aligned}$$

□

- $\zeta > 1$ Overdamped response:

$$v = -(v(0^-)\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + (v(0^-)\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta+\sqrt{}}$$

$$i_R = -(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + (\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta+\sqrt{\zeta}}$$

$$i_L = (\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - (\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}}$$

$$i_C = (\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - (\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}}$$

Proof.

$$v = Ae^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + Be^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$v(0^-) = A + B$$

$$i_R = \frac{A}{R}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + \frac{B}{R}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$

$$\begin{aligned}
i_C &= C\frac{\mathrm{d}v}{\mathrm{d}t} \\
&= -\frac{A}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{B}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t} \\
i_L &= -i_R - i_C \\
&= -\frac{A}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} - \frac{B}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}
\end{aligned}$$

$$\begin{aligned}
i_L(0^-) &= -\frac{A}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta} - \frac{v(0^-)-A}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta} \\
&= -\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta} - \frac{A}{R}\frac{\sqrt{\zeta^2-1}}{\zeta}
\end{aligned}$$

$$A = -(v(0^-)\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})$$

$$B = v(0^-)\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}}$$

$$v = -(v(0^-)\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}+(v(0^-)\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta+\sqrt{}}$$

$$i_L = (\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}-(\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-}}$$

Check $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$:

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = -(v(0^-)\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t} + (v(0^-)\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})$$

$$i_R = -(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}+(\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta+\sqrt{\zeta}}$$

$$i_C = (\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}-(\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-}}$$

□

- $\zeta = 1$ Critically damped response:

$$v = (v(0^-) - (v(0^-) + 2i_L(0^-)R)\omega_0 t)e^{-\omega_0 t}$$

$$i_R = (\frac{v(0^-)}{R} - (\frac{v(0^-)}{R} + 2i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_L = (i_L(0^-) + (\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_C = (-\frac{v(0^-)}{R} - i_L(0^-) + (\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

Proof.

$$v = (A + Bt)e^{-\omega_0 t}$$

$$A = v(0^-)$$

$$v = (v(0^-) + Bt)e^{-\omega_0 t}$$

$$i_R = \frac{v(0^-) + Bt}{R}e^{-\omega_0 t}$$

$$\begin{aligned} i_C &= C\frac{\mathrm{d}v}{\mathrm{d}t} \\ &= \frac{\frac{B}{\omega_0} - v(0^-) - Bt}{2R}e^{-\omega_0 t} \end{aligned}$$

$$\begin{aligned} i_L &= -i_R - i_C \\ &= -\frac{\frac{B}{\omega_0} + v(0^-) + Bt}{2R}e^{-\omega_0 t} \end{aligned}$$

$$i_L(0^-) = -\frac{\frac{B}{\omega_0} + v(0^-)}{2R}$$

$$B = -(v(0^-) + 2i_L(0^-)R)\omega_0$$

$$v = (v(0^-) - (v(0^-) + 2i_L(0^-)R)\omega_0 t)e^{-\omega_0 t}$$

$$i_L = (i_L(0^-) + (\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

Check $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$:

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = (v(0^-) - (v(0^-) + 2i_L(0^-)R)\omega_0 t)e^{-\omega_0 t}$$

$$i_R = (\frac{v(0^-)}{R} - (\frac{v(0^-)}{R} + 2i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_C = (-\frac{v(0^-)}{R} - i_L(0^-) + (\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

□

- $0 < \zeta < 1$ Underdamped response:

$$v = e^{-\omega_0\zeta t}(v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (v(0^-) + 2i_L(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_R = e^{-\omega_0\zeta t}(\frac{v(0^-)}{R}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (\frac{v(0^-)}{R} + 2i_L(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_L = e^{-\omega_0\zeta t}(i_L(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (\frac{1}{2}\frac{v(0^-)}{R\zeta\sqrt{1-\zeta^2}} + i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_C = e^{-\omega_0\zeta t}((\frac{v(0^-)}{R} + i_L(0^-))\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (\frac{1}{2}\frac{v(0^-)}{R}\frac{1-2\zeta^2}{\zeta\sqrt{1-\zeta^2}} - i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$\omega = \omega_0\sqrt{1-\zeta^2}$$

$$t_n = \frac{2\pi n}{\omega_0\sqrt{1-\zeta^2}}$$

$$v(t_n) = e^{-\omega_0\zeta t_n}v(0^-)$$

$$i_L(t_n) = e^{-\omega_0\zeta t} i_L(0^-)$$
$$E_C(t_n) = \frac{1}{2}C(v(0^-))^2 e^{-2\omega_0\zeta t_n}$$
$$E_L(t_n) = \frac{1}{2}L(i_L(0^-))^2 e^{-2\omega_0\zeta t_n}$$
$$E(t_n) = (\frac{1}{2}L(i_L(0^-))^2 + \frac{1}{2}C(v(0^-))^2)e^{-2\omega_0\zeta t_n}$$
$$E_R(t_n) = (\frac{1}{2}L(i_L(0^-))^2 + \frac{1}{2}C(v(0^-))^2)(1 - e^{-2\omega_0\zeta t_n})$$

Proof.

$$v = e^{-\omega_0\zeta t}(A\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$A = i(0^-)$$
$$v = e^{-\omega_0\zeta t}(v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$i_R = \frac{1}{R}e^{-\omega_0\zeta t}(v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$\begin{aligned}
i_C &= C\frac{\mathrm{d}v}{\mathrm{d}t} \\
&= \frac{1}{2R\omega_0\zeta}e^{-\omega_0\zeta t}(-\omega_0\zeta v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \omega_0\zeta B\sin\left(\omega_0\sqrt{1-\zeta^2}t\right) - \omega_0\sqrt{1-\zeta^2}v(0^-)\sin\left(\omega_0\sqrt{1\right. \\
&= -\frac{1}{2R}e^{-\omega_0\zeta t}((v(0^-) - B\frac{\sqrt{1-\zeta^2}}{\zeta})\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (v(0^-)\frac{\sqrt{1-\zeta^2}}{\zeta} + B)\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))
\end{aligned}$$

$$\begin{aligned}
i_L &= -i_R - i_C \\
&= \frac{1}{2R}e^{-\omega_0\zeta t}(-(v(0^-) + B\frac{\sqrt{1-\zeta^2}}{\zeta})\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (v(0^-)\frac{\sqrt{1-\zeta^2}}{\zeta} - B)\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))
\end{aligned}$$

$$i_L(0^-) = -\frac{1}{2R}(v(0^-) + B\frac{\sqrt{1-\zeta^2}}{\zeta})$$
$$B = -(v(0^-) + 2i_L(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}$$
$$v = e^{-\omega_0\zeta t}(v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (v(0^-) + 2i_L(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$
$$i_L = e^{-\omega_0\zeta t}(i_L(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (\frac{1}{2}\frac{v(0^-)}{R\zeta\sqrt{1-\zeta^2}} + i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

Check $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$:

$$\begin{aligned}
L\frac{\mathrm{d}i_L}{\mathrm{d}t} &= 2R\zeta e^{-\omega_0\zeta t}(-i_L(0^-)\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (\frac{1}{2}\frac{v(0^-)}{R\sqrt{1-\zeta^2}} + i_L(0^-)\frac{\zeta^2}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right) - i_L(\\
&= e^{-\omega_0\zeta t}(v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (v(0^-) + 2i_L(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))
\end{aligned}$$

$$i_R = e^{-\omega_0\zeta t}(\frac{v(0^-)}{R}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - (\frac{v(0^-)}{R} + 2i_L(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_C = e^{-\omega_0\zeta t}((\frac{v(0^-)}{R} + i_L(0^-))\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (\frac{1}{2}\frac{v(0^-)}{R}\frac{1-2\zeta^2}{\zeta\sqrt{1-\zeta^2}} - i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

□

Natural response (Laplace domain):

$$V = \frac{v(0^-)RLCs - i_L(0^-)RL}{RLCs^2 + Ls + R}$$
$$= \frac{v(0^-)s - 2i_L(0^-)R\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$I_R = \frac{v(0^-)LCs - i_L(0^-)L}{RLCs^2 + Ls + R}$$
$$= \frac{\frac{v(0^-)s}{R} - 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$I_L = \frac{i_L(0^-)RLCs + v(0^-)RC + i_L(0^-)L}{RLCs^2 + Ls + R}$$
$$= \frac{i_L(0^-)s + \frac{v(0^-)\omega_0}{2R\zeta} + 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$I_C = -\frac{(v(0^-) + i_L(0^-)R)LCs + v(0^-)RC}{RLCs^2 + Ls + R}$$
$$= -\frac{(\frac{v(0^-)}{R} + i_L(0^-))s + \frac{v(0^-)\omega_0}{2R\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

Proof.

$$V = \frac{v(0^-)C - \frac{i_L(0^-)}{s}}{Cs + \frac{1}{R} + \frac{1}{Ls}}$$
$$= \frac{v(0^-)RLCs - i_L(0^-)RL}{RLCs^2 + Ls + R}$$
$$= \frac{v(0^-)s - 2i_L(0^-)R\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$I_R = \frac{v(0^-)LCs - i_L(0^-)L}{RLCs^2 + Ls + R}$$
$$= \frac{\frac{v(0^-)s}{R} - 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$I_L = \frac{V}{Ls} + \frac{i_L(0^-)}{s}$$
$$= \frac{v(0^-)RC - \frac{i_L(0^-)R}{s} + i_L(0^-)RLCs + i_L(0^-)L + \frac{i_L(0^-)R}{s}}{RLCs^2 + Ls + R}$$
$$= \frac{i_L(0^-)RLCs + v(0^-)RC + i_L(0^-)L}{RLCs^2 + Ls + R}$$
$$= \frac{i_L(0^-)s + \frac{v(0^-)\omega_0}{2R\zeta} + 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

$$
\begin{aligned}
I_C &= CsV - Cv(0^-) \\
&= \frac{v(0^-)RLC^2s^2 - i_L(0^-)RLCs - v(0^-)RLC^2s^2 - v(0^-)LCs - v(0^-)RC}{RLCs^2 + Ls + R} \\
&= -\frac{(v(0^-) + i_L(0^-)R)LCs + v(0^-)RC}{RLCs^2 + Ls + R} \\
&= -\frac{(\frac{v(0^-)}{R} + i_L(0^-))s + \frac{v(0^-)\omega_0}{2R\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

Check consistency:

- $\zeta > 1$ Overdamped response:

$$
\begin{aligned}
\mathcal{L}\{v\} &= -(v(0^-)\frac{\zeta - \sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{1}{s+\omega_0(\zeta - \sqrt{\zeta^2-1})} + (v(0^-)\frac{\zeta + \sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{}{\sqrt{\zeta}} \\
&= \frac{-(v(0^-)\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})(s+\omega_0(\zeta + \sqrt{\zeta^2-1})) + (v(0^-)\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})(s+\omega_0}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= \frac{v(0^-)s - 2i_L(0^-)R\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}\{i_L\} &= (\frac{v(0^-)}{R}\frac{\zeta - \sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta + \sqrt{\zeta^2-1}}{2\zeta}\frac{1}{s+\omega_0(\zeta - \sqrt{\zeta^2-1})} - (\frac{v(0^-)}{R}\frac{\zeta + \sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i \\
&= \frac{(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta+\sqrt{\zeta^2-1}}{2\zeta}(s+\omega_0(\zeta + \sqrt{\zeta^2-1})) - (\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta-\sqrt{\zeta}}{2\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= \frac{i_L(0^-)s + \frac{v(0^-)\omega_0}{2R\zeta} + 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}\{i_C\} &= (\frac{v(0^-)}{R}\frac{\zeta - \sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta - \sqrt{\zeta^2-1}}{2\zeta}\frac{1}{s+\omega_0(\zeta - \sqrt{\zeta^2-1})} - (\frac{v(0^-)}{R}\frac{\zeta + \sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + \\
&= \frac{(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta-\sqrt{\zeta^2-1}}{2\zeta}(s+\omega_0(\zeta + \sqrt{\zeta^2-1})) - (\frac{v(0^-)}{R}\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta+\sqrt{\zeta}}{2\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= -\frac{(\frac{v(0^-)}{R} + i_L(0^-))s + \frac{v(0^-)\omega_0}{2R\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

- $\zeta = 1$ Critically damped response:

$$
\begin{aligned}
\mathcal{L}\{v\} &= \frac{v(0^-)}{s+\omega_0} - \frac{(v(0^-) + 2i_L(0^-)R)\omega_0}{s^2 + 2\omega_0 s + {\omega_0}^2} \\
&= \frac{v(0^-)s - 2i_L(0^-)R\omega_0}{s^2 + 2\omega_0 s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}\{i_L\} &= \frac{i_L(0^-)}{s+\omega_0} + \frac{(\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0}{s^2 + 2\omega_0 s + {\omega_0}^2} \\
&= \frac{i_L(0^-)s + \frac{v(0^-)\omega_0}{2R} + 2i_L(0^-)\omega_0}{s^2 + 2\omega_0 s + {\omega_0}^2}
\end{aligned}
$$

$$\begin{aligned}
\mathcal{L}\{i_C\} &= -\frac{\frac{v(0^-)}{R} + i_L(0^-)}{s+\omega_0} + \frac{(\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0}{s^2 + 2\omega_0 s + {\omega_0}^2} \\
&= -\frac{(\frac{v(0^-)}{R} + i_L(0^-))s + \frac{v(0^-)\omega_0}{2R}}{s^2 + 2\omega_0 s + {\omega_0}^2}
\end{aligned}$$

- $0 < \zeta < 1$ Underdamped response:

$$\begin{aligned}
\mathcal{L}\{v\} &= \frac{v(0^-)(s+\omega\zeta) - (v(0^-) + 2i_L(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\omega_0\sqrt{1-\zeta^2}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= \frac{v(0^-)s - 2i_L(0^-)R\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
\mathcal{L}\{i_L\} &= \frac{i_L(0^-)(s+\omega\zeta) + (\frac{1}{2}\frac{v(0^-)}{R\zeta\sqrt{1-\zeta^2}} + i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\omega_0\sqrt{1-\zeta^2}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= \frac{i_L(0^-)s + \frac{v(0^-)\omega_0}{2R\zeta} + 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
\mathcal{L}\{i_C\} &= \frac{(\frac{v(0^-)}{R} + i_L(0^-))(s+\omega\zeta) + (\frac{1}{2}\frac{v(0^-)}{R}\frac{1-2\zeta^2}{\zeta\sqrt{1-\zeta^2}} - i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\omega_0\sqrt{1-\zeta^2}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \\
&= -\frac{(\frac{v(0^-)}{R} + i_L(0^-))s + \frac{v(0^-)\omega_0}{2R\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

□

Forced response (time domain):

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{LC} = \frac{1}{C}\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}v}{\mathrm{d}t} + {\omega_0}^2 v = 2R\omega_0\zeta\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

Proof.

$$i_R + i_L + i_C = i_{in}$$

$$v = Ri_R$$

$$i_C = C\frac{\mathrm{d}v}{\mathrm{d}t}$$

$$v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$$

$$\frac{1}{R}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{L} + C\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} = \frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + \frac{1}{RC}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{LC} = \frac{1}{C}\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2 v}{\mathrm{d}t^2} + 2\omega_0\zeta\frac{\mathrm{d}v}{\mathrm{d}t} + {\omega_0}^2 v = 2R\omega_0\zeta\frac{\mathrm{d}i_{in}}{\mathrm{d}t}$$

□

- $\zeta > 1$ Overdamped response:

$$v = 2R\omega_0\zeta \int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_R = 2\omega_0\zeta \int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_L = \frac{\omega_0}{\sqrt{\zeta^2-1}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_C = i_{in} - \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

Proof.

$$v_1 = e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}$$
$$v_2 = e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})t}$$
$$\frac{1}{v_1v_2' - v_2v_1'} = -\frac{1}{2\omega_0\sqrt{\zeta^2-1}}e^{2\omega_0\zeta t}$$

$$\begin{aligned}
v &= \frac{R\zeta}{\sqrt{\zeta^2-1}}(e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}\int_0^t e^{\omega_0(\zeta-\sqrt{\zeta^2-1})\tau}\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau - e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}\int_0^t e^{\omega_0(\zeta+\sqrt{\zeta^2-1})\tau}\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau) \\
&= \frac{R\zeta}{\sqrt{\zeta^2-1}}(e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}(e^{\omega_0(\zeta-\sqrt{\zeta^2-1})t}(i_{in} - i_{in}(0^-)) - \omega_0(\zeta-\sqrt{\zeta^2-1})\int_0^t e^{\omega_0(\zeta-\sqrt{\zeta^2-1})\tau}i_{in}(\tau)\,\mathrm{d}\tau) - e^{-\omega} \\
&= \frac{R\omega_0\zeta}{\sqrt{\zeta^2-1}}(-e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}(\zeta-\sqrt{\zeta^2-1})\int_0^t e^{\omega_0(\zeta-\sqrt{\zeta^2-1})\tau}i_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}(\zeta+\sqrt{\zeta^2-1})\int_0^t e^{\omega} \\
&= \frac{R\omega_0\zeta}{\sqrt{\zeta^2-1}}\int_0^t(-(\zeta-\sqrt{\zeta^2-1})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})(t-\tau)} + (\zeta+\sqrt{\zeta^2-1})e^{-\omega_0(\zeta+\sqrt{\zeta^2-1})(t-\tau)})i_{in}(\tau)\,\mathrm{d}\tau \\
&= \frac{R\omega_0\zeta}{\sqrt{\zeta^2-1}}\int_0^t(-\zeta e^{-\omega_0\zeta(t-\tau)}(e^{\omega_0\sqrt{\zeta^2-1}(t-\tau)} - e^{-\omega_0\sqrt{\zeta^2-1}(t-\tau)}) + \sqrt{\zeta^2-1}e^{-\omega_0\zeta(t-\tau)}(e^{\omega_0\sqrt{\zeta^2-1}(t-\tau)} + e^{-\omega_0\sqrt{\zeta^2-}} \\
&= 2R\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$i_R = 2\omega_0\zeta \int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$\begin{aligned}
i_C &= C\frac{\mathrm{d}v}{\mathrm{d}t} \\
&= i_{in} + \int_0^t \frac{\partial}{\partial t}(e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)))i_{in}(\tau)\,\mathrm{d}\tau \\
&= i_{in} + \omega_0\int_0^t(-\zeta e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)) + \sqrt{\zeta^2-1}e \\
&= i_{in} - \omega_0\int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$\begin{aligned}
i_L &= i_{in} - i_R - i_C \\
&= \frac{\omega_0}{\sqrt{\zeta^2-1}}\int_0^t e^{-\omega_0\zeta(t-\tau)}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

Check $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$:

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = 2R\omega_0\zeta \int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

□

- $\zeta = 1$ Critically damped response:

$$v = 2R\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_R = 2\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_L = {\omega_0}^2 \int_0^t (t-\tau)e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_C = i_{in} - \omega_0 \int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau$$

Proof.

$$v_1 = e^{-\omega_0 t}$$

$$v_2 = te^{-\omega_0 t}$$

$$\frac{1}{v_1v_2' - v_2v_1'} = e^{2\omega_0 t}$$

$$\begin{aligned}
v &= 2R\omega_0(-e^{-\omega_0 t}\int_0^t te^{\omega_0\tau}\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau + te^{-\omega_0 t}\int_0^t e^{\omega_0\tau}\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau) \\
&= 2R\omega_0(-e^{-\omega_0 t}(te^{\omega_0 t}(i_{in} - i_{in}(0^-)) - \int_0^t (1+\omega_0\tau)e^{\omega_0\tau} i_{in}(\tau)\,\mathrm{d}\tau) + te^{-\omega_0 t}(e^{\omega_0 t}(i_{in} - i_{in}(0^-)) - \int_0^t \omega_0 e^{\omega_0\tau} i_{in}(\\
&= 2R\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$i_R = 2\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau$$

$$\begin{aligned}
i_C &= C\frac{\mathrm{d}v}{\mathrm{d}t} \\
&= i_{in} + \int_0^t \frac{\partial}{\partial t}((1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)})i_{in}(\tau)\,\mathrm{d}\tau \\
&= i_{in} - \omega_0 \int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

$$\begin{aligned}
i_L &= i_{in} - i_R - i_C \\
&= {\omega_0}^2 \int_0^t (t-\tau)e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

Check $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$:

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = 2R\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau$$

□

- $0 < \zeta < 1$ Underdamped response:

$$v = 2R\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_R = 2\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_L = \frac{\omega_0}{\sqrt{1-\zeta^2}} \int_0^t e^{-\omega_0\zeta(t-\tau)} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_C = i_{in} - \omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

Proof.

$$v_1 = e^{-\omega_0\zeta t} \cos\left(\omega_0\sqrt{1-\zeta^2}t\right)$$

$$v_2 = e^{-\omega_0\zeta t} \sin\left(\omega_0\sqrt{1-\zeta^2}t\right)$$

$$v_1 v_2' = e^{-2\omega_0\zeta t}(-\omega_0\zeta \sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \omega_0\sqrt{1-\zeta^2}\cos^2(\omega_0\sqrt{1-\zeta^2}t))$$

$$v_2 v_1' = e^{-2\omega_0\zeta t}(-\omega_0\zeta \sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) - \omega_0\sqrt{1-\zeta^2}\sin^2(\omega_0\sqrt{1-\zeta^2}t))$$

$$\frac{1}{v_1v_2' - v_2v_1'} = \frac{1}{e^{-2\omega_0\zeta t}(-\omega_0\zeta \sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + \omega_0\sqrt{1-\zeta^2}\cos^2(\omega_0\sqrt{1-\zeta^2}t) + \omega_0\zeta \text{ s}}$$

$$= \frac{1}{\omega_0\sqrt{1-\zeta^2}}e^{2\omega_0\zeta t}$$

$$v = \frac{2R\zeta}{\sqrt{1-\zeta^2}}(-e^{-\omega_0\zeta t}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\int_0^t e^{\omega_0\zeta\tau}\sin\left(\omega_0\sqrt{1-\zeta^2}\tau\right)\frac{\mathrm{d}i_{in}}{\mathrm{d}\tau}\,\mathrm{d}\tau + e^{-\omega_0\zeta t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)\int_0^t$$

$$= \frac{2R\zeta}{\sqrt{1-\zeta^2}}(-e^{-\omega_0\zeta t}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)(e^{\omega_0\zeta t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)(i_{in} - i_{in}(0^-)) - \omega_0\int_0^t e^{\omega_0\zeta\tau}(\sqrt{1-\zeta^2}\cos($$

$$= \frac{2R\omega_0\zeta}{\sqrt{1-\zeta^2}}(e^{-\omega_0\zeta t}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\int_0^t e^{\omega_0\zeta\tau}(\sqrt{1-\zeta^2}\cos\left(\omega_0\sqrt{1-\zeta^2}\tau\right) + \zeta\sin\left(\omega_0\sqrt{1-\zeta^2}\tau\right))i_{in}(\tau)$$

$$= \frac{2R\omega_0\zeta}{\sqrt{1-\zeta^2}}\int_0^t e^{-\omega_0\zeta(t-\tau)}(\sqrt{1-\zeta^2}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\cos\left(\omega_0\sqrt{1-\zeta^2}\tau\right) + \zeta\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)\sin\left(\omega_0\sqrt{}\right.$$

$$= 2R\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_R = 2\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$i_C = C\frac{\mathrm{d}v}{\mathrm{d}t}$$

$$= i_{in} + \int_0^t \frac{\partial}{\partial t}(e^{-\omega_0\zeta(t-\tau)}(\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)))i_{in}(\tau)\,\mathrm{d}\tau$$

$$= i_{in} - \omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) + \sqrt{1-\zeta^2}\sin(\omega$$

$$= i_{in} - \omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))i_{in}(\tau)\,\mathrm{d}\tau$$

$$\begin{aligned}
i_L &= i_{in} - i_R - i_C \\
&= \frac{\omega_0}{\sqrt{1-\zeta^2}} \int_0^t e^{-\omega_0\zeta(t-\tau)} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) i_{in}(\tau)\,\mathrm{d}\tau
\end{aligned}$$

Check $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$:

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = 2R\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)} (\zeta \cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}} \sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)) i_{in}(\tau)\,\mathrm{d}\tau$$

□

Forced response (Laplace domain):

$$\begin{aligned}
V &= \frac{I_{in}RLs}{RLCs^2 + Ls + R} \\
&= \frac{2I_{in}R\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
I_R &= \frac{I_{in}Ls}{RLCs^2 + Ls + R} \\
&= \frac{2I_{in}\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
I_L &= \frac{I_{in}R}{RLCs^2 + Ls + R} \\
&= \frac{I_{in}{\omega_0}^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
I_C &= \frac{I_{in}RLCs^2}{RLCs^2 + Ls + R} \\
&= \frac{I_{in}s^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

Proof.

$$\begin{aligned}
V &= \frac{I_{in}}{Cs + \frac{1}{R} + \frac{1}{Ls}} \\
&= \frac{I_{in}RLs}{RLCs^2 + Ls + R} \\
&= \frac{I_{in}\frac{s}{C}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \\
&= \frac{2I_{in}R\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
I_R &= \frac{I_{in}Ls}{RLCs^2 + Ls + R} \\
&= \frac{2I_{in}\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$
\begin{aligned}
I_L &= \frac{V}{Ls} \\
&= \frac{I_{in}R}{RLCs^2 + Ls + R} \\
&= \frac{I_{in}{\omega_0}^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

$$
\begin{aligned}
I_C &= VCs \\
&= \frac{I_{in}RLCs^2}{RLCs^2 + Ls + R} \\
&= \frac{I_{in}s^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

Check consistency:

- $\zeta > 1$ Overdamped response:

$$
\begin{aligned}
\mathcal{L}\{v\} &= 2R\omega_0\zeta\frac{s + \omega_0\zeta - \frac{\zeta}{\sqrt{\zeta^2-1}}\omega_0\sqrt{\zeta^2 - 1}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}I_{in} \\
&= \frac{2I_{in}R\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

- $\zeta = 1$ Critically damped response:

$$
\begin{aligned}
\mathcal{L}\{v\} &= 2R\omega_0(\frac{1}{s + \omega_0} - \frac{\omega_0}{s^2 + 2\omega_0 s + {\omega_0}^2})I_{in} \\
&= \frac{2I_{in}R\omega_0 s}{s^2 + 2\omega_0 s + {\omega_0}^2}
\end{aligned}
$$

- $0 < \zeta < 1$ Underdamped response:

$$
\begin{aligned}
\mathcal{L}\{v\} &= 2R\omega_0\frac{\zeta(s + \omega_0\zeta) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\omega_0\sqrt{1 - \zeta^2}}{s^2 + 2\omega_0 s + {\omega_0}^2})I_{in} \\
&= \frac{2I_{in}R\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}
$$

□

Total response (time domain):

- $\zeta > 1$ Overdamped response:

$$
v = 2R\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2 - 1}(t - \tau)\right) - \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sinh\left(\omega_0\sqrt{\zeta^2 - 1}(t - \tau)\right))i_{in}(\tau)\,\mathrm{d}\tau - (v(0^-)\frac{\zeta -}{2}
$$

$$
i_R = 2\omega_0\zeta\int_0^t e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2 - 1}(t - \tau)\right) - \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sinh\left(\omega_0\sqrt{\zeta^2 - 1}(t - \tau)\right))i_{in}(\tau)\,\mathrm{d}\tau - (\frac{v(0^-)}{R}\frac{\zeta -}{2}
$$

$$i_L = \frac{\omega_0}{\sqrt{\zeta^2-1}} \int_0^t e^{-\omega_0\zeta(t-\tau)} \sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) i_{in}(\tau)\,\mathrm{d}\tau + (\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} + i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta+\sqrt{\cdots}}{2\cdots}$$

$$i_C = i_{in} - \omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta\cosh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big) - \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\Big(\omega_0\sqrt{\zeta^2-1}(t-\tau)\Big)) i_{in}(\tau)\,\mathrm{d}\tau + (\frac{v(0^-\cdots}{R}\cdots$$

- $\zeta = 1$ Critically damped response:

$$v = 2R\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau + (v(0^-) - (v(0^-) + 2i_L(0^-)R)\omega_0 t)e^{-\omega_0 t}$$

$$i_R = 2\omega_0 \int_0^t (1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau + (\frac{v(0^-)}{R} - (\frac{v(0^-)}{R} + 2i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_L = {\omega_0}^2 \int_0^t (t-\tau)e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau + (i_L(0^-) + (\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_C = i_{in} - \omega_0 \int_0^t (2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)} i_{in}(\tau)\,\mathrm{d}\tau + (-\frac{v(0^-)}{R} - i_L(0^-) + (\frac{1}{2}\frac{v(0^-)}{R} + i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

- $0 < \zeta < 1$ Underdamped response:

$$v = 2R\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big)) i_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}(v(0^-\cdots$$

$$i_R = 2\omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big)) i_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}(\frac{v(0^-)}{R}\cdots$$

$$i_L = \frac{\omega_0}{\sqrt{1-\zeta^2}} \int_0^t e^{-\omega_0\zeta(t-\tau)} \sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) i_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}(i_L(0^-)\cos\Big(\omega_0\sqrt{1-\zeta^2}t\Big) + (\frac{1}{2}\frac{v(0^-)}{R\zeta\sqrt{1-\cdots}}\cdots$$

$$i_C = i_{in} - \omega_0 \int_0^t e^{-\omega_0\zeta(t-\tau)}(2\zeta\cos\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big) - \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}}\sin\Big(\omega_0\sqrt{1-\zeta^2}(t-\tau)\Big)) i_{in}(\tau)\,\mathrm{d}\tau + e^{-\omega_0\zeta t}((\cdots$$

Total response (Laplace domain):

$$\begin{aligned} V &= \frac{I_{in}RLs + v(0^-)RLCs - i_L(0^-)RL}{RLCs^2 + Ls + R} \\ &= \frac{2I_{in}R\omega_0\zeta s + v(0^-)s - 2i_L(0^-)R\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \end{aligned}$$

$$\begin{aligned} I_R &= \frac{I_{in}Ls + v(0^-)LCs - i_L(0^-)L}{RLCs^2 + Ls + R} \\ &= \frac{2I_{in}\omega_0\zeta s + \frac{v(0^-)s}{R} - 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \end{aligned}$$

$$\begin{aligned} I_L &= \frac{I_{in}R + i_L(0^-)RLCs + v(0^-)RC + i_L(0^-)L}{RLCs^2 + Ls + R} \\ &= \frac{I_{in}{\omega_0}^2 + i_L(0^-)s + \frac{v(0^-)\omega_0}{2R\zeta} + 2i_L(0^-)\omega_0\zeta}{s^2 + 2\omega_0\zeta s + {\omega_0}^2} \end{aligned}$$

$$I_C = \frac{I_{in}RLCs^2 - (v(0^-) + i_L(0^-)R)LCs - v(0^-)RC}{RLCs^2 + Ls + R}$$
$$= \frac{I_{in}s^2 - (\frac{v(0^-)}{R} + i_L(0^-))s - \frac{v(0^-)\omega_0}{2R\zeta}}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}$$

Steady state (time domain):

$$i_L = i_{in}$$
$$i_R = i_L = 0$$
$$v = 0$$

Constant source forced response (time domain):

- $\zeta > 1$ Overdamped response:

$$v = 2i_{in}Re^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2 - 1}}\sinh\left(\omega_0\sqrt{\zeta^2 - 1}t\right)$$
$$i_R = 2i_{in}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2 - 1}}\sinh\left(\omega_0\sqrt{\zeta^2 - 1}t\right)$$
$$i_L = i_{in}(1 - e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2 - 1}t\right) + \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sinh\left(\omega_0\sqrt{\zeta^2 - 1}t\right)))$$
$$i_C = i_{in}e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2 - 1}t\right) - \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sinh\left(\omega_0\sqrt{\zeta^2 - 1}t\right))$$

- $\zeta = 1$ Critically damped response:

$$v = 2i_{in}R\omega_0te^{-\omega_0t}$$
$$i_R = 2i_{in}\omega_0te^{-\omega_0t}$$
$$i_L = i_{in}(1 - (1 + \omega_0t)e^{-\omega_0t})$$
$$i_C = i_{in}(1 - \omega_0t)e^{-\omega_0t}$$

- $0 < \zeta < 1$ Underdamped response:

$$v = 2i_{in}Re^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{1 - \zeta^2}}\sin\left(\omega_0\sqrt{1 - \zeta^2}t\right)$$
$$i_R = 2i_{in}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{1 - \zeta^2}}\sin\left(\omega_0\sqrt{1 - \zeta^2}t\right)$$
$$i_L = i_{in}(1 - e^{-\omega_0\zeta t}(\cos\left(\omega_0\sqrt{\zeta^2 - 1}t\right) + \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sin\left(\omega_0\sqrt{\zeta^2 - 1}t\right)))$$
$$i_C = i_{in}e^{-\omega_0\zeta t}(\cos\left(\omega_0\sqrt{\zeta^2 - 1}t\right) - \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sin\left(\omega_0\sqrt{\zeta^2 - 1}t\right))$$

Constant source total response (time domain):

- $\zeta > 1$ Overdamped response:

$$v = 2i_{in}Re^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right)-(v(0^-)\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)R\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}+(v(0^-)$$

$$i_R = 2i_{in}e^{-\omega_0\zeta t}\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right)-(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})e^{-\omega_0(\zeta-\sqrt{\zeta^2-1})t}+(\frac{v(0^-)}{R}\frac{\zeta}{}$$

$$i_L = i_{in}(1-e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right)+\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right)))+(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-}}$$

$$i_C = i_{in}e^{-\omega_0\zeta t}(\cosh\left(\omega_0\sqrt{\zeta^2-1}t\right)-\frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}t\right))+(\frac{v(0^-)}{R}\frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}+i_L(0^-)\frac{\zeta}{\sqrt{\zeta^2-1}})\frac{\zeta}{}$$

- $\zeta = 1$ Critically damped response:

$$v = (v(0^-) + (2i_{in}R - v(0^-) - 2i_L(0^-)R)\omega_0 t)e^{-\omega_0 t}$$

$$i_R = (\frac{v(0^-)}{R} + (2i_{in} - \frac{v(0^-)}{R} - 2i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_L = i_{in} - (i_{in} - i_L(0^-) + (i_{in} - \frac{1}{2}\frac{v(0^-)}{R} - i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

$$i_C = (i_{in} - \frac{v(0^-)}{R} - i_L(0^-) - (i_{in} - \frac{1}{2}\frac{v(0^-)}{R} - i_L(0^-))\omega_0 t)e^{-\omega_0 t}$$

- $0 < \zeta < 1$ Underdamped response:

$$v = e^{-\omega_0\zeta t}(v(0^-)\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (2i_{in}R - v(0^-) - 2i_L(0^-)R)\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_R = e^{-\omega_0\zeta t}(\frac{v(0^-)}{R}\cos\left(\omega_0\sqrt{1-\zeta^2}t\right) + (i_{in} - \frac{v(0^-)}{R} - 2i_L(0^-))\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_L = i_{in}-e^{-\omega_0\zeta t}((i_{in}-i_L(0^-))\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)+(i_{in}-\frac{1}{2}\frac{v(0^-)}{R\zeta\sqrt{1-\zeta^2}}-i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}t\right))$$

$$i_C = e^{-\omega_0\zeta t}((i_{in}+\frac{v(0^-)}{R}+i_L(0^-))\cos\left(\omega_0\sqrt{1-\zeta^2}t\right)-(i_{in}-\frac{v(0^-)}{2R}\frac{1-2\zeta^2}{\zeta\sqrt{1-\zeta^2}}+i_L(0^-)\frac{\zeta}{\sqrt{1-\zeta^2}})\sin\left(\omega_0\sqrt{1-\zeta^2}\right.$$

Resonance state (time domain): Maximum energy stored E_m, total energy lost per period E_l, quality factor Q.

$$i_{in} = i_m\cos\left(\omega_0 t + \theta\right)$$

$$v(0^-) = i_m R$$

$$i_L(0^-) = 0$$

$$v = i_m R\cos\left(\omega_0 t + \theta\right)$$

$$i_C = -i_L = -i_m R\omega_0\sin\left(\omega_0 t + \theta\right) = \frac{i_m R}{L\omega_0}\sin\left(\omega_0 t + \theta\right)$$

$$E_m = \frac{C{i_m}^2R^2}{2}$$

$$E_l = \frac{{i_m}^2 R\pi}{\omega_0}$$

$$Q = CR\omega_0 = R\sqrt{\frac{C}{L}}$$

Resonance state (phasor domain):

$$\mathbf{I}_{in} = i_m e^{j\theta}$$

$$\mathbf{V} = i_m R e^{j\theta}$$

$$\mathbf{I}_C = -\mathbf{I}_L = -j i_m RC\omega_0 e^{j\theta} = -\frac{j i_m R}{L\omega_0} e^{j\theta}$$

Impulse response:

- $\zeta > 1$ Overdamped response:

$$h_v = 2R\omega_0\zeta e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))$$

$$h_{i_R} = 2\omega_0\zeta e^{-\omega_0\zeta(t-\tau)}(\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))$$

$$h_{i_L} = \frac{\omega_0}{\sqrt{\zeta^2-1}} e^{-\omega_0\zeta(t-\tau)}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right)$$

$$h_{i_C} = \delta(t) - \omega_0 e^{-\omega_0\zeta(t-\tau)}(2\zeta\cosh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right) - \frac{2\zeta^2-1}{\sqrt{\zeta^2-1}}\sinh\left(\omega_0\sqrt{\zeta^2-1}(t-\tau)\right))$$

- $\zeta = 1$ Critically damped response:

$$h_v = 2R\omega_0(1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}$$

$$h_{i_R} = 2\omega_0(1-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}$$

$$h_{i_L} = {\omega_0}^2(t-\tau)e^{-\omega_0(t-\tau)}$$

$$h_{i_C} = \delta(t) - \omega_0(2-\omega_0(t-\tau))e^{-\omega_0(t-\tau)}$$

- $0 < \zeta < 1$ Underdamped response:

$$h_v = 2R\omega_0 e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))$$

$$h_{i_R} = 2\omega_0 e^{-\omega_0\zeta(t-\tau)}(\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{\zeta^2}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))$$

$$h_{i_L} = \frac{\omega_0}{\sqrt{1-\zeta^2}} e^{-\omega_0\zeta(t-\tau)}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right)$$

$$h_{i_C} = \delta(t) - \omega_0 e^{-\omega_0\zeta(t-\tau)}(2\zeta\cos\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right) - \frac{2\zeta^2-1}{\sqrt{1-\zeta^2}}\sin\left(\omega_0\sqrt{1-\zeta^2}(t-\tau)\right))$$

Transfer function:

$$\begin{aligned}
H_V &= \frac{RLs}{RLCs^2 + Ls + R} \\
&= \frac{2R\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
H_{I_R} &= \frac{Ls}{RLCs^2 + Ls + R} \\
&= \frac{2\omega_0\zeta s}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
H_{I_L} &= \frac{R}{RLCs^2 + Ls + R} \\
&= \frac{{\omega_0}^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

$$\begin{aligned}
H_{I_C} &= \frac{RLCs^2}{RLCs^2 + Ls + R} \\
&= \frac{s^2}{s^2 + 2\omega_0\zeta s + {\omega_0}^2}
\end{aligned}$$

I_L being low-pass filter:

$$\lim_{\omega\to 0} \left| H_{I_L}(j\omega) \right| = 1$$

$$\lim_{\omega\to \infty} \left| H_{I_L}(j\omega) \right| = 0$$

I_C being high-pass filter:

$$\lim_{\omega\to 0} \left| H_{I_C}(j\omega) \right| = 0$$

$$\lim_{\omega\to \infty} \left| H_{I_C}(j\omega) \right| = 1$$

XIII Transformer

i Ideal transformer

An ideal transformer is a two-port, linear, time-invariant, lossless network consists of two magnetically coupled windings, arbitrarily assigned as primary and secondary coil respectively, where the primary coil has N_1 turns, voltage v_1, and current i_1, and the secondar coil has N_2 turns, voltage v_2, and current i_2. Here, the voltage and current of the two coils are assumed to be measured from opposite directions, that is, dots on the same side. The turn ratio n is defined as

$$n = \frac{N_1}{N_2}.$$

Magnetic coupling of same magnetic flux per turn enforces

$$\frac{v_1}{N_1} = \frac{v_2}{N_2}.$$

Power consistency enforces

$$v_1 i_1 = v_2 i_2.$$

Thus

$$\frac{i_1}{N_2} = \frac{i_2}{N_1}.$$

That is,

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix} \begin{bmatrix} v_2 \\ i_2 \end{bmatrix}.$$

ii Diagram

Transformer Transformer with Core

iii Dot convention

On diagram, a dot is drawn on one terminal of each coil. If dots of the two coils are on the same side, the voltages (and currents) of them as measured in the same direction are in-phase; if dots of the two coils are on the opposite side, the voltages (and currents) of them as measured in the same direction are out-of-phase. Equivalently, if the voltage (and current) of one coil is measured from the dotted terminal to the undotted terminal and the voltage (and current) of the other coil is measured from the undotted terminal to the dotted terminal, the measured voltages (and currents) of them in-phase.

iv Coupled inductors model

Primary and secondary coils are inductors with self inductance L_1 and L_2 and mutual inductance M, that is,

$$\begin{aligned} v_1 &= L_1 \frac{\mathrm{d}i_1}{\mathrm{d}dt} + M \frac{\mathrm{d}i_2}{\mathrm{d}t} \\ v_2 &= L_2 \frac{\mathrm{d}i_2}{\mathrm{d}dt} + M \frac{\mathrm{d}i_1}{\mathrm{d}t}, \end{aligned}$$

with constraints that inductive coupling factor is one,

$$\frac{M}{\sqrt{L_1 L_2}} = 1$$

and

$$L_1 = n^2 L_2$$

i.e.,

$$\begin{aligned} M &= n L_2 \\ L_1 &= n^2 L_2 \end{aligned}$$

i.e.,

$$\begin{aligned} M &= \frac{L_1}{n} \\ L_2 &= \frac{L_1}{n^2}. \end{aligned}$$

This leaves one degree of freedom, thus an arbitrary positive L_1 or L_2 can be assigned.

v VCVS and CCVS model

Primary coil is replaced with a VCVS

$$v_1 = nv_2.$$

Second coil is replaced with a CCCS

$$i_2 = ni_1.$$

vi Impedance reflection model

Only works if the two coils' circuit are only coupled by the transformer.

If a load impedance Z_s is on secondary coil, seen from primary coil, it's a load impedance

$$Z_p = n^2 Z_s.$$

XIV Operational Amplifier (op amp or op-amp)

i Definition

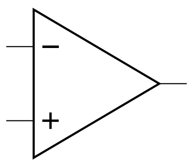
An op amp has five terminals:

- v_+: non-inverting input voltage of the terminal at the + sign;
- v_-: inverting input voltage of the terminal at the − sign;
- v_o: output voltage of the terminal at the opposite side of inputs;
- v_{CC+}: positive supply voltage, denoted on the + side of an op-amp or omitted;
- v_{CC-}: negative supply voltage, denoted on the − side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os}),$$

where A is open-loop gain, and v_{os} is input offset voltage.

ii Diagram



iii Ideal op amp

- Infinite open-loop gain, infinite bandwidth, and zero output impedance: $A = \infty$ and is independent of frequency and output voltage.
- Zero input offset voltage: $v_{os} = 0$.
- No current flow into either of the input terminals.

This is a linear element with inputs v_+ and v_- and output v_o.

iv General amplifier with inverting configuration

A general amplifier with inverting configuration consists of an op amp where input voltage v_i is connected to v_- through an input impedance Z_i, v_+ is connected to ground, and v_o is fed back to v_- through a feedback impedance Z_f.

$$I_i = \frac{V_i}{Z_i} = I_f = \frac{-V_o}{Z_f},$$
$$V_o = -\frac{Z_f}{Z_i}V_i.$$

When $\left|\Re(\frac{Z_f}{Z_i})\right| < 1$, it's called an attenuator.

v Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i, v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f.

$$i_i = \frac{v_i}{R_i} = i_f = \frac{-v_o}{R_f},$$
$$v_o = -\frac{R_f}{R_i}v_i.$$

vi General amplifier with noninverting configuration

A general amplifier with noninverting configuration consist of an op amp where input voltage v_i is connected to v_+, v_- is connected ground through a ground impedance Z_g, and v_o is fed back to v_- through a feedback impedance Z_f.

$$V_i = V_o\frac{Z_g}{Z_f + Z_g},$$
$$V_o = (1 + \frac{Z_f}{Z_g})V_i.$$

vii Noninverting amplifier

A noninverting amplifier consist of an op amp where input voltage v_i is connected to v_+, v_- is connected ground through a ground resistor R_g, and v_o is fed back to v_- through a feedback resistor R_f.

$$v_i = v_o\frac{R_g}{R_f + R_g},$$
$$v_o = (1 + \frac{R_f}{R_g})v_i.$$

viii Unity gain amplifier, unity gain buffer, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+ and v_o is fed back to v_-.

$$v_o = v_i.$$

ix Summing amplifier

A summing amplifier consists of an op amp, input voltages $v_1, v_2, \ldots, v_n$, n input resistors $R_1, R_2, \ldots, R_n$, and a feedback resistor R_f, where for any $k \in \mathbb{N} \wedge k \le n$, v_k is connected to v_- through R_k, v_+ is connected to ground, and v_o is fed back to v_- through R_f.

$$i_k = \frac{v_k}{R_k},$$

$$\sum_{k=1}^{n} i_k = i_f = \frac{-v_o}{R_f},$$

$$v_o = -R_f \sum_{k=1}^{n} \frac{v_k}{R_k}.$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \le n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^{n} v_k.$$

x Difference amplifier or differential amplifier

A difference amplifier consists of a minuend v_1, a subtrahend v_2, and four resistors R_1, R_2, R_3, R_4 satisfying the condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4},$$

where v_1, v_2 are connected to v_-, v_+ through R_1, R_2 respectively, v_o is fed back to v_- through R_3, and v_+ is connected to ground through R_4.

$$v_+ = v_2 \frac{R_4}{R_2 + R_4},$$

$$\frac{v_1 - v_-}{R_1} = \frac{v_- - v_o}{R_3},$$

$$\frac{v_1}{R_2} - v_2 \frac{R_4}{R_2 + R_4} \frac{R_2 + R_4}{R_2 R_4} + \frac{v_o}{R_4} = 0,$$

$$v_o = \frac{R_4}{R_2}(v_2 - v_1).$$

When $R_1 = R_2 = R_3 = R_4$:

$$v_o = (v_2 - v_1).$$

xi Comparator

A comparator is an op amp with v_+ and v_- connected to the voltage sources to be compared, in which v_o is high when $v_+ > v_-$ and low when $v_+ < v_-$.

xii Inverting integrator

An inverting integrator consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i, v_+ is connected to ground, and v_o is fed back to v_- through a feedback capacitor C_f.

Time domain:

$$i_i = \frac{v_i}{R_i} = i_f = -C_f \frac{\mathrm{d}v_o}{\mathrm{d}t},$$

$$v_o = v_o(0^-) - \int_0^t \frac{v_i(\tau)}{R_i C_f}\,\mathrm{d}\tau\,.$$

Laplace domain:

$$I_i = \frac{V_i}{R_i} = I_f = -C_f s V_o + C_f v_o(0^-),$$

$$V_o = \frac{v_o(0^-)}{s} - \frac{V_i}{R_i C_f s}.$$

xiii Inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_i is connected to v_- through an input capacitor C_i, v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f.

Time domain:

$$i_i = C_i \frac{\mathrm{d}v_i}{\mathrm{d}t} = i_f = \frac{-v_o}{R_f},$$

$$v_o = R_f C_i (v_i(0^-) - \int_0^t v_i(\tau)\,\mathrm{d}\tau).$$

Laplace domain:

$$I_i = C_i s V_i - C_i v_i(0^-) = I_f = \frac{-V_o}{R_f},$$

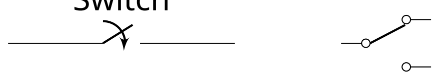
$$V_o = R_f C_i (v_i(0^-) - s V_i).$$

xiv Active filter

Given a (passive or active) filter v_f of an input v_s and an inverting amplifier, noninverting amplifier, or unit gain buffer enforcing $v_o = K v_i$, by connecting the ground and v_i of the amplifier to the two terminals of the v_f, the two terminals ground and v_o become an active filter of v_s enforcing $v_o = K v_f$.

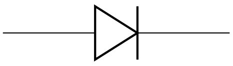
xv Switches

Switch SPDT Switch

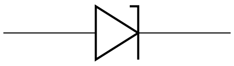


xvi Diodes

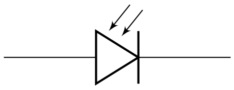
Diode



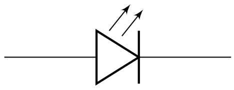
Zener Diode



Photodiode

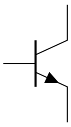


LED

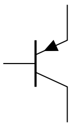


xvii Transistors

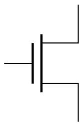
NPN Transistor



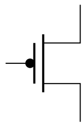
PNP Transistor



N-MOSFET

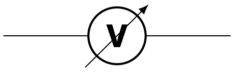


P-MOSFET

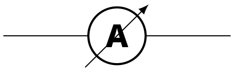


xviii Measuring Instruments

Voltmeter



Ammeter



Ohmmeter